

Fundamental Physics with PTAs

Andrea Mitridate

VIPER summer school | July 27, 2026



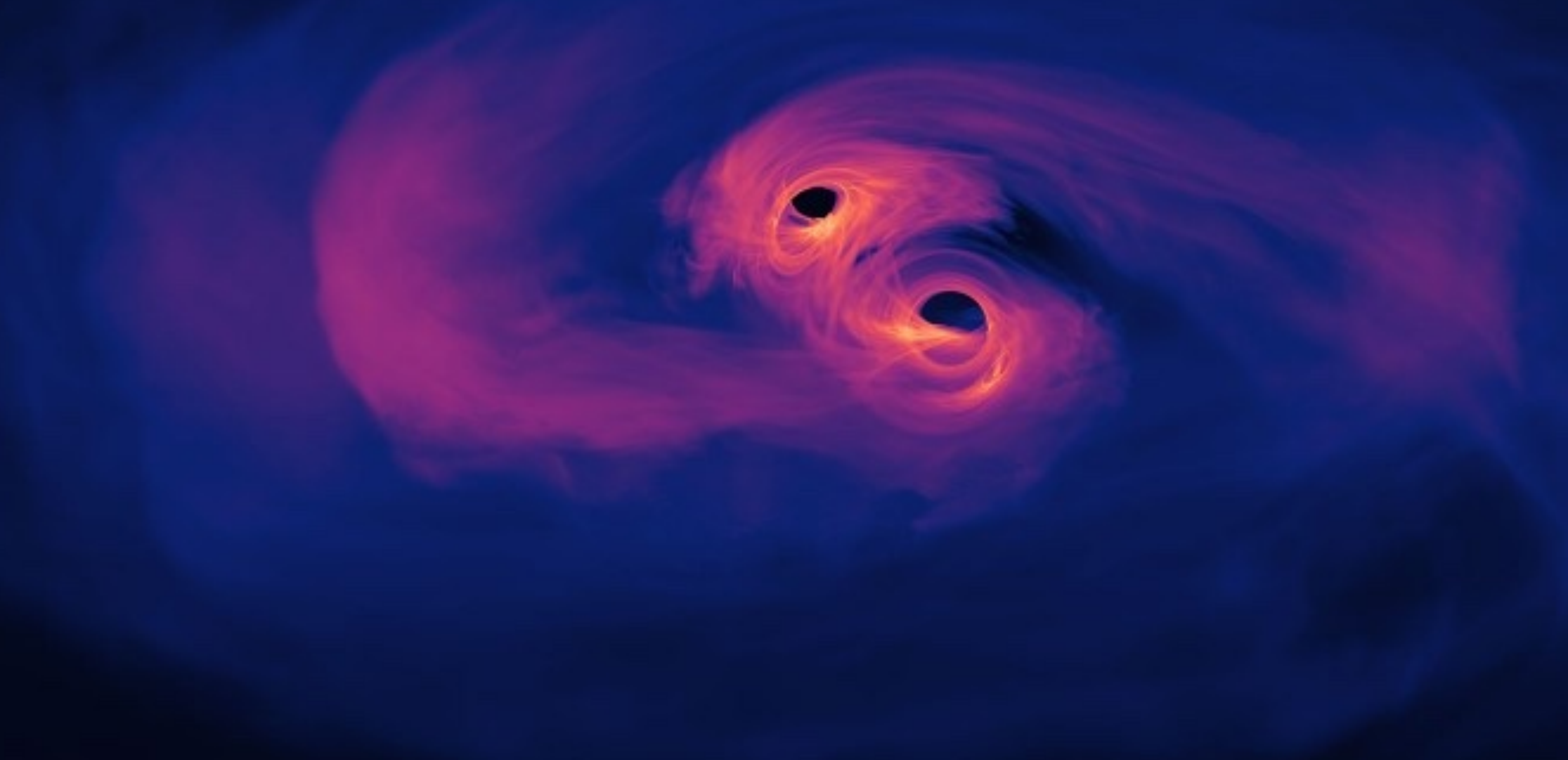
THE ROYAL SOCIETY

IMPERIAL

WHAT YOU HAVE ALREADY LEARNED

SMBHB are a great source of GWs at nHz frequencies

PTA observations *can* teach us a lot about their properties



WHAT YOU HAVE ALREADY LEARNED

SMBHB are a great source of GWs at nHz frequencies

PTA observations *can* teach us a lot about their properties



is there anything else that PTAs can teach us?

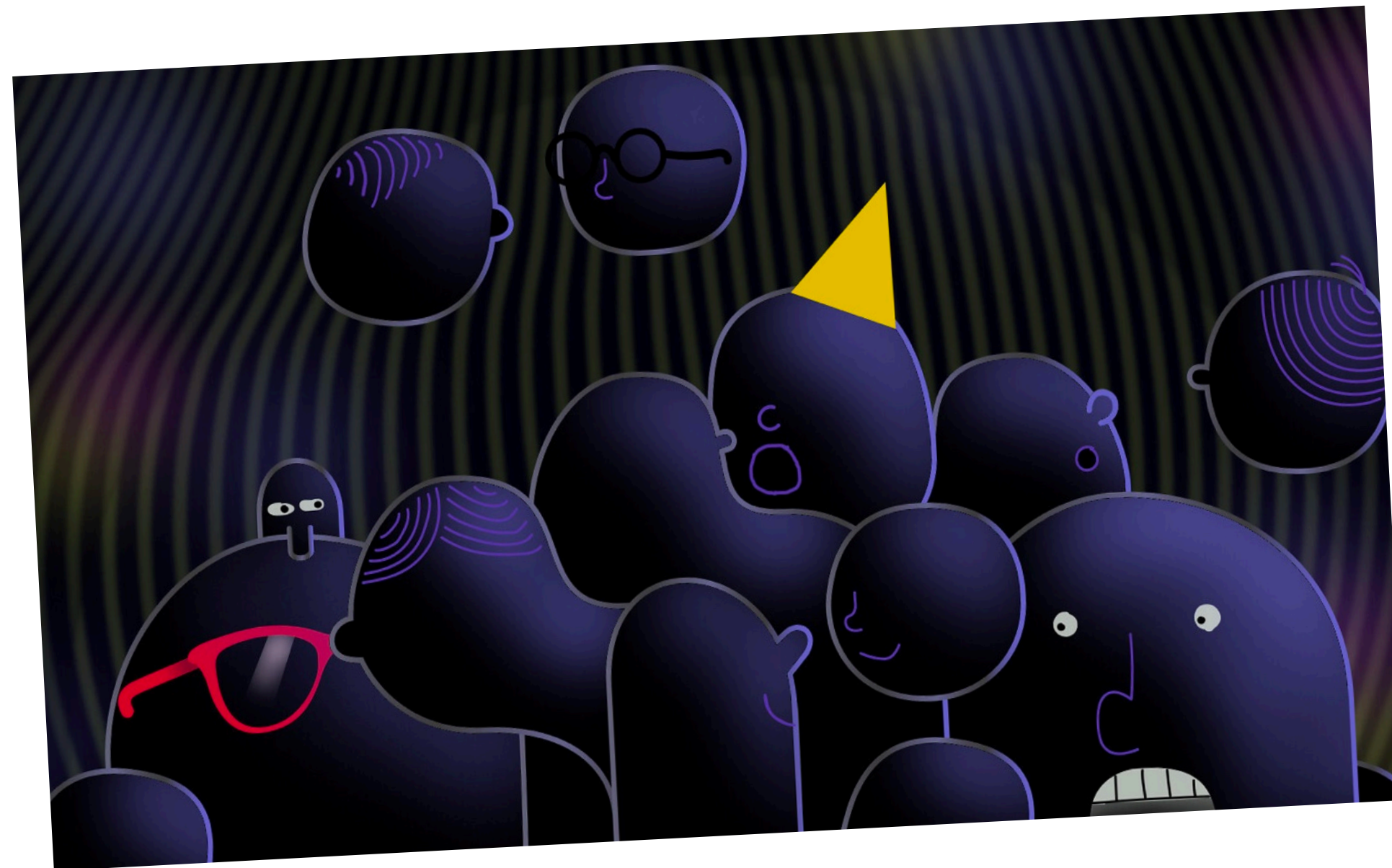
WHAT YOU WILL LEARN TODAY

How we can use PTAs to learn something about:

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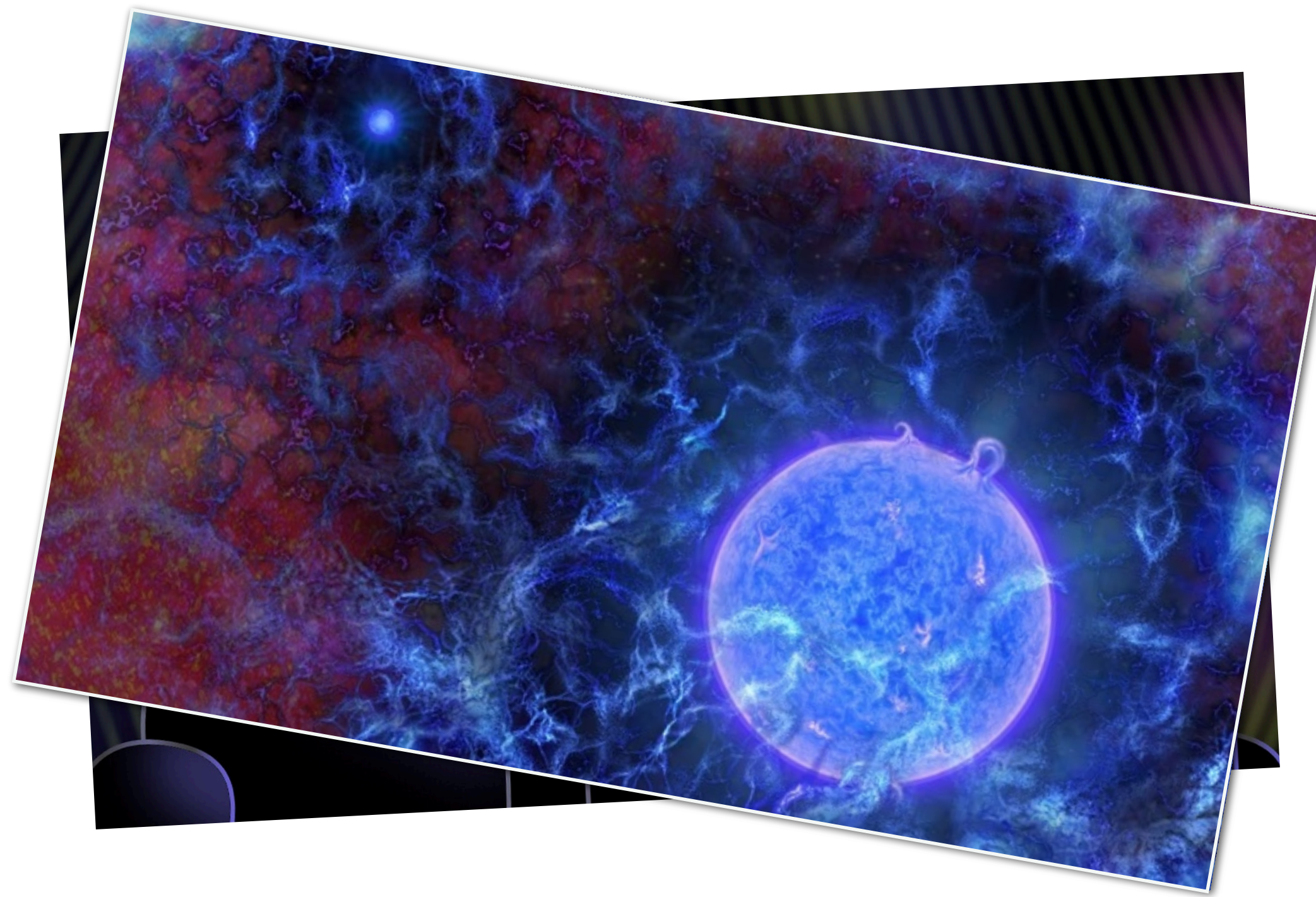
- dark matter



WHAT YOU WILL LEARN TODAY

How we can use PTAs to learn something about:

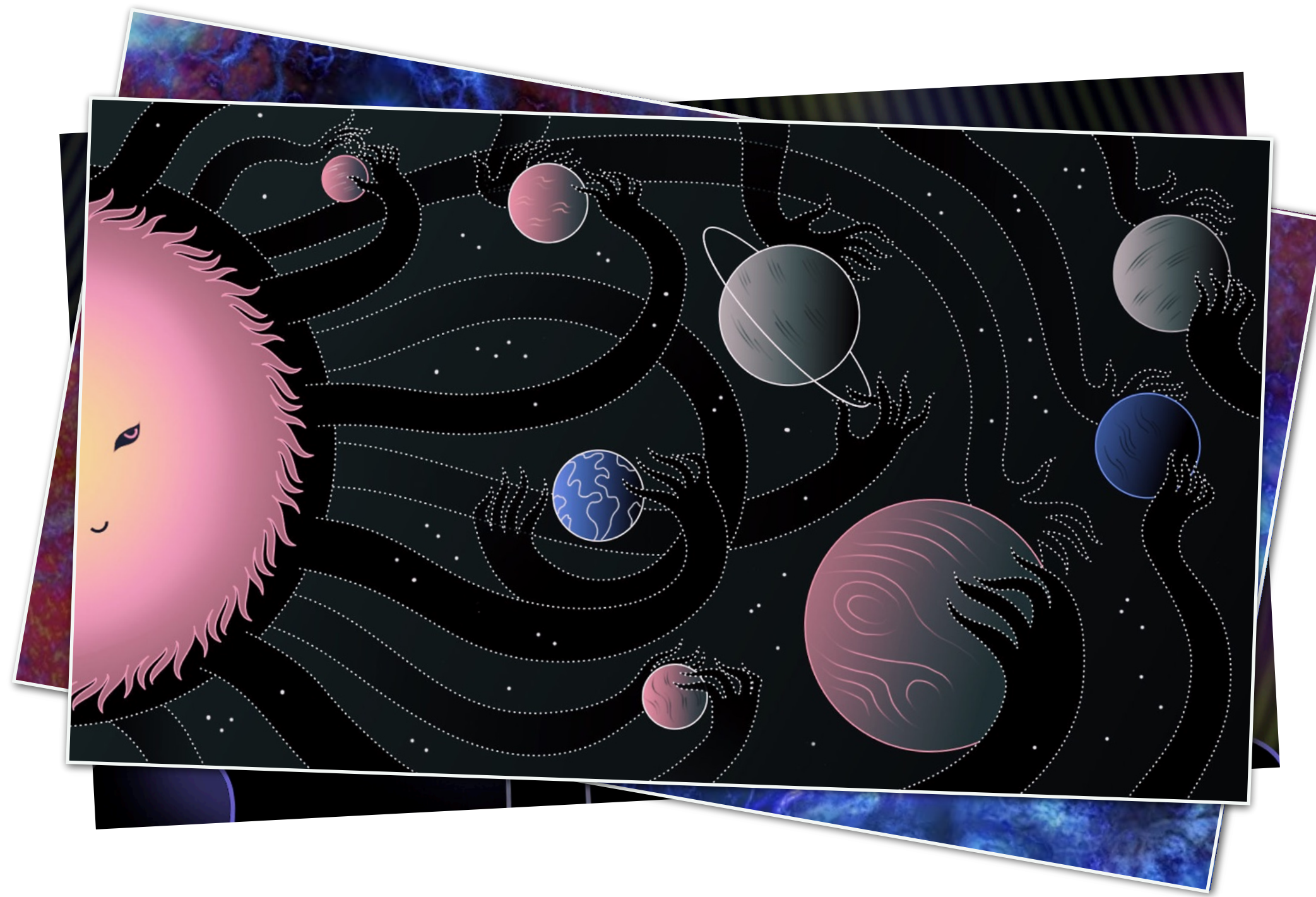
- dark matter
- the early universe



WHAT YOU WILL LEARN TODAY

How we can use PTAs to learn something about:

- dark matter
- the early universe
- the theory of gravity



dark matter

ISM PROPAGATION

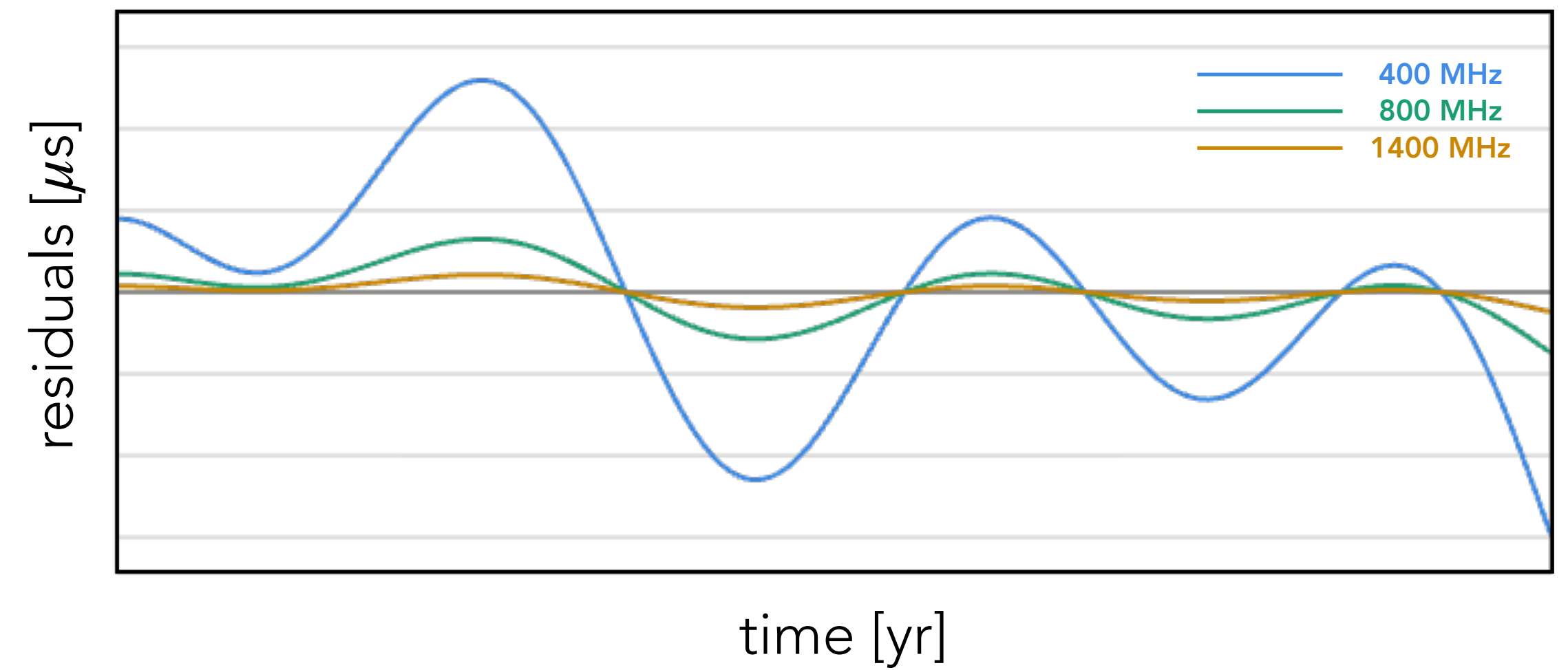


ISM PROPAGATION

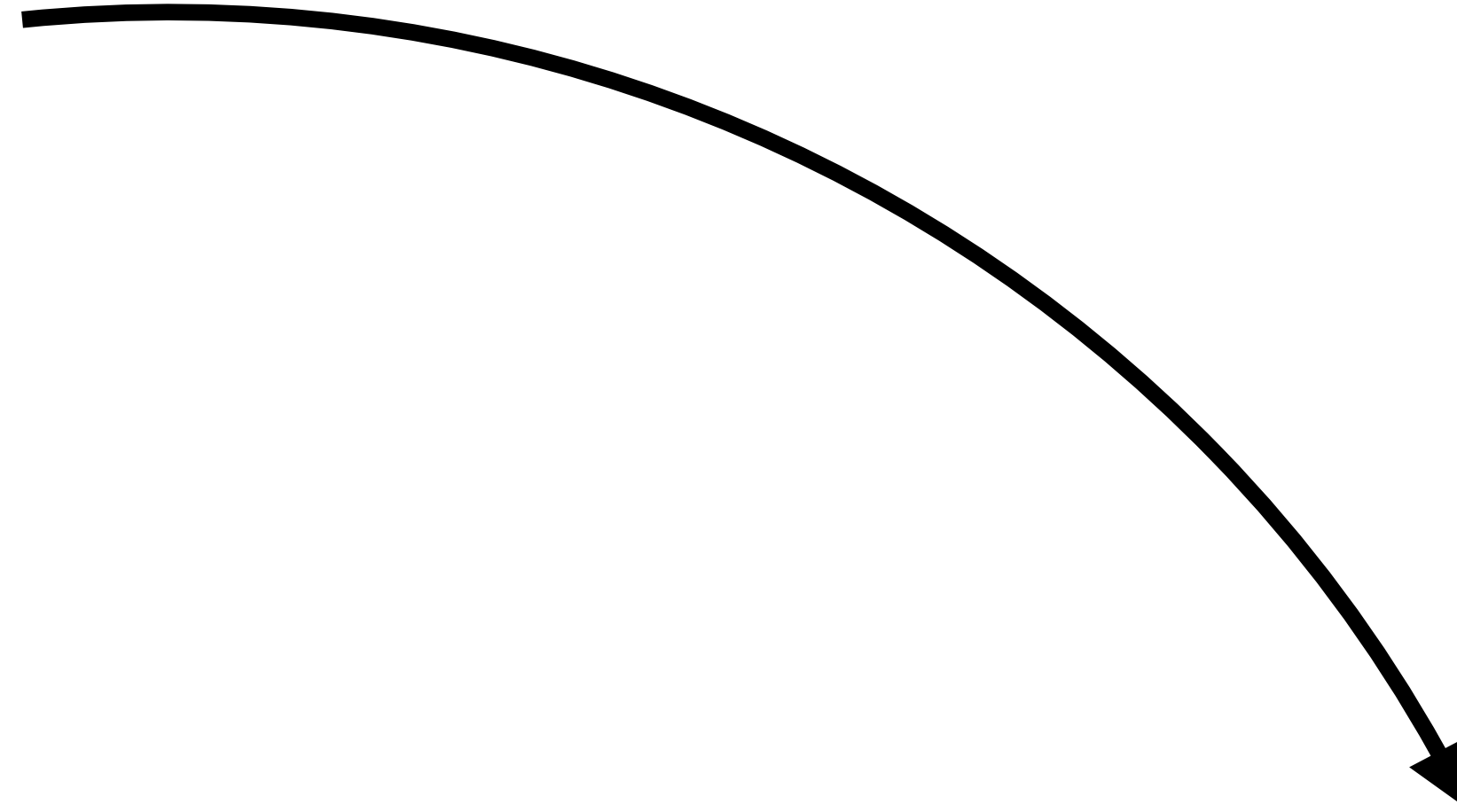
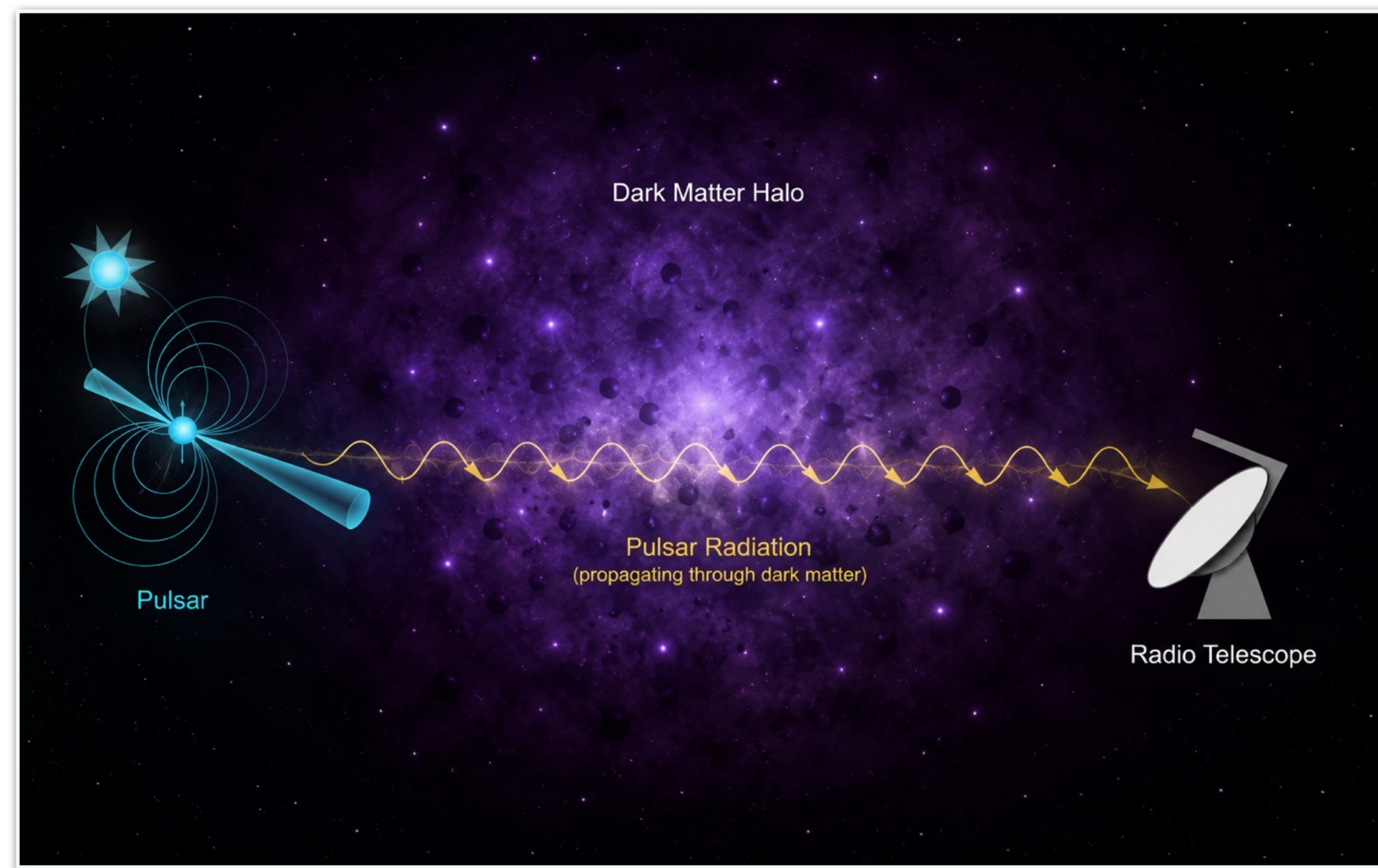


$$\delta t_a \propto \frac{\text{DM}}{\nu^2}$$

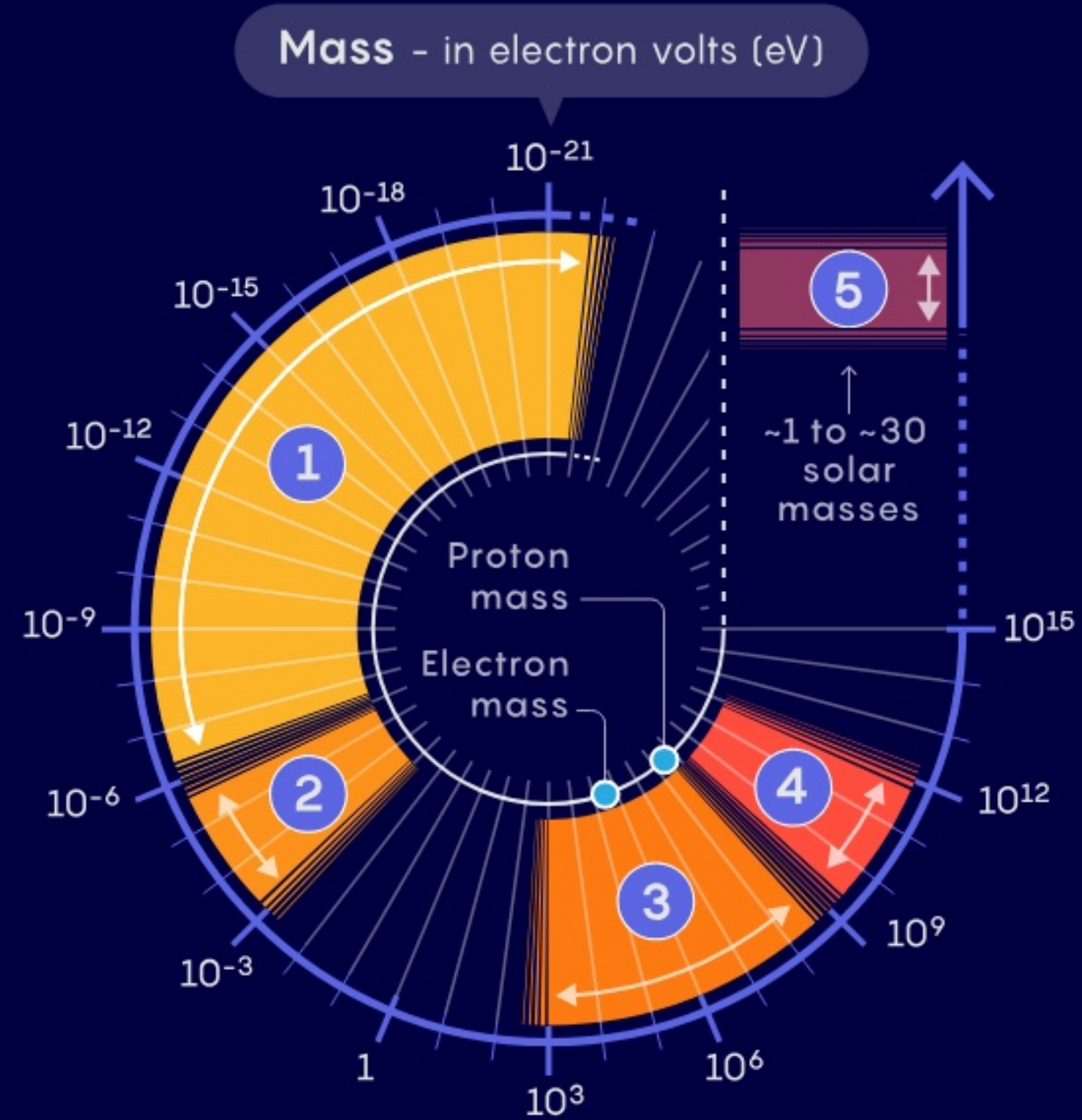
$$\text{DM} \equiv \int_0^{L_a} dx \, n_e(\hat{n}_a x)$$



DARK MATTER PROPAGATION

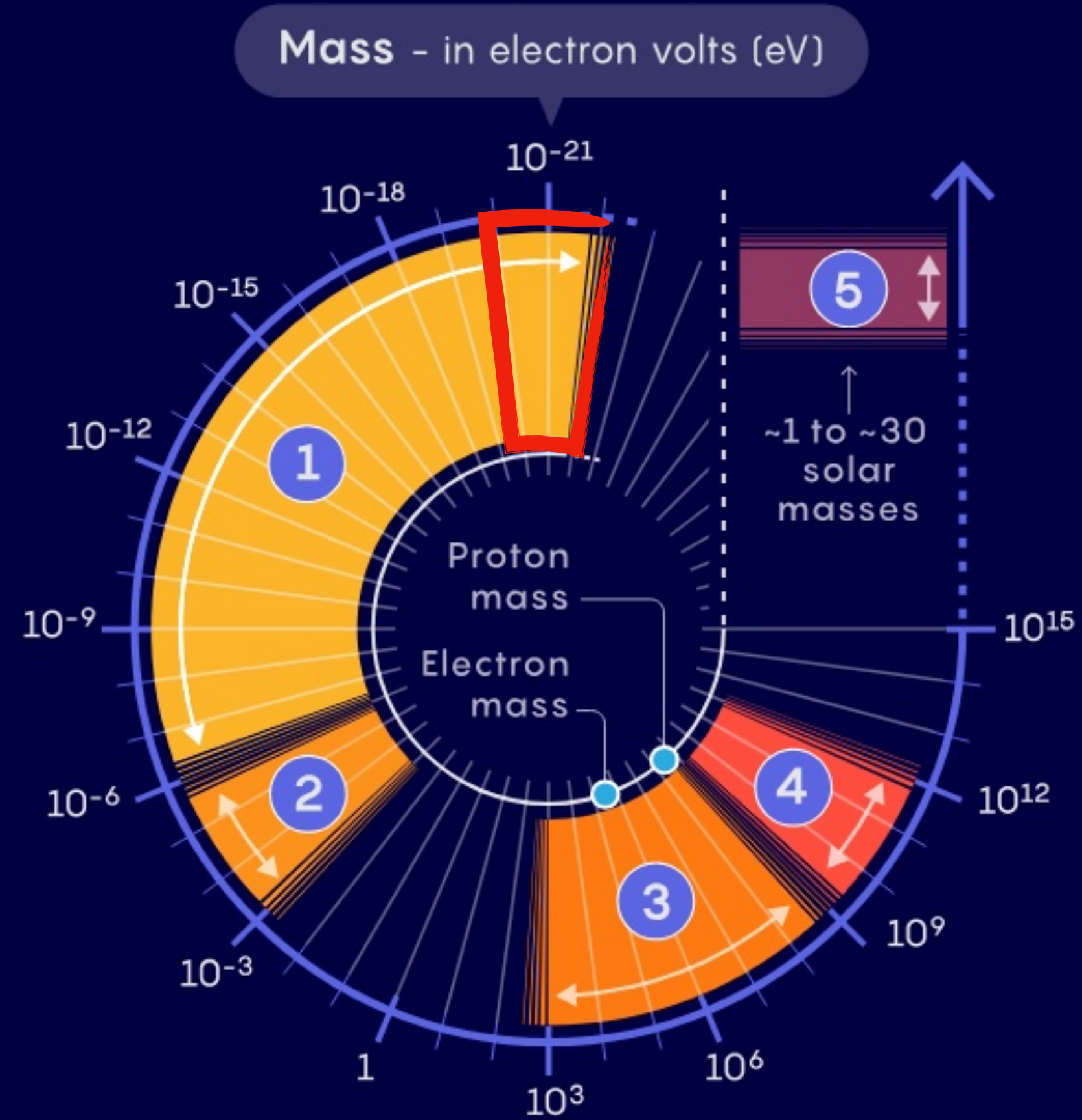


ULDM



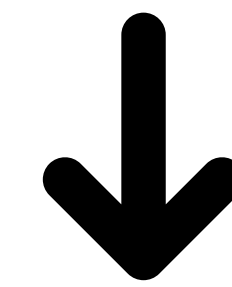
- 1 ULTRALIGHT DARK MATTER
- 2 AXIONS
- 3 SUB-GEV DARK MATTER
- 4 WIMPS
- 5 PRIMORDIAL BLACK HOLES

ULDM



- 1 ULTRALIGHT DARK MATTER
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$$m_{\text{ULDM}} \ll 1 \text{ eV}$$



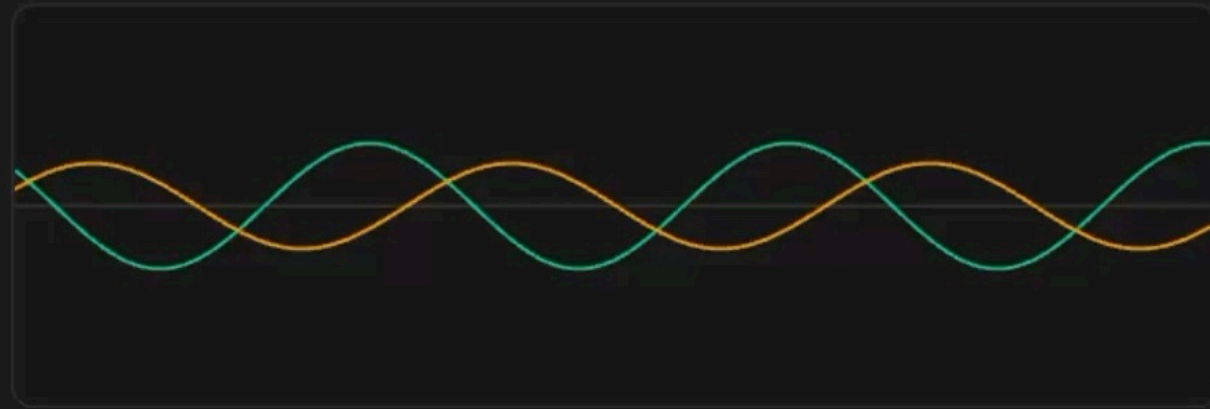
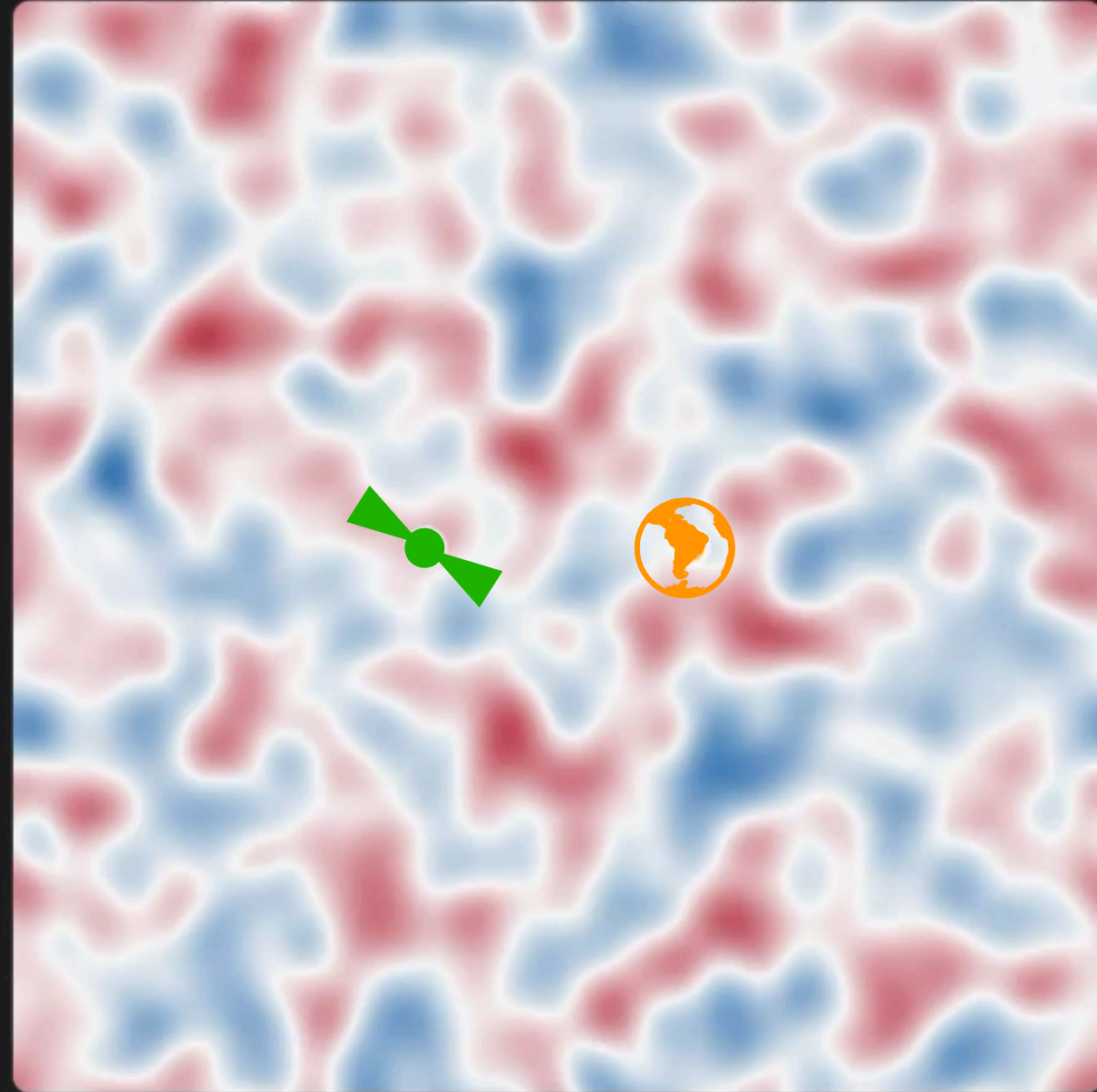
$$\rho_{\text{DM}} \sim 0.4 \text{ GeV cm}^{-3}$$

$$n_{\text{ULDM}} \gg 10^9 \text{ cm}^{-3}$$

within the Galactic halo
we can treat ULDM as a **classical field**

ULDM

$$\phi(t, \vec{x})$$



t

$$\phi(t, \vec{x}_1)$$

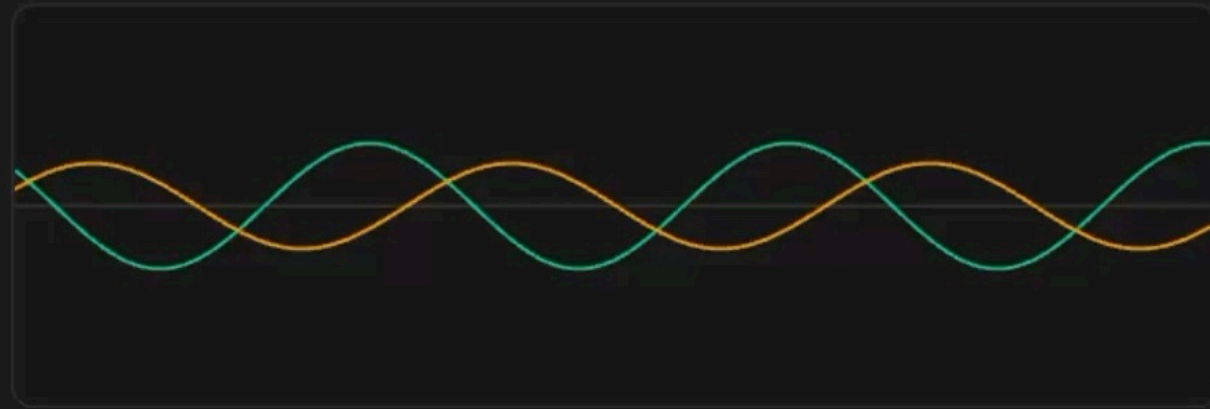
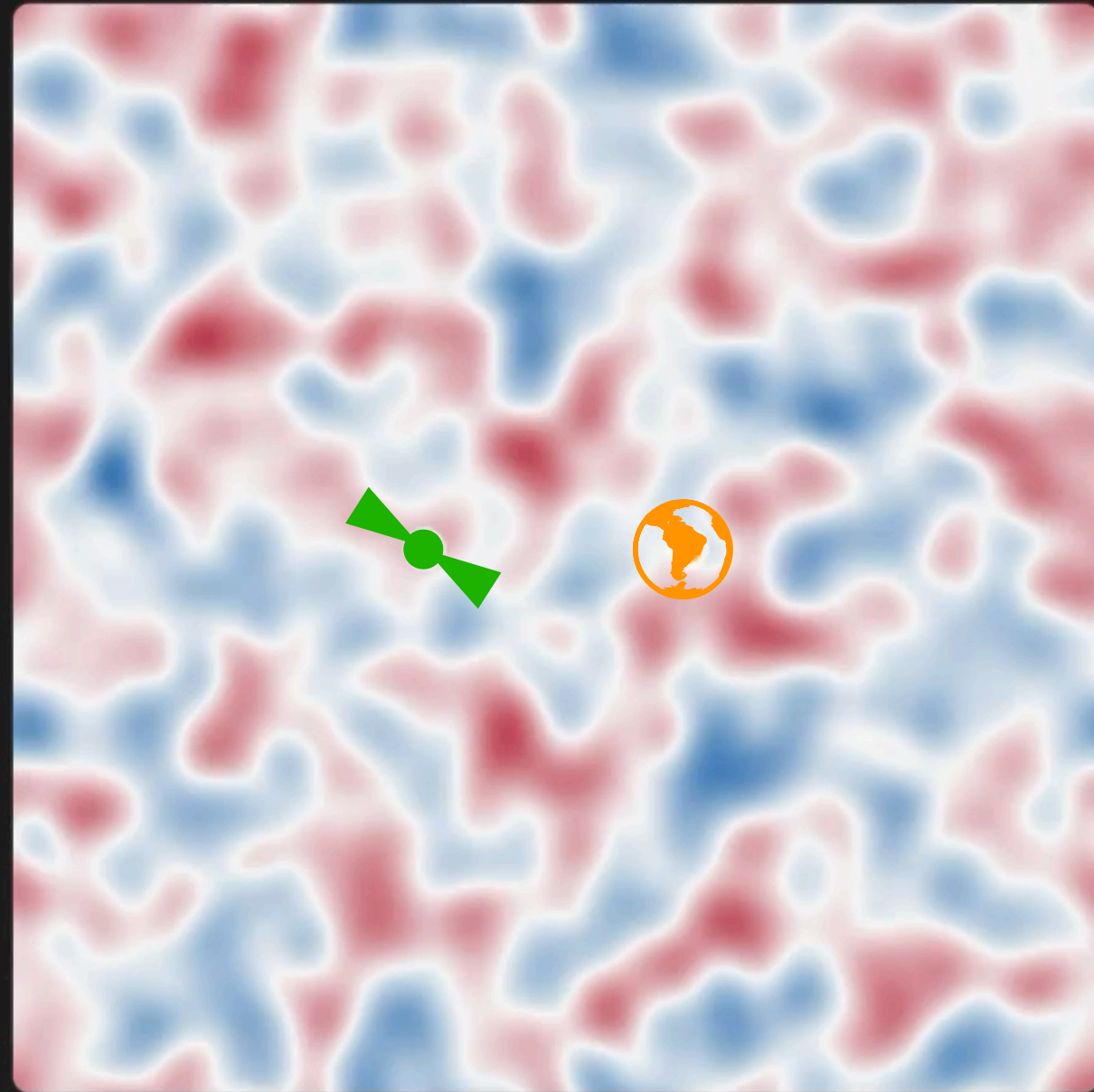
$$\phi(t, \vec{x}_2)$$

amplitude

$$\phi(t, \vec{x}) = \boxed{\frac{\sqrt{2\rho_\phi}}{m_\phi}} \hat{\phi}(\vec{x}) \cos \left[m_\phi t - m_\phi \vec{v} \cdot \vec{x} + \theta \right]$$

ULDM

$$\phi(t, \vec{x})$$



$$\phi(t, \vec{x}_1)$$

$$\phi(t, \vec{x}_2)$$

t

$$\phi(t, \vec{x}) = \frac{\sqrt{2\rho_\phi}}{m_\phi} \hat{\phi}(\vec{x}) \cos \left[m_\phi t - m_\phi \vec{v} \cdot \vec{x} + \theta \right]$$

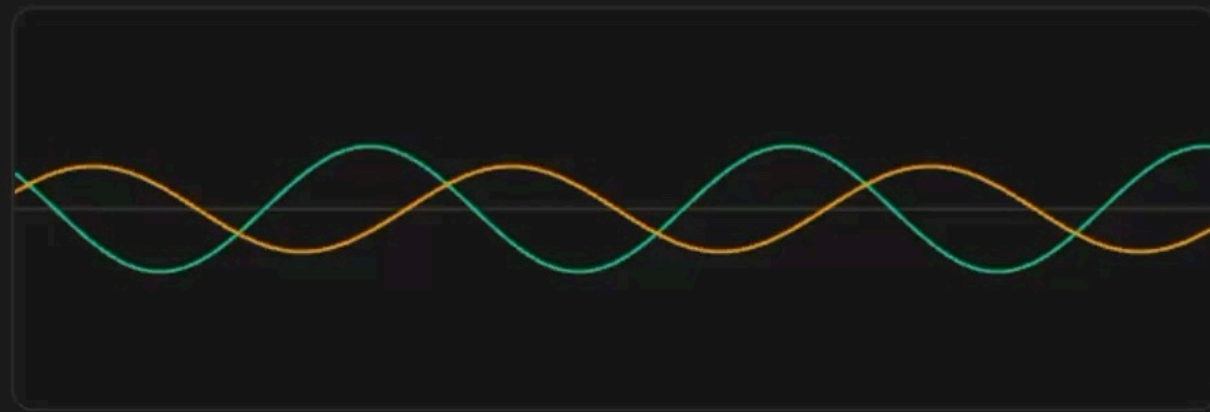
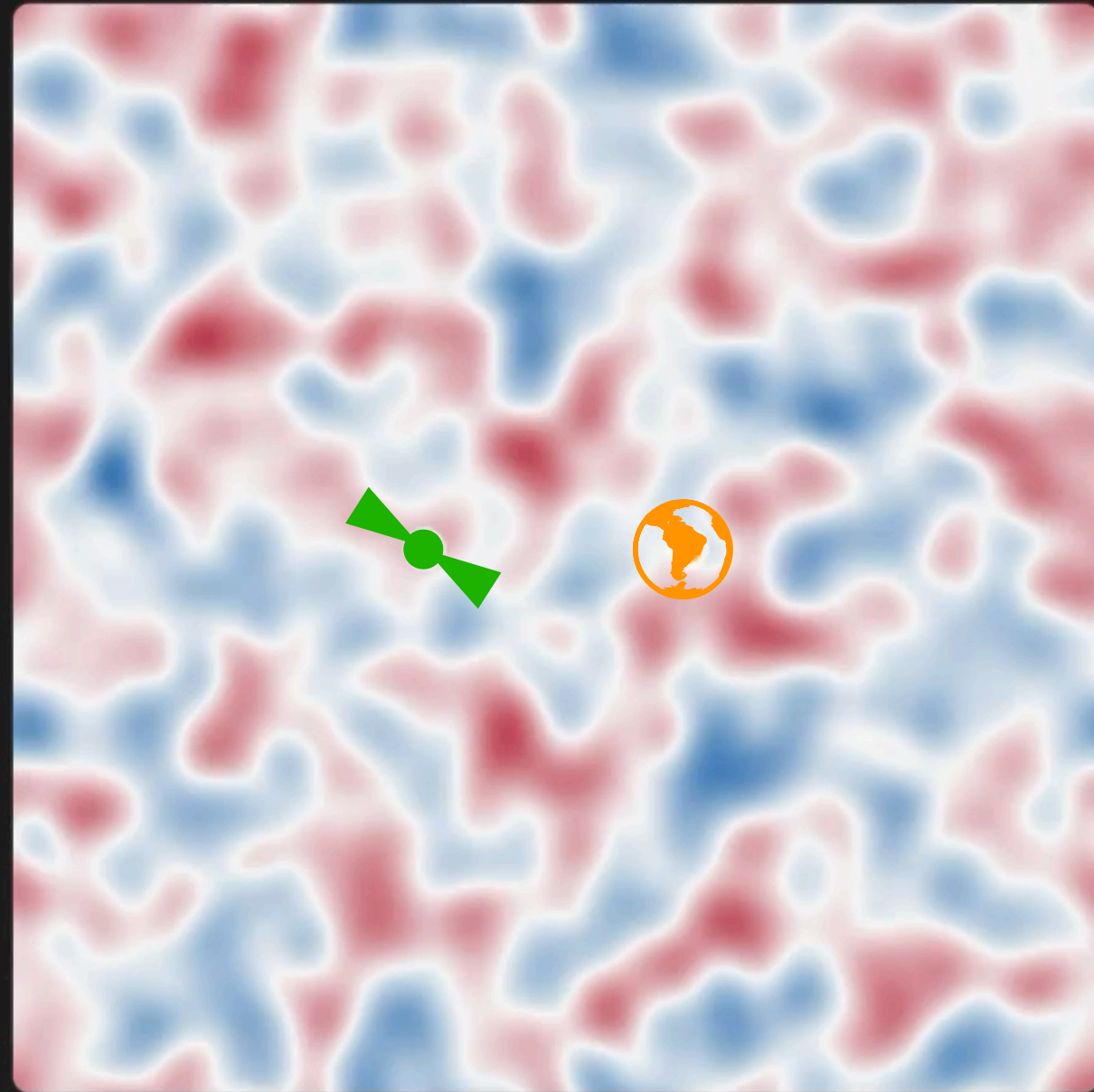
amplitude

time fluctuations

$$f \sim 2 \text{ nHz} \left(\frac{m_\phi}{10^{-23} \text{ eV}} \right)$$

ULDM

$$\phi(t, \vec{x})$$



$$\phi(t, \vec{x}_1)$$

$$\phi(t, \vec{x}_2)$$

t

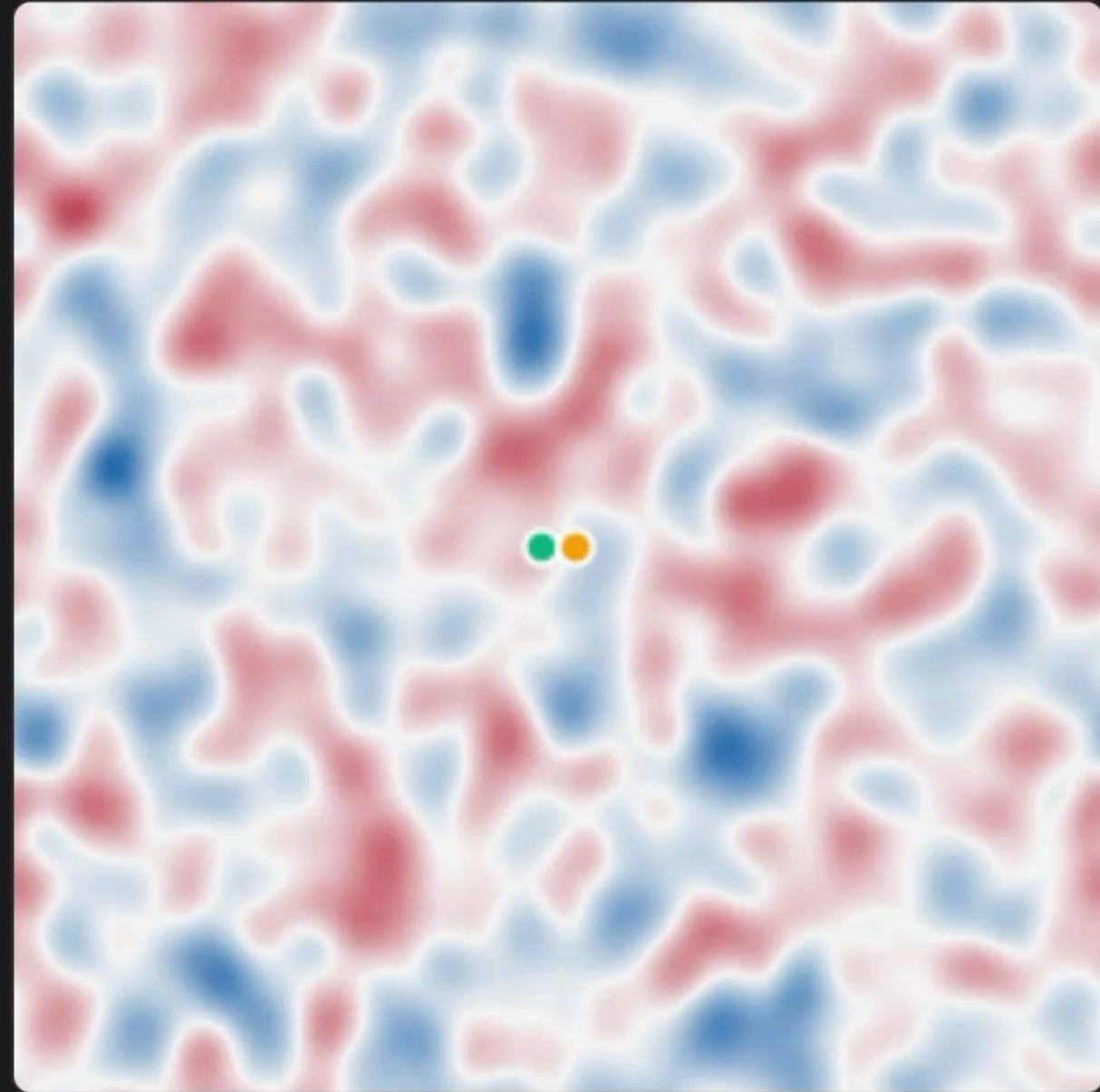
$$\phi(t, \vec{x}) = \underbrace{\frac{\sqrt{2\rho_\phi}}{m_\phi}}_{\text{amplitude}} \hat{\phi}(\vec{x}) \cos \left[\underbrace{m_\phi t}_{\text{time fluctuations}} - \underbrace{m_\phi \vec{v} \cdot \vec{x} + \theta}_{\text{spatial fluctuations}} \right]$$

$$\lambda \sim 1.2 \text{ kpc} \left(\frac{10^{-23} \text{ eV}}{m_\phi} \right)$$

$$f \sim 2 \text{ nHz} \left(\frac{m_\phi}{10^{-23} \text{ eV}} \right)$$

ULDM

$$\phi(t, \vec{x})$$



t

$$\phi(t, \vec{x}_1)$$

$$\phi(t, \vec{x}_2)$$

$$\phi(t, \vec{x}) = \frac{\sqrt{2\rho_\phi}}{m_\phi} \boxed{\hat{\phi}(\vec{x})} \cos \left[m_\phi t - m_\phi \vec{v} \cdot \vec{x} + \theta \right]$$



random variable that captures the random nature of the ULDM field

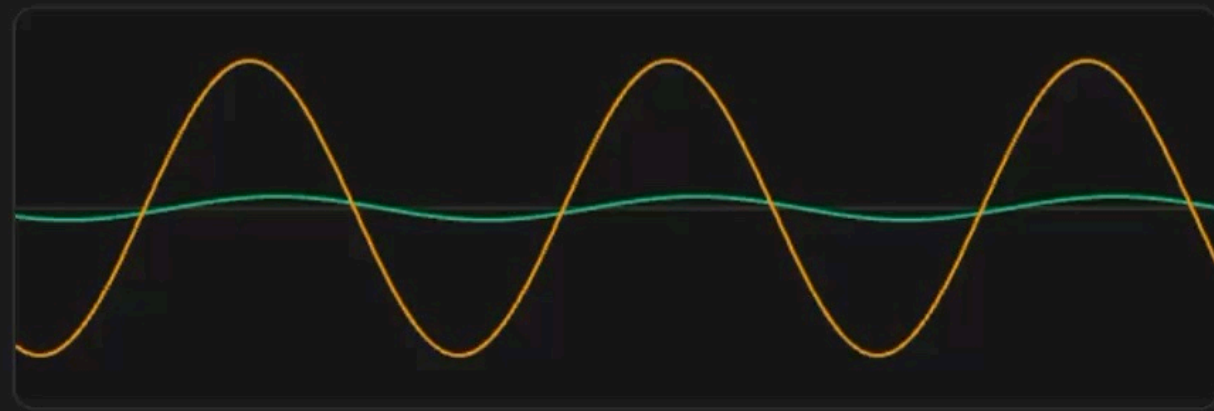
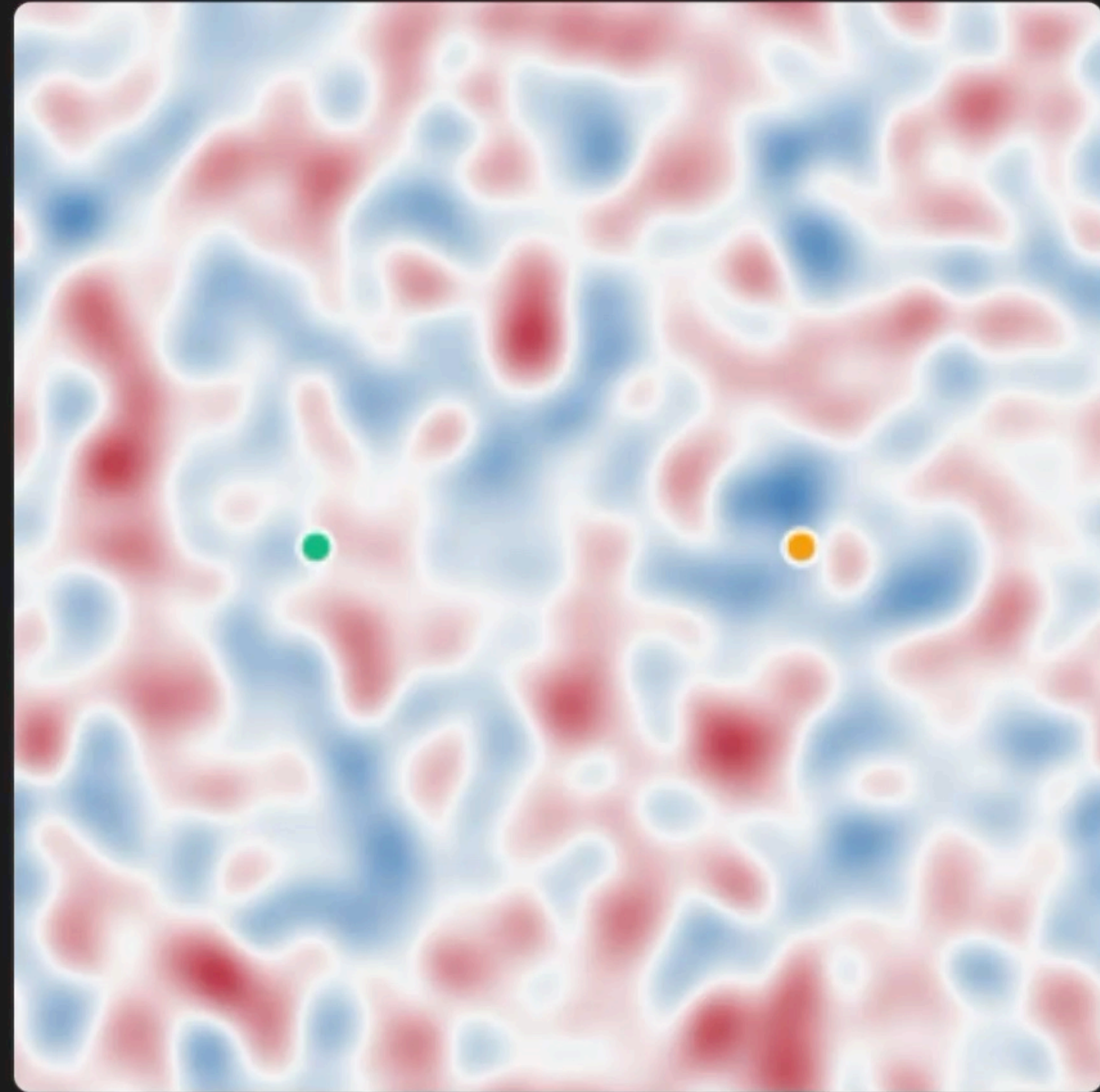
$$\langle \hat{\phi}(\vec{x}_1) \hat{\phi}(\vec{x}_2) \rangle \sim 1$$

if

$$|\vec{x}_1 - \vec{x}_2| \ll \ell_c \sim 1.2 \text{ kpc} \left(\frac{10^{-23} \text{ eV}}{m_\phi} \right)$$

ULDM

$$\phi(t, \vec{x})$$



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random variable that captures the random nature of the ULDM field

$$\langle \hat{\phi}(\vec{x}_1) \hat{\phi}(\vec{x}_2) \rangle \sim 0$$

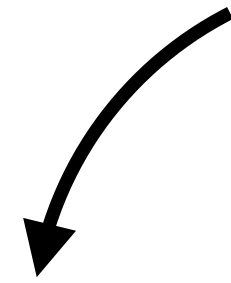
if

$$|\vec{x}_1 - \vec{x}_2| \gg \ell_c \sim 1.2 \text{ kpc} \left(\frac{10^{-23} \text{ eV}}{m_\phi} \right)$$

ULDM

$$ds^2 = dt^2 - \left[1 - 2\Psi(t, \vec{x})\right] dx^2$$

$$\Psi(t, \vec{x}) = \pi G \phi^2(t, \vec{x})$$



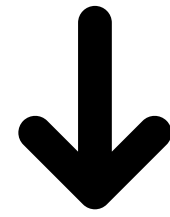
GWs

$$ds^2 = dt^2 - \left[1 - h_{ij}(t, \vec{x})\right] dx^i dx^j$$

$$\Psi(t, \vec{x}) = \pi G \phi^2(t, \vec{x})$$

ULDM

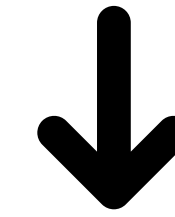
$$ds^2 = dt^2 - \left[1 - 2\Psi(t, \vec{x})\right] dx^2$$



$$\delta t_a(t) = \frac{\pi G}{2m_\phi} \left[\phi^2(t, \vec{x}_e) - \phi^2(t - L_a, \vec{x}_a) \right]$$

GWs

$$ds^2 = dt^2 - \left[1 - h_{ij}(t, \vec{x})\right] dx^i dx^j$$

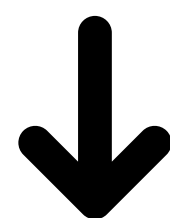


$$\delta t_a(t) \propto \frac{p_a^i p_a^j}{2(1 + \hat{\Omega} \cdot \hat{p}_a)} \left[h_{ij}(t, \vec{x}_e) - h_{ij}(t - L_a, \vec{x}_a) \right]$$

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ULDM

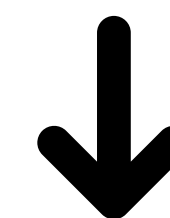
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Search parameters:

- **1 amplitude parameter (DM density)**
- 1 frequency parameter (DM mass)
- $1 + N_p$ phase parameters
- $1 + N_p$ scale parameters (with non trivial correlations)

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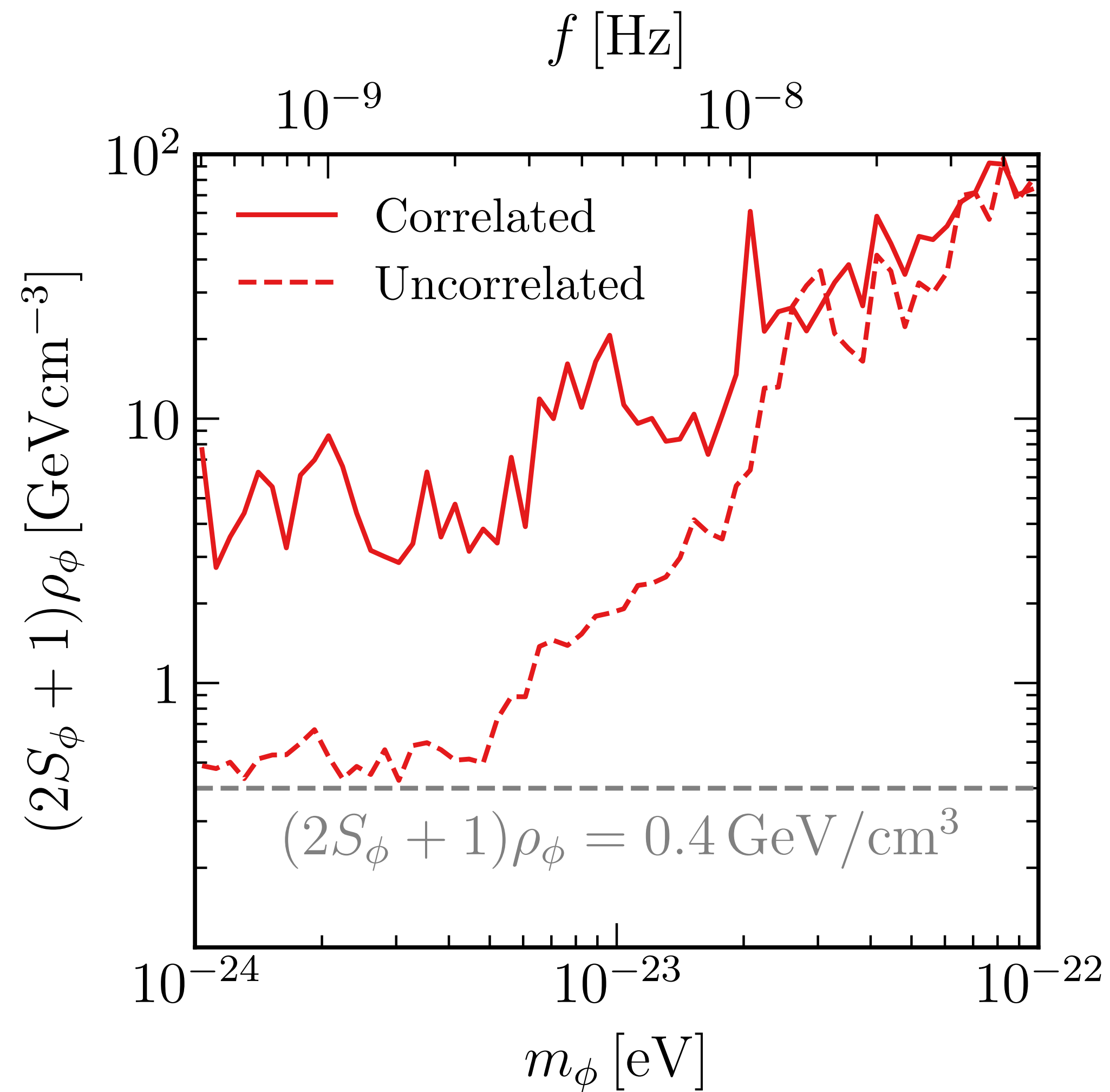
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ULDM: GRAVITATIONAL SIGNALS

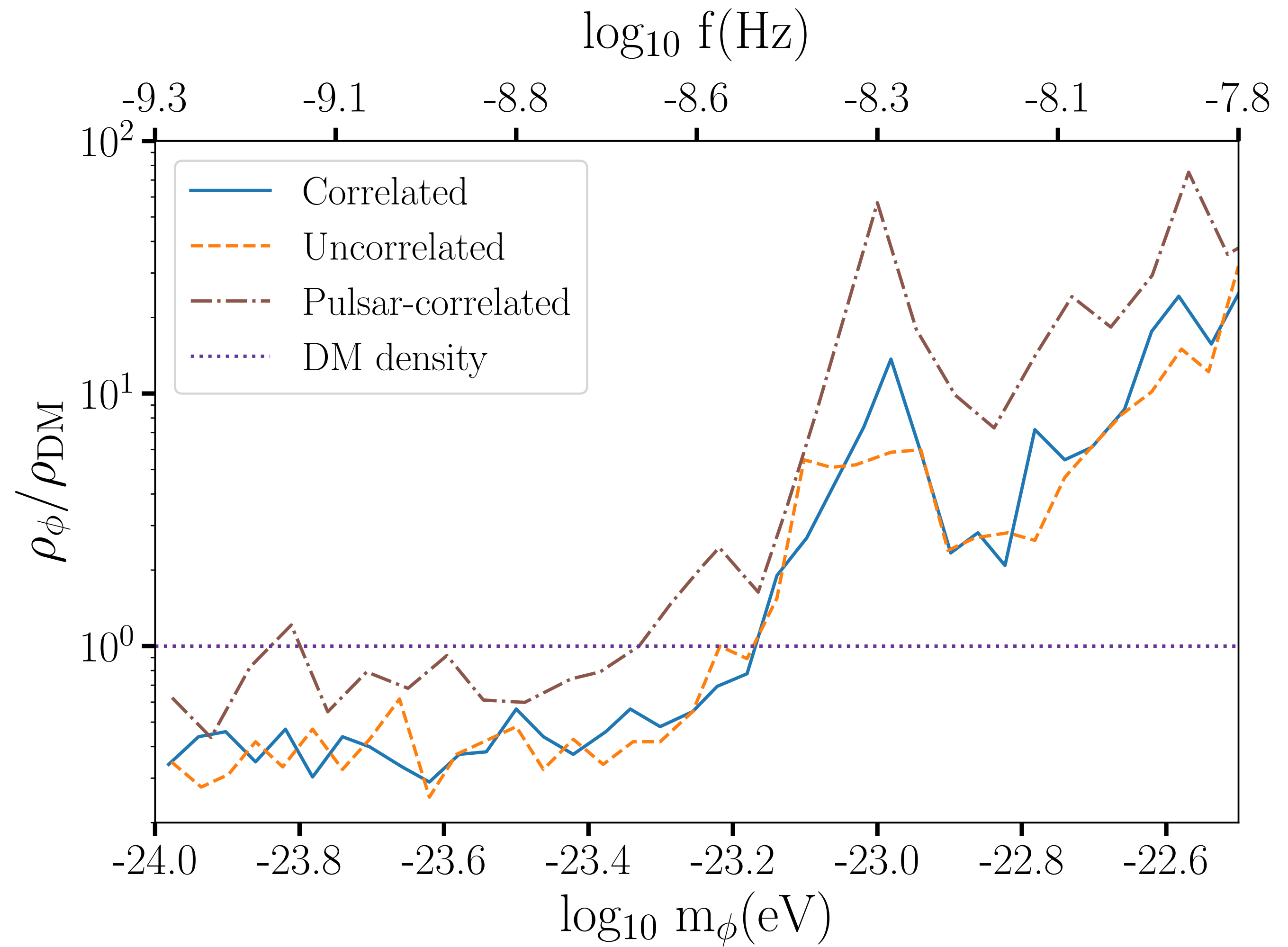
NANOGrav 15-year

Afzal et al. [2306.16219]



EPTA DR2

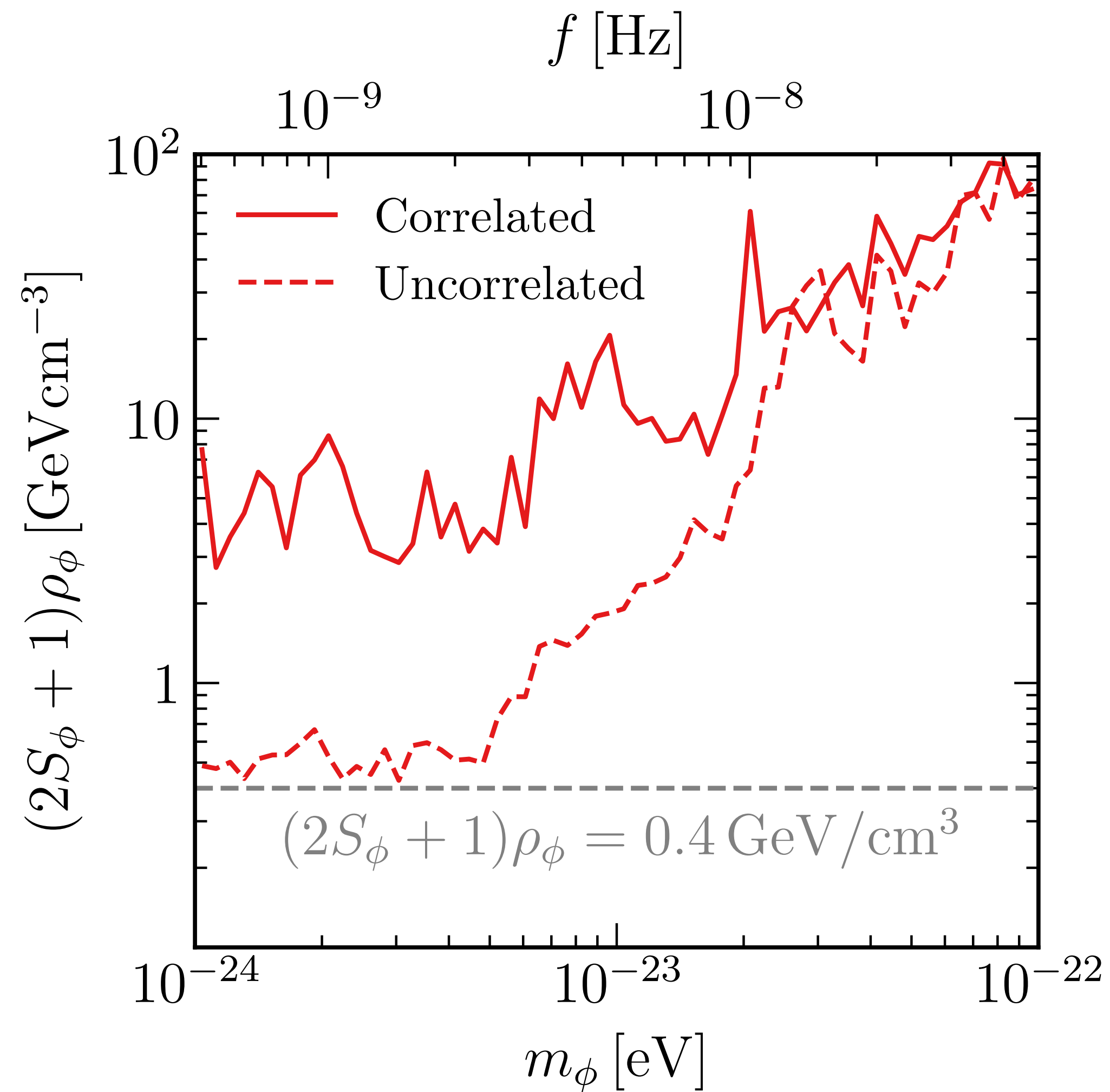
Smarra et al. [2306.16228]



ULDM: GRAVITATIONAL SIGNALS

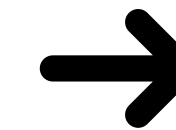
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Correlated

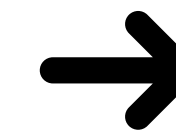
$$\langle \hat{\phi}(\vec{x}_1) \hat{\phi}(\vec{x}_2) \rangle = 1$$



1 scale parameter

Uncorrelated

$$\langle \hat{\phi}(\vec{x}_1) \hat{\phi}(\vec{x}_2) \rangle = 0$$



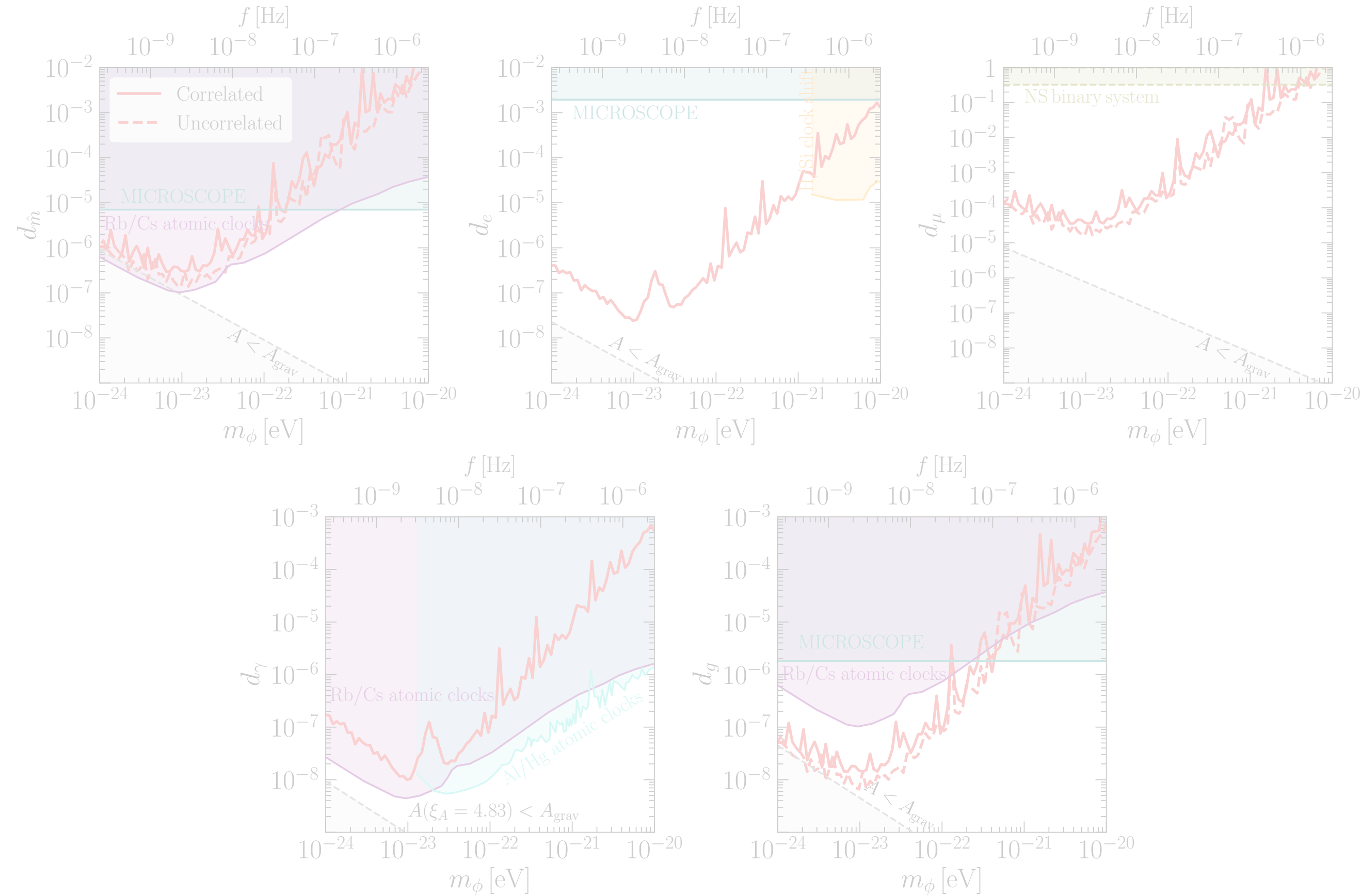
$1 + N_p$ scale parameters

dark matter: a quick survey of other effects

ULDM direct couplings

If ULDM couples to ordinary matter and fields via interactions other than gravity, other effects in PTA observations arise:

- fluctuations in the frequency of the atomic clocks used to time pulsars
- fluctuations in the pulsars' angular frequency



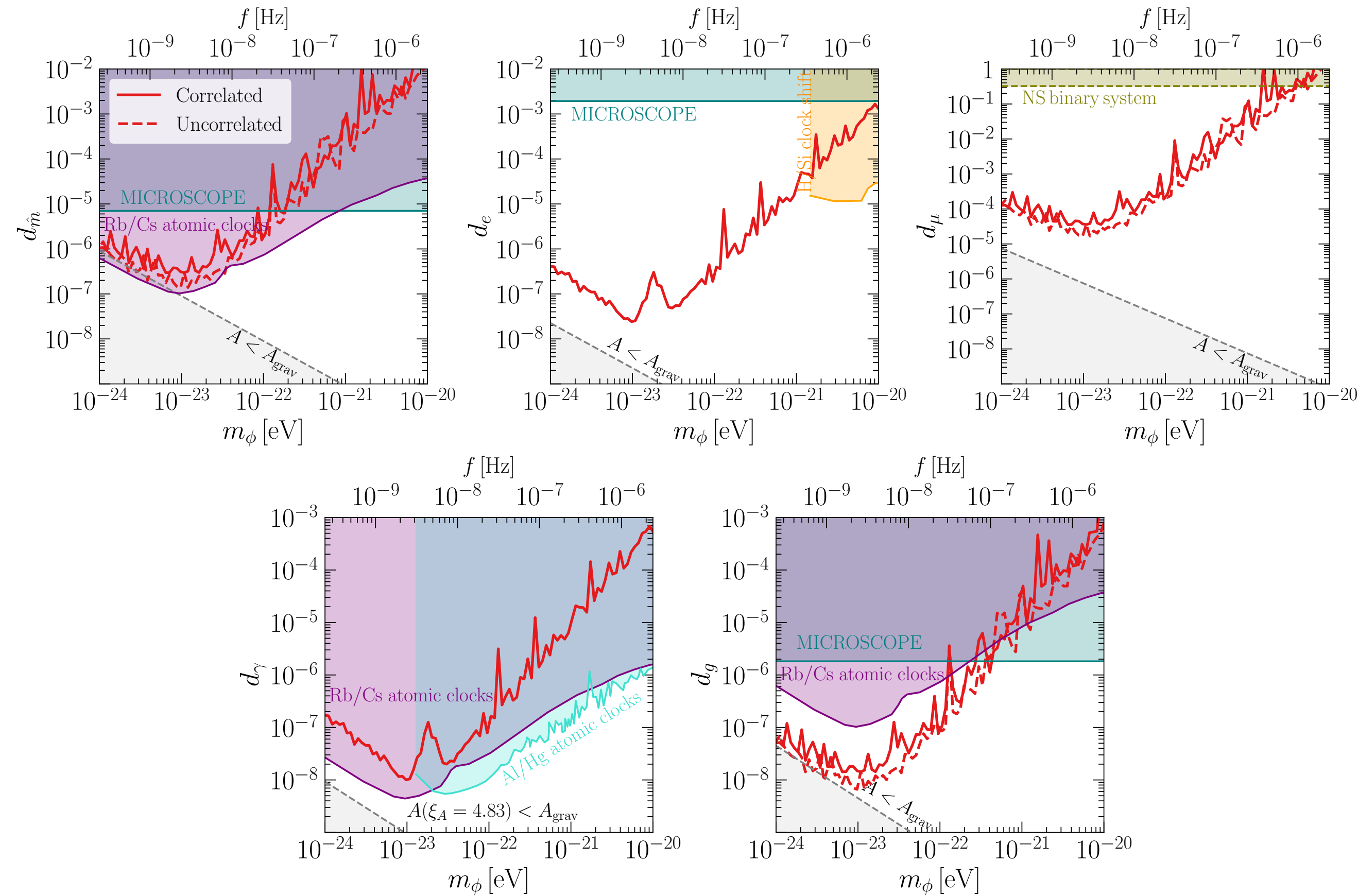
Kaplan, **AM**, Trickle, *"Constraining fundamental constant variations from ultralight dark matter with pulsar timing arrays"*, PRD 106 (2022)

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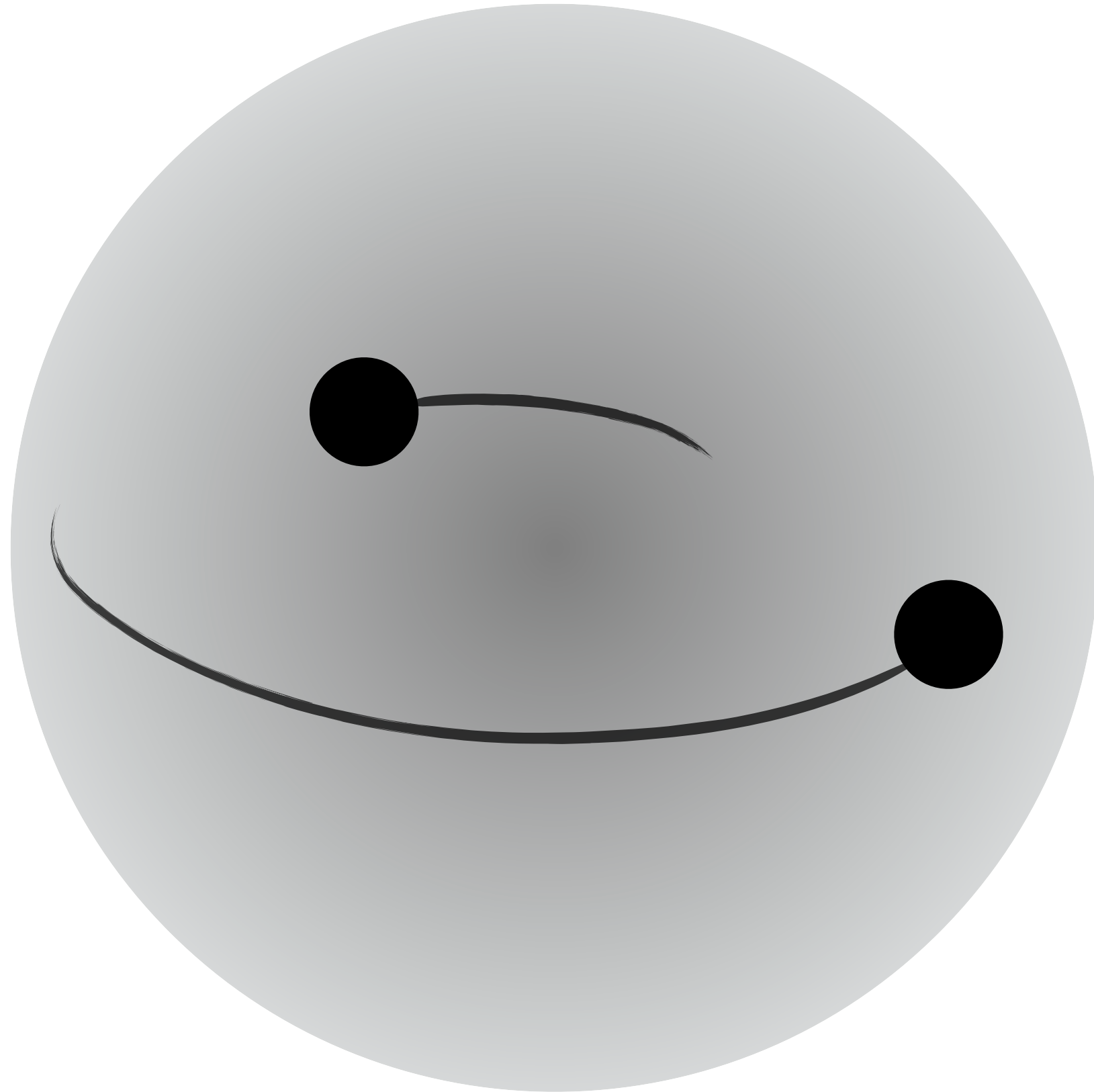


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DM DRIVEN SMBH HARDENING

interactions with the DM halo can drive SMBHB orbital evolution, suppressing the GWB power at low frequencies



$$\frac{da}{dt} = \left(\frac{da}{dt}\right)_{\text{GW}} + \left(\frac{da}{dt}\right)_{\text{bar}} + \left(\frac{da}{dt}\right)_{\text{DM}}$$

star + gas contribution

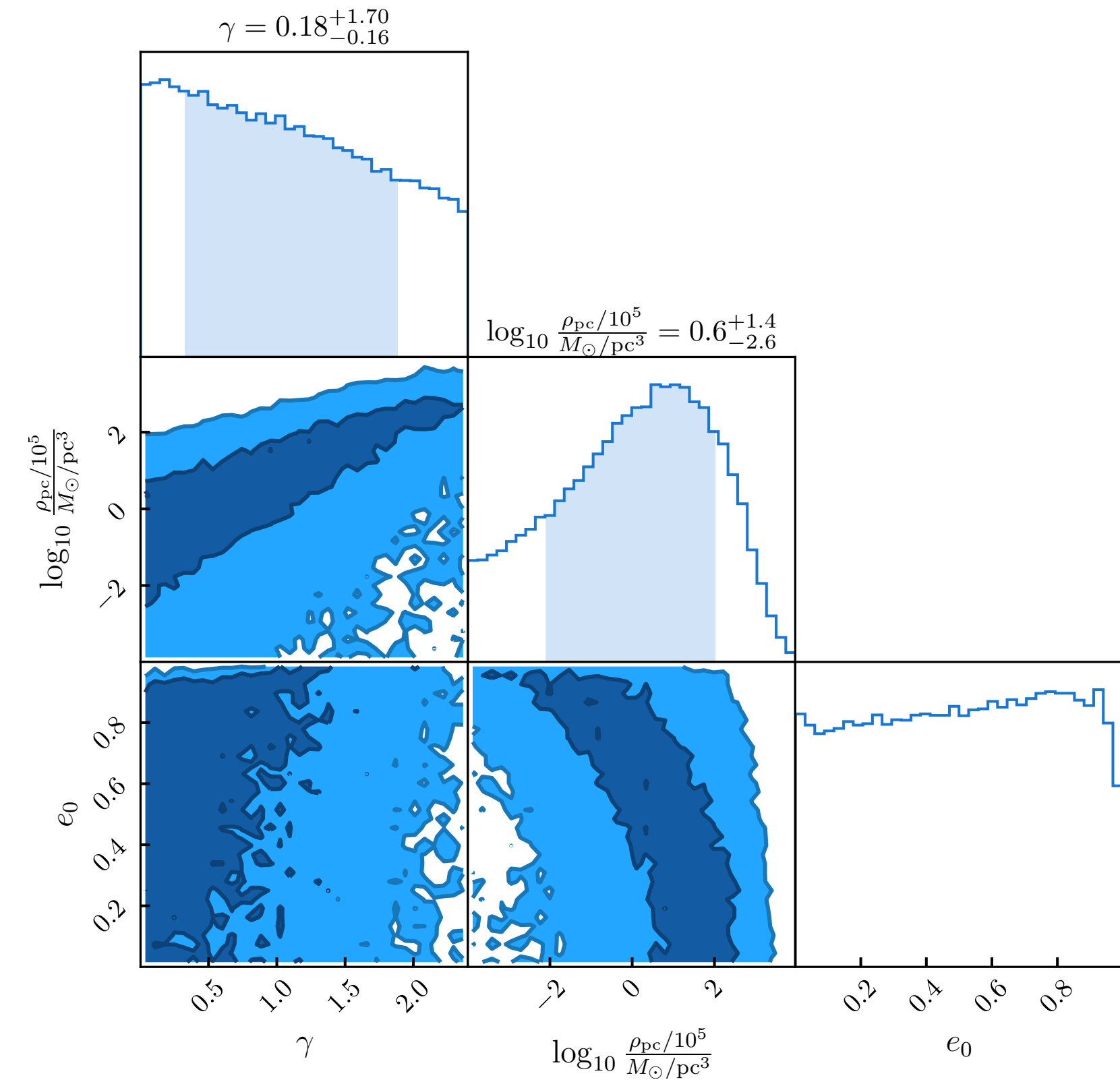
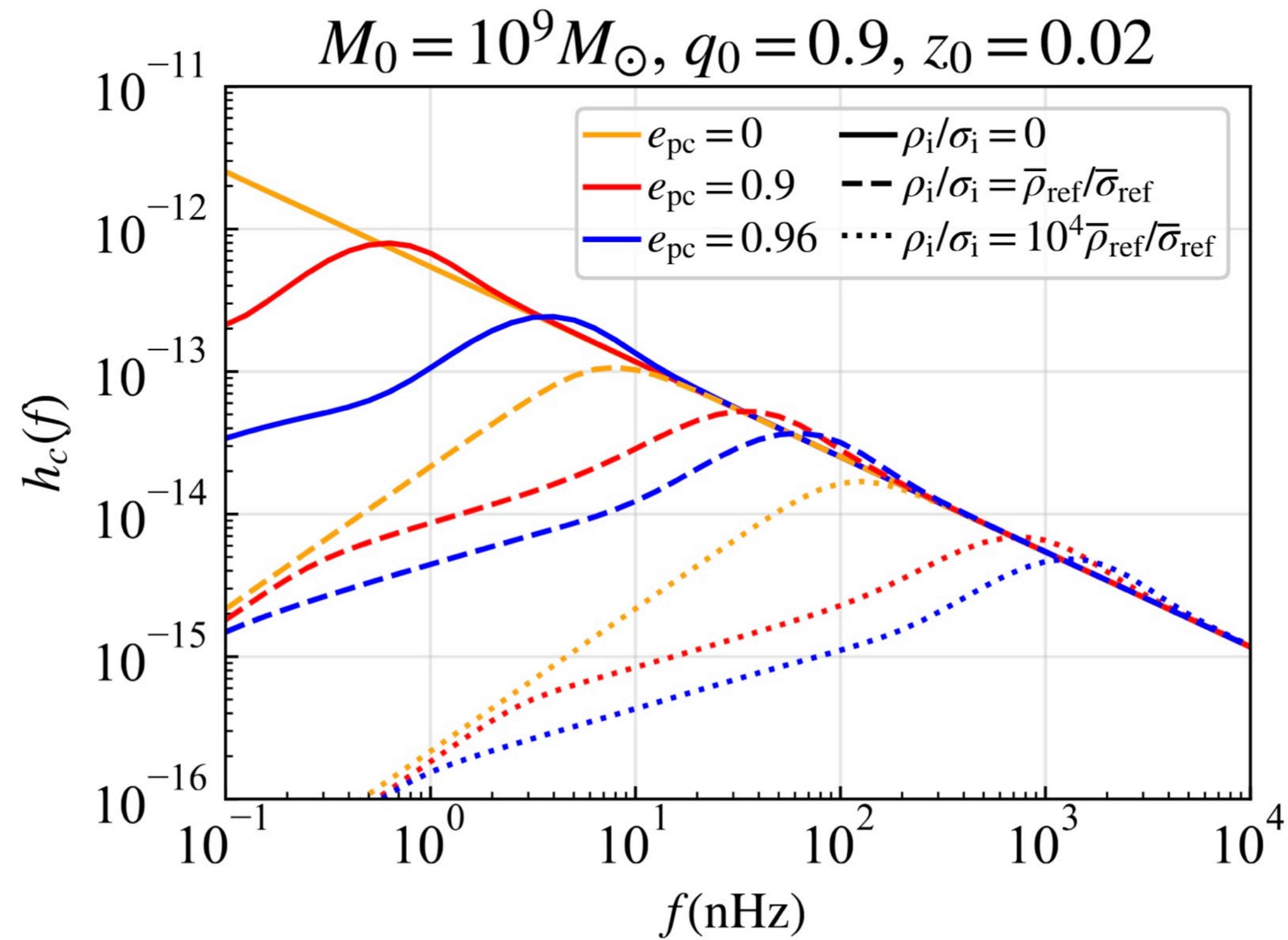
GW contribution

DM contribution

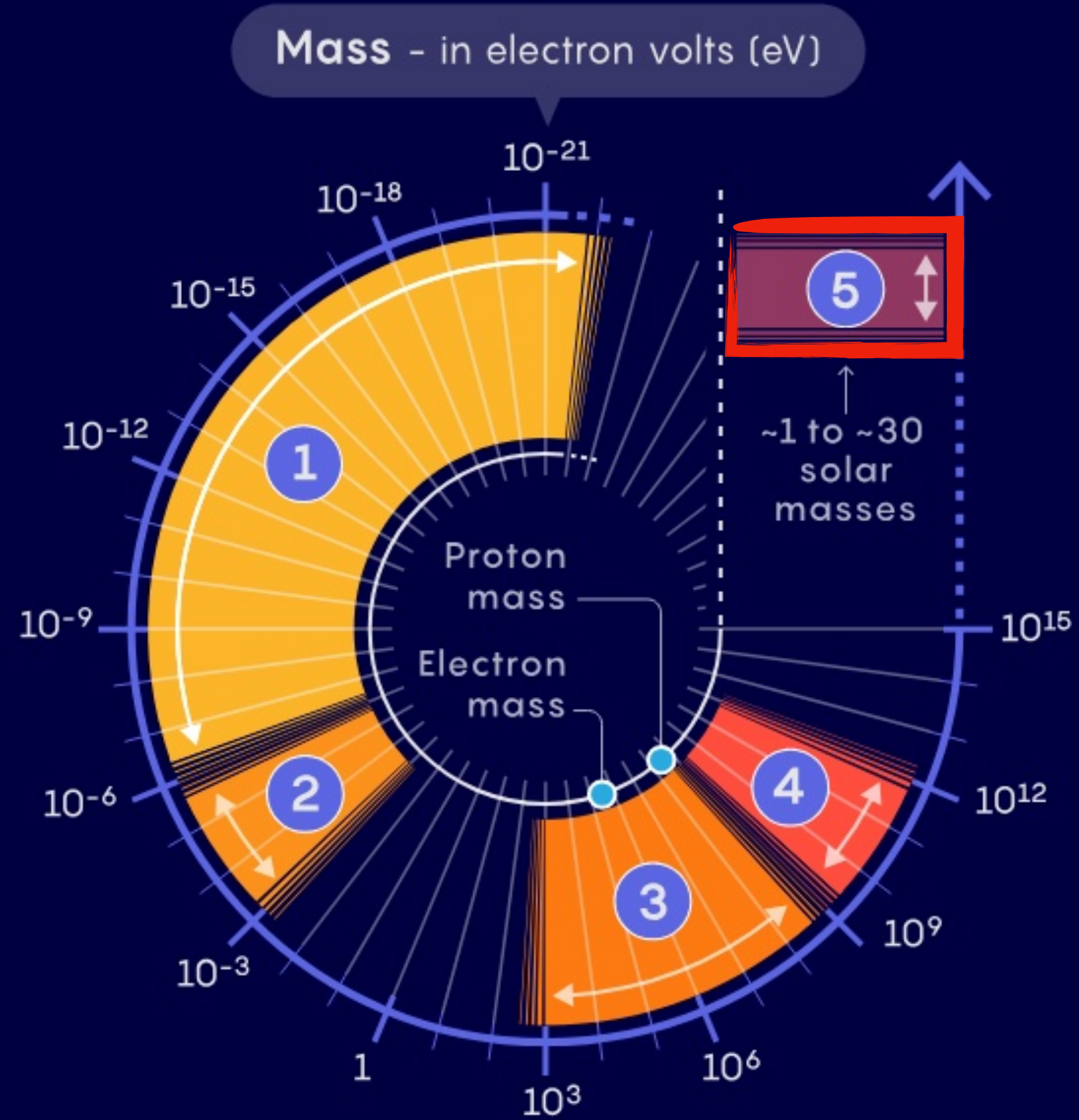
The diagram shows the equation for the rate of change of the semi-major axis, $\frac{da}{dt}$, which is the sum of three contributions: $\left(\frac{da}{dt}\right)_{\text{GW}}$ (gravitational waves), $\left(\frac{da}{dt}\right)_{\text{bar}}$ (stellar and gas bar), and $\left(\frac{da}{dt}\right)_{\text{DM}}$ (dark matter). Arrows point from the labels 'GW contribution' and 'DM contribution' to their respective terms in the equation. An arrow points from 'star + gas contribution' to the $\left(\frac{da}{dt}\right)_{\text{bar}}$ term.

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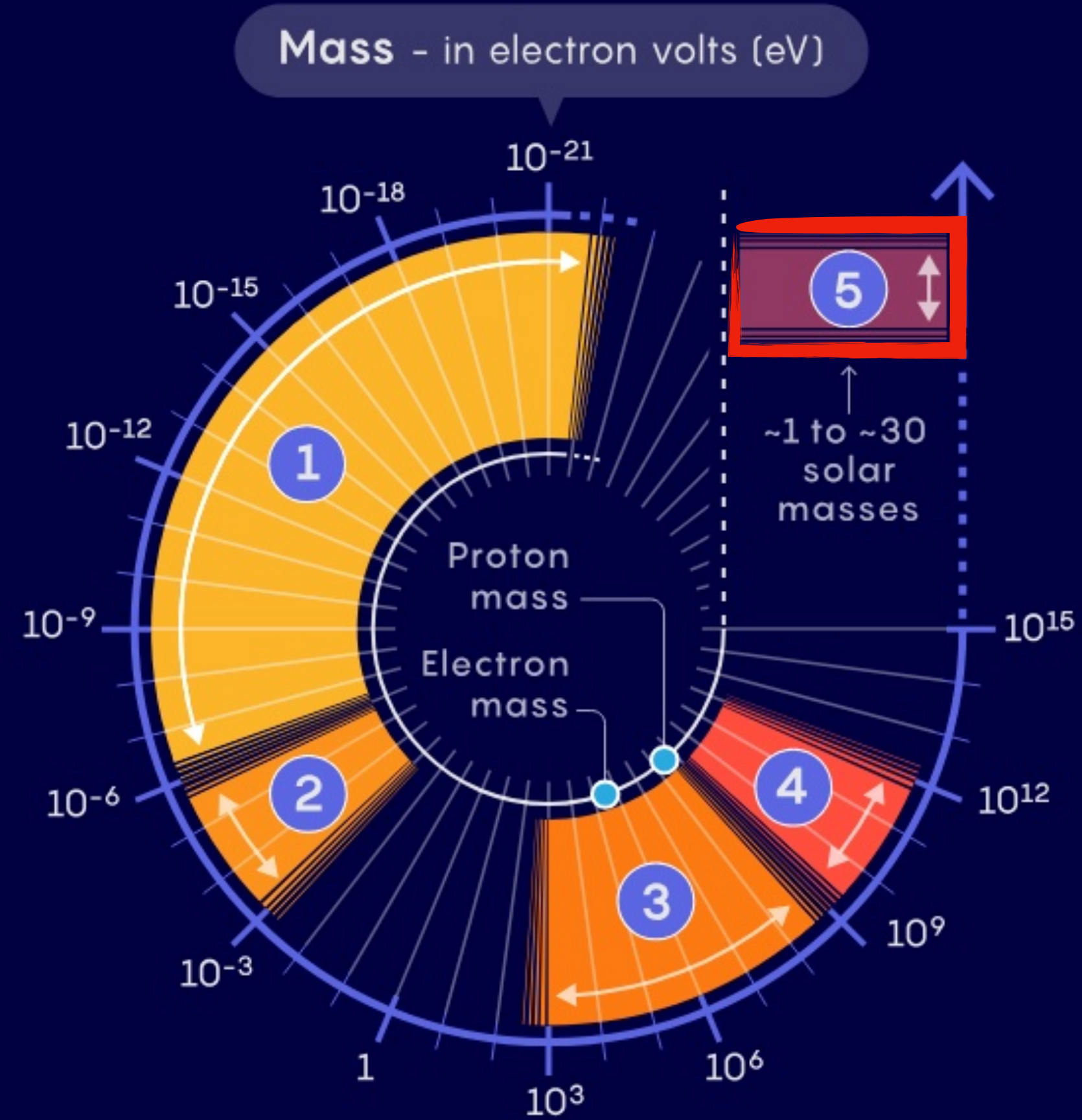


DM SUBSTRUCTURES



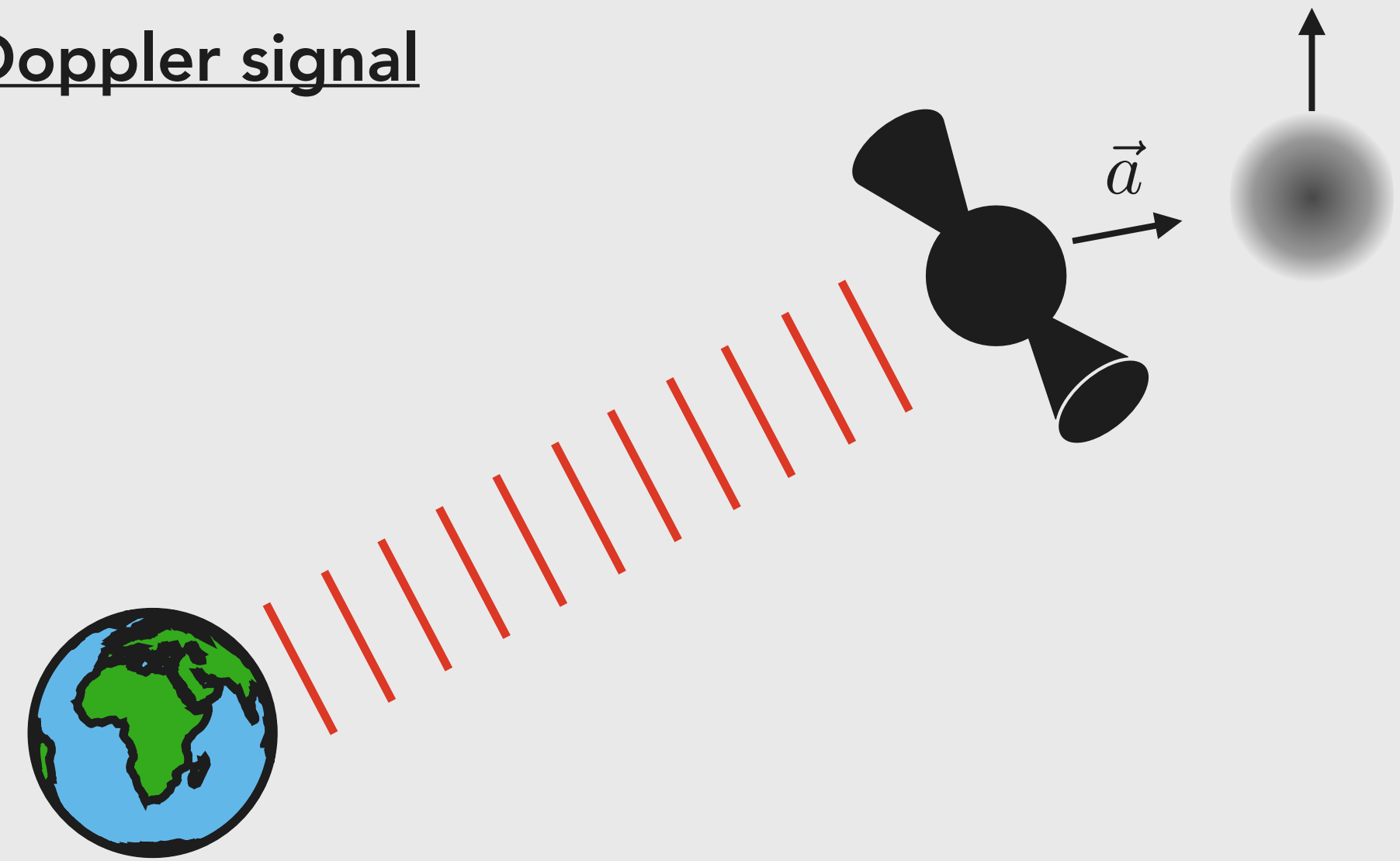
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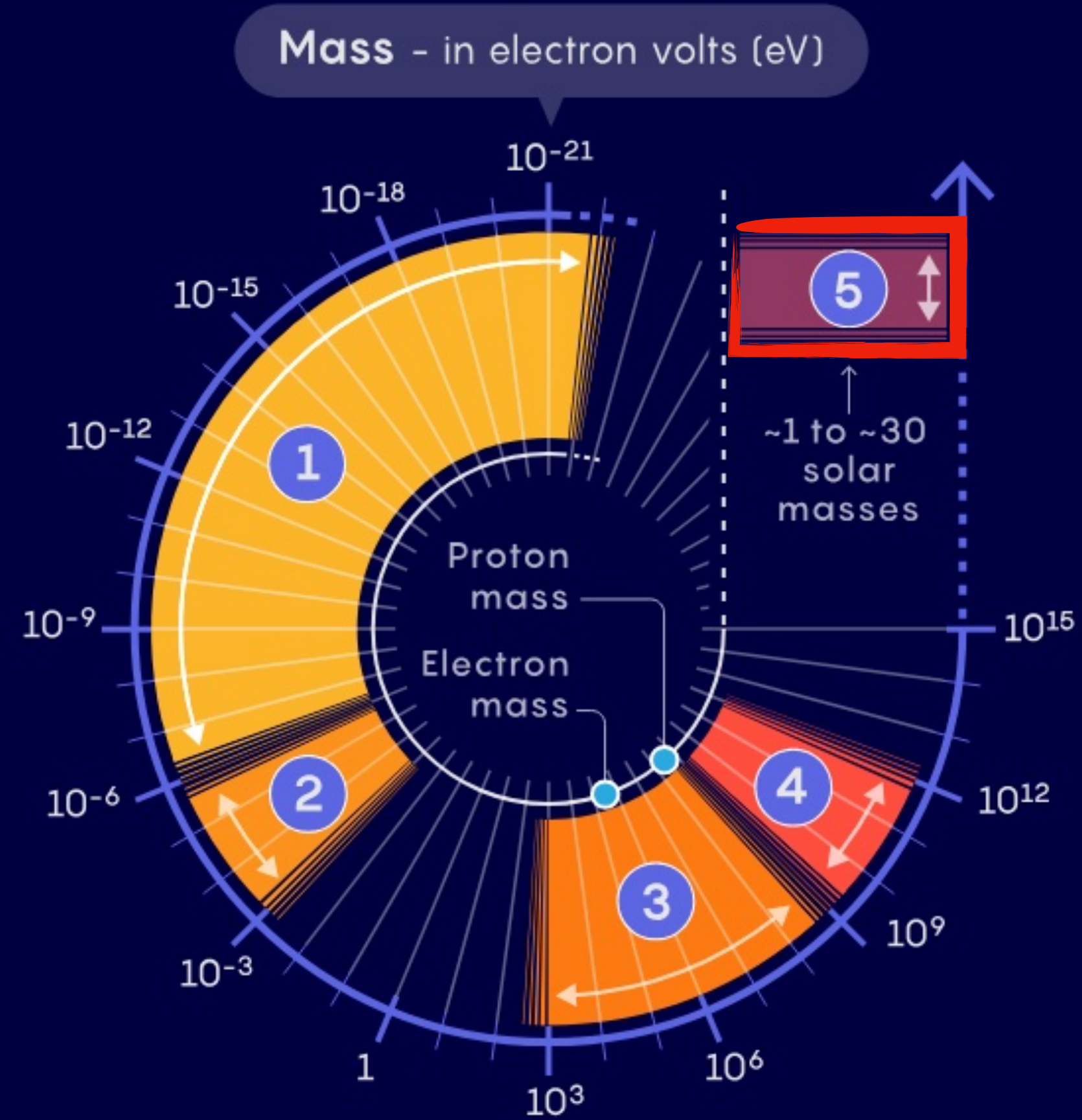


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Doppler signal

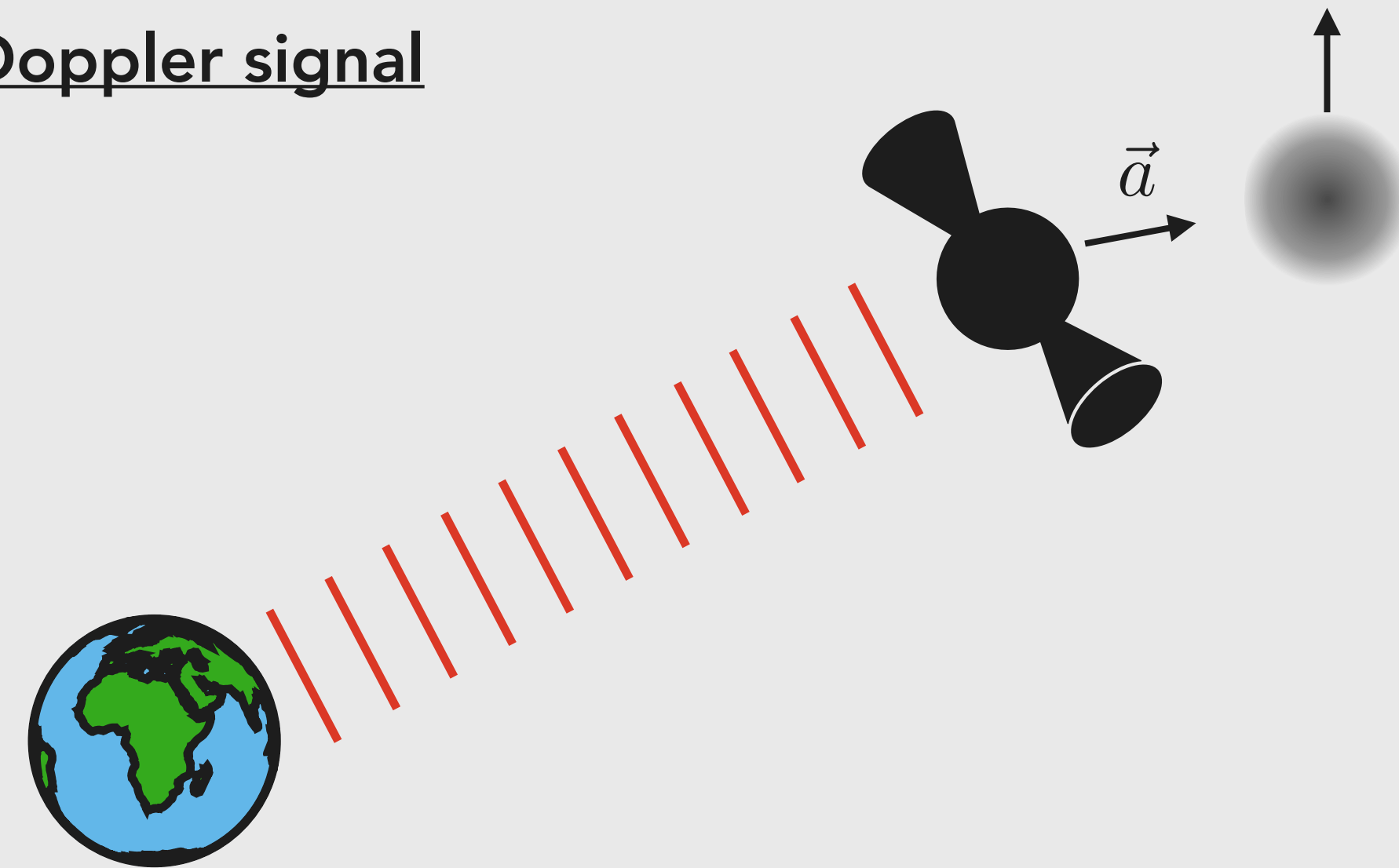


DM SUBSTRUCTURES

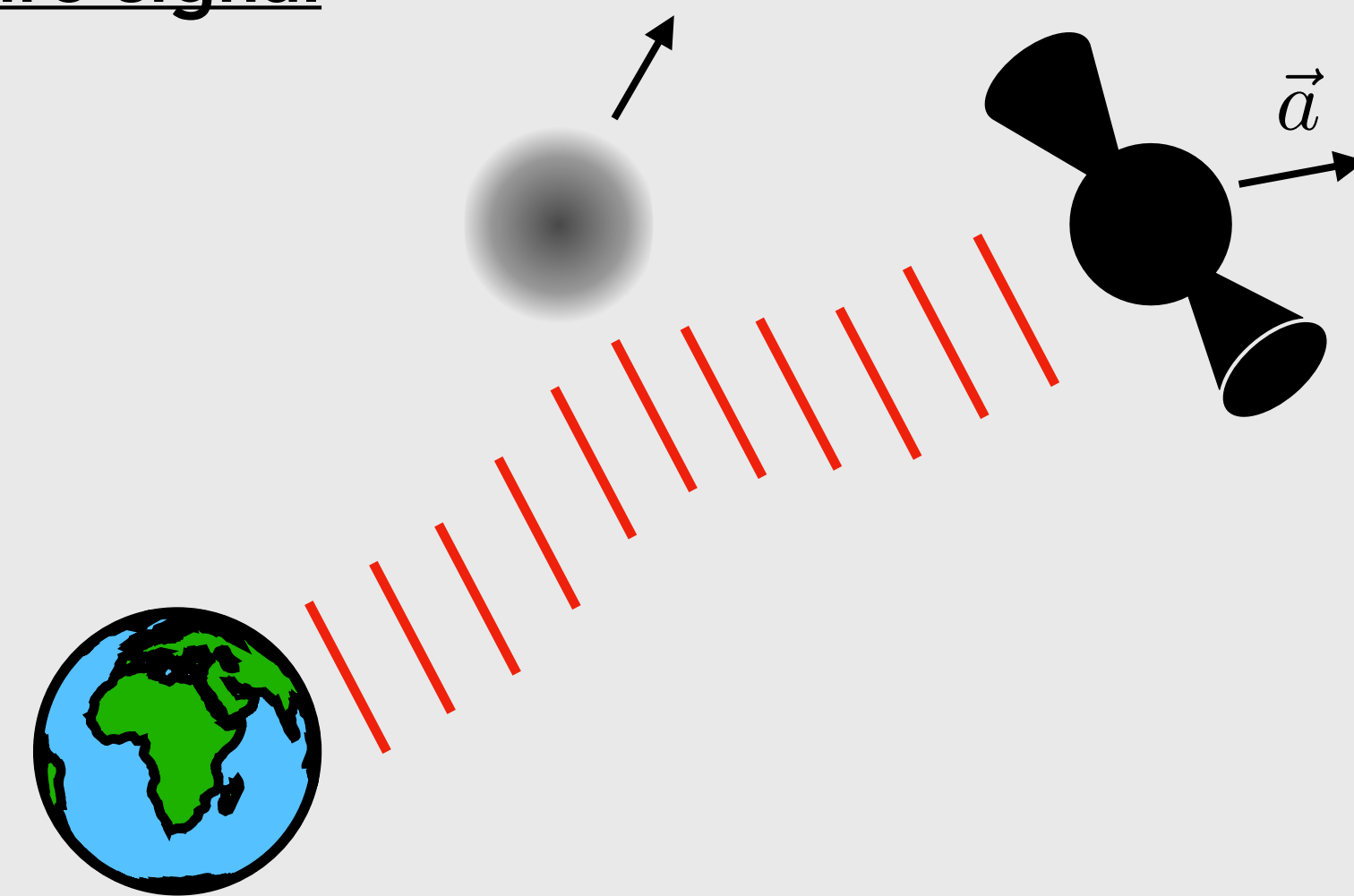


- | | |
|--------------------------|----------|
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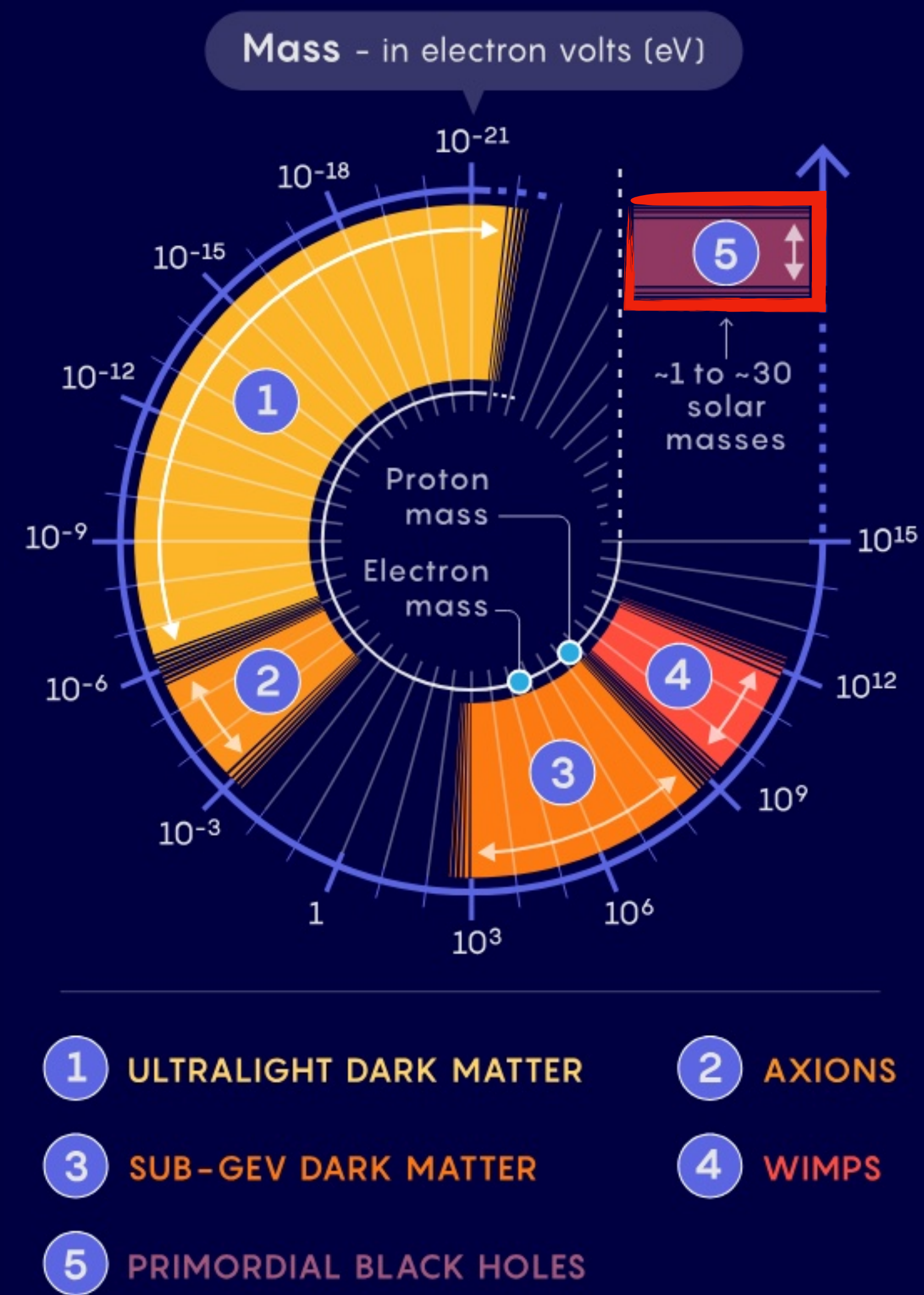
Doppler signal



Shapiro signal

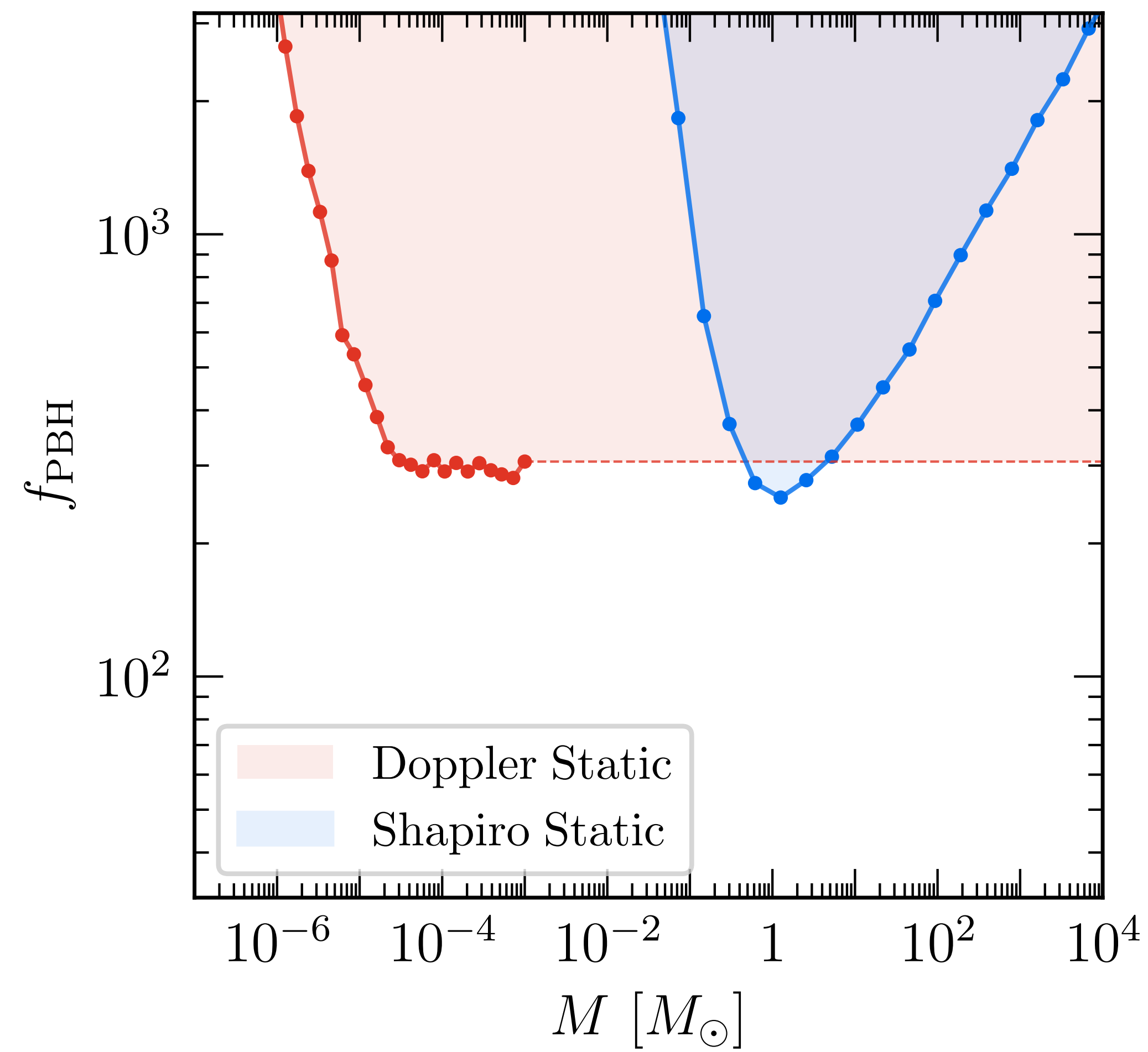


DM SUBSTRUCTURES

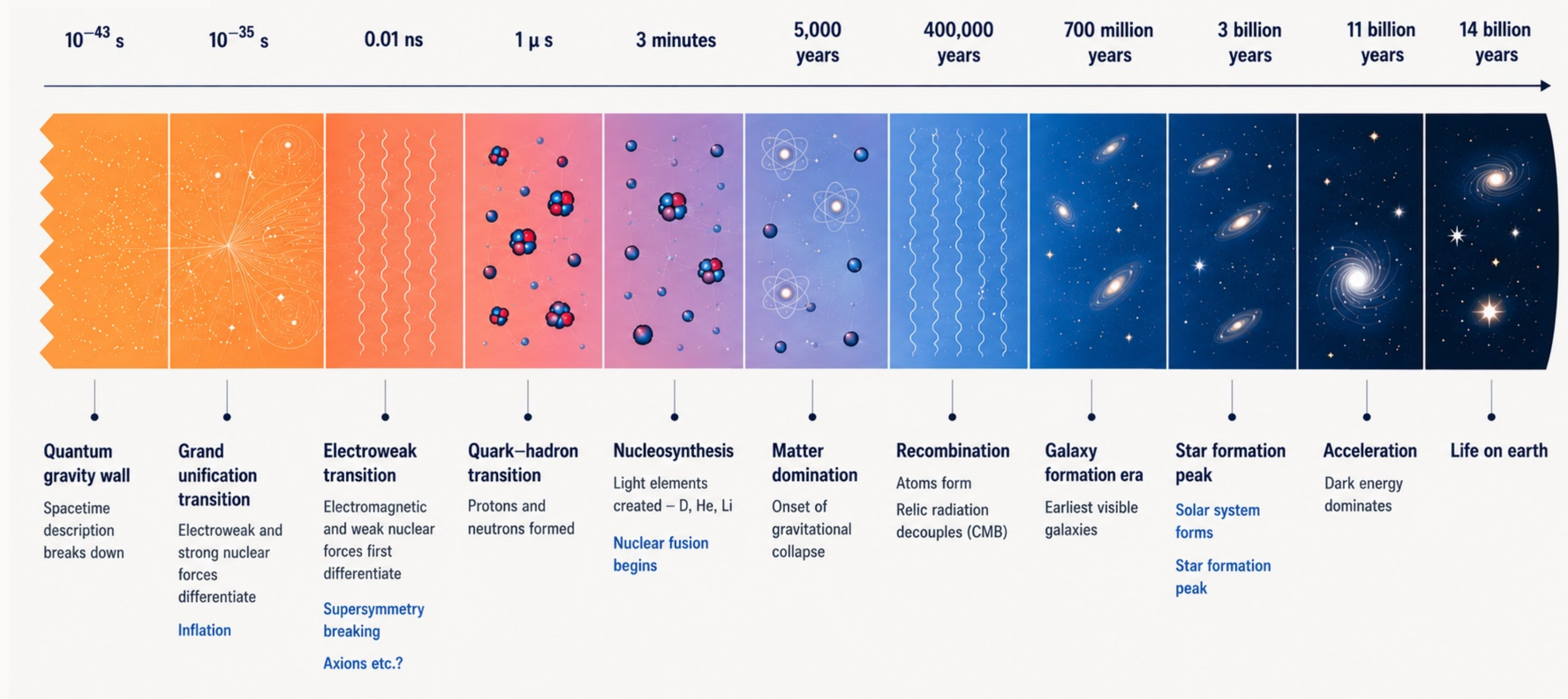


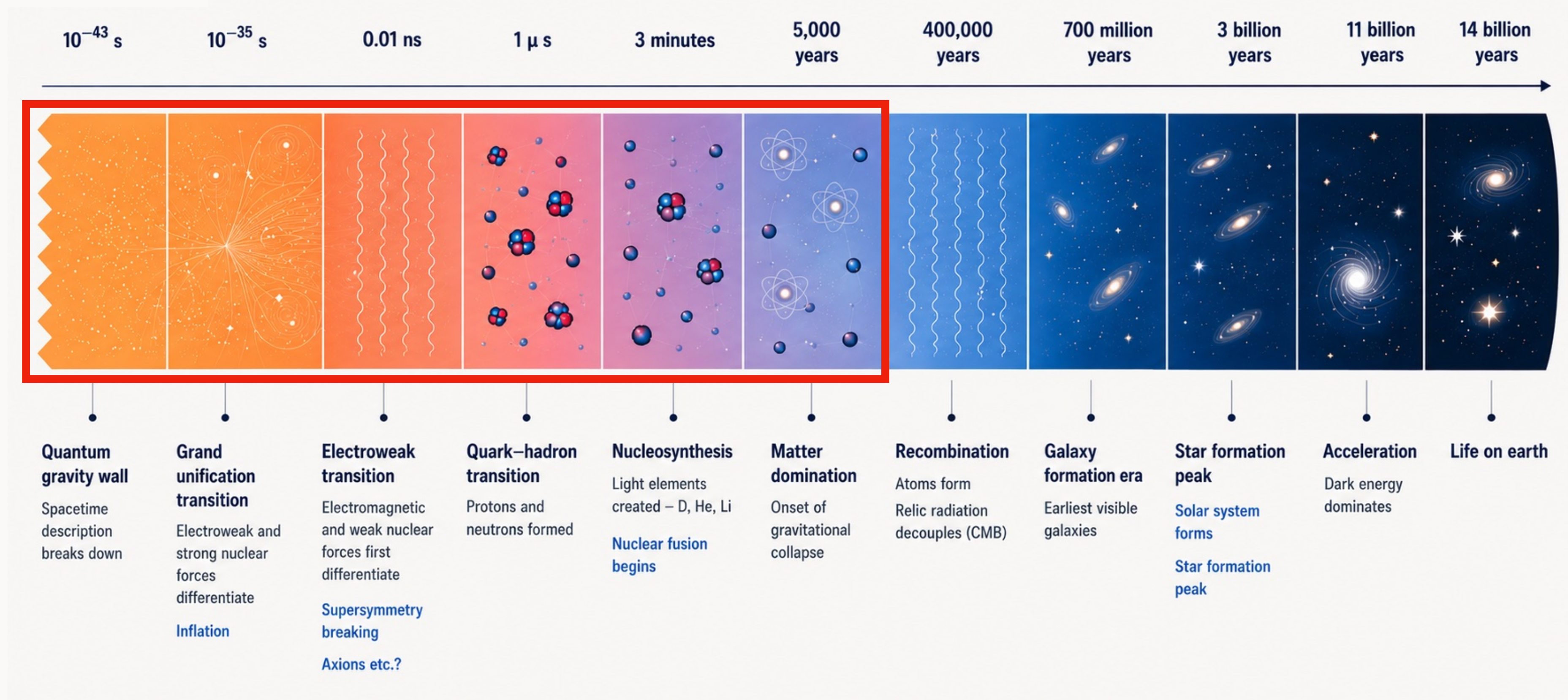
for slow transiting objects ($\tau > T_{\text{obs}}$) both events induce a deterministic timing delay of the form

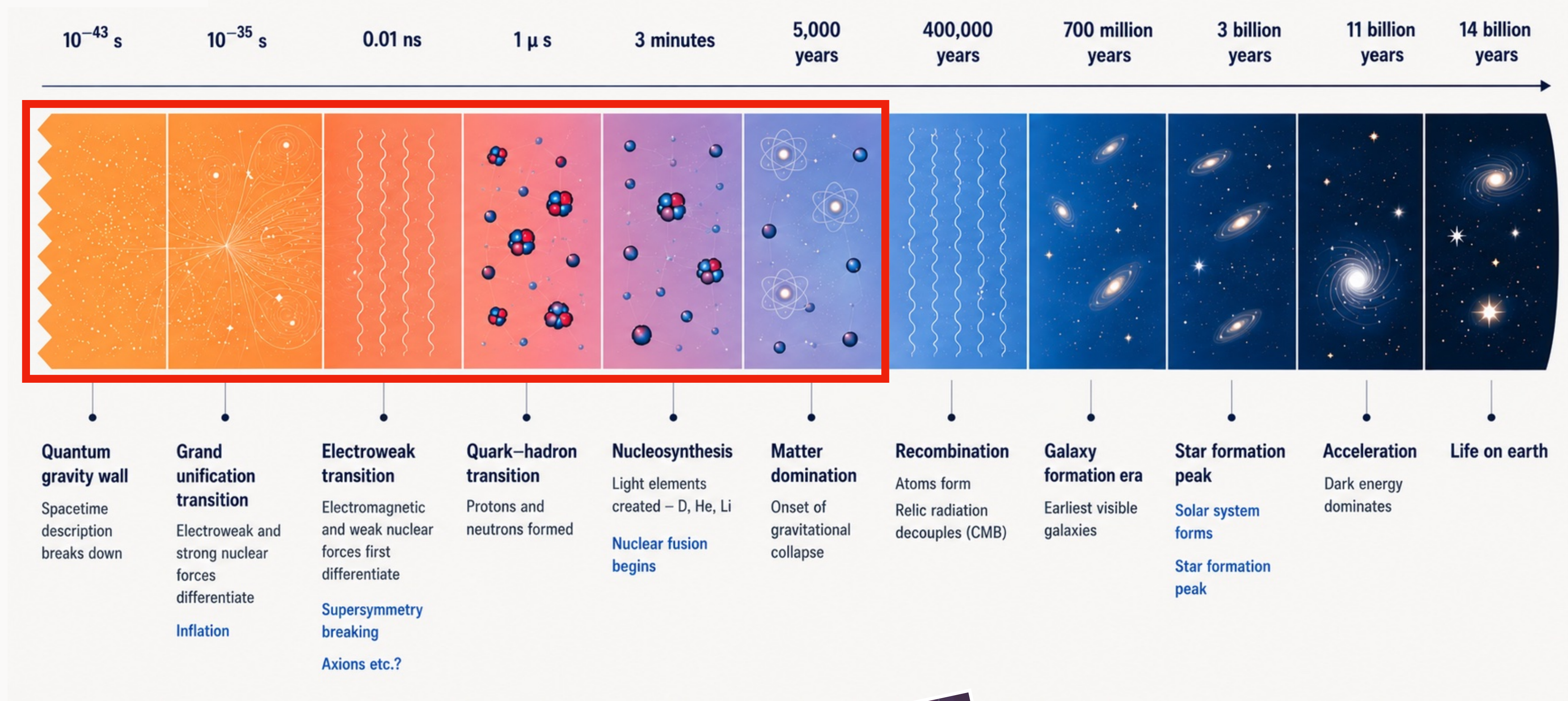
$$\delta t(t) = A \frac{t^3}{\text{yr}^2}$$

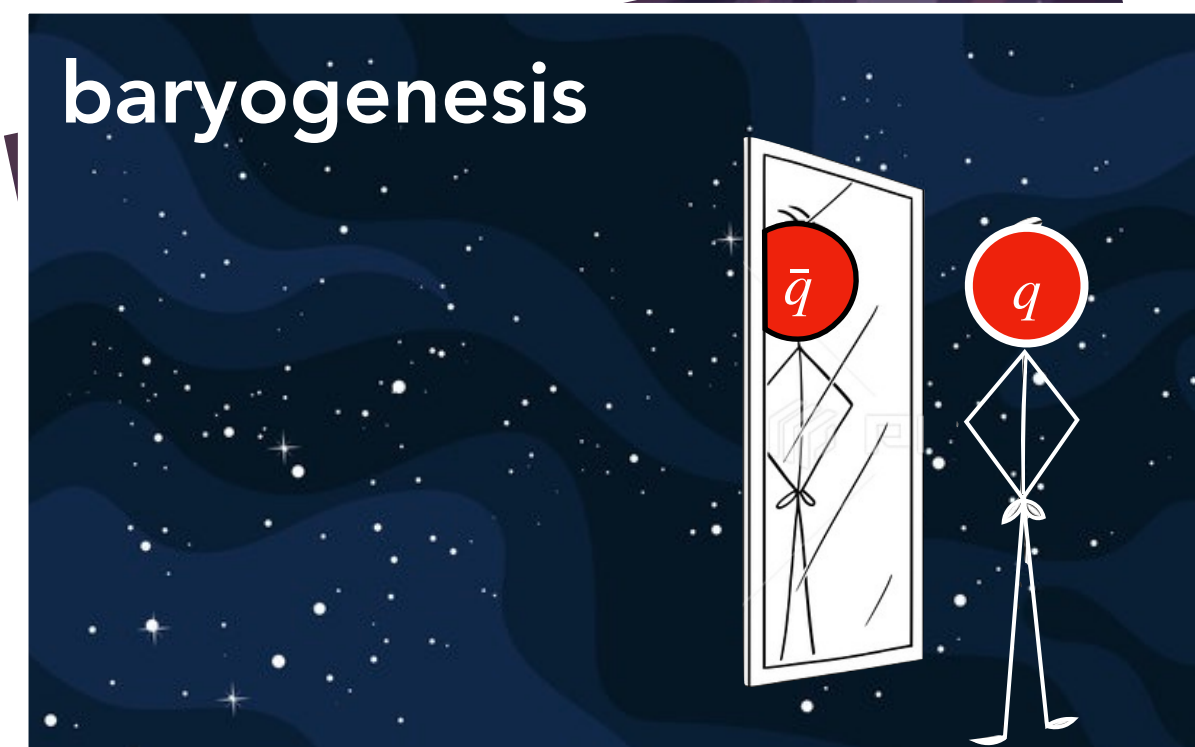
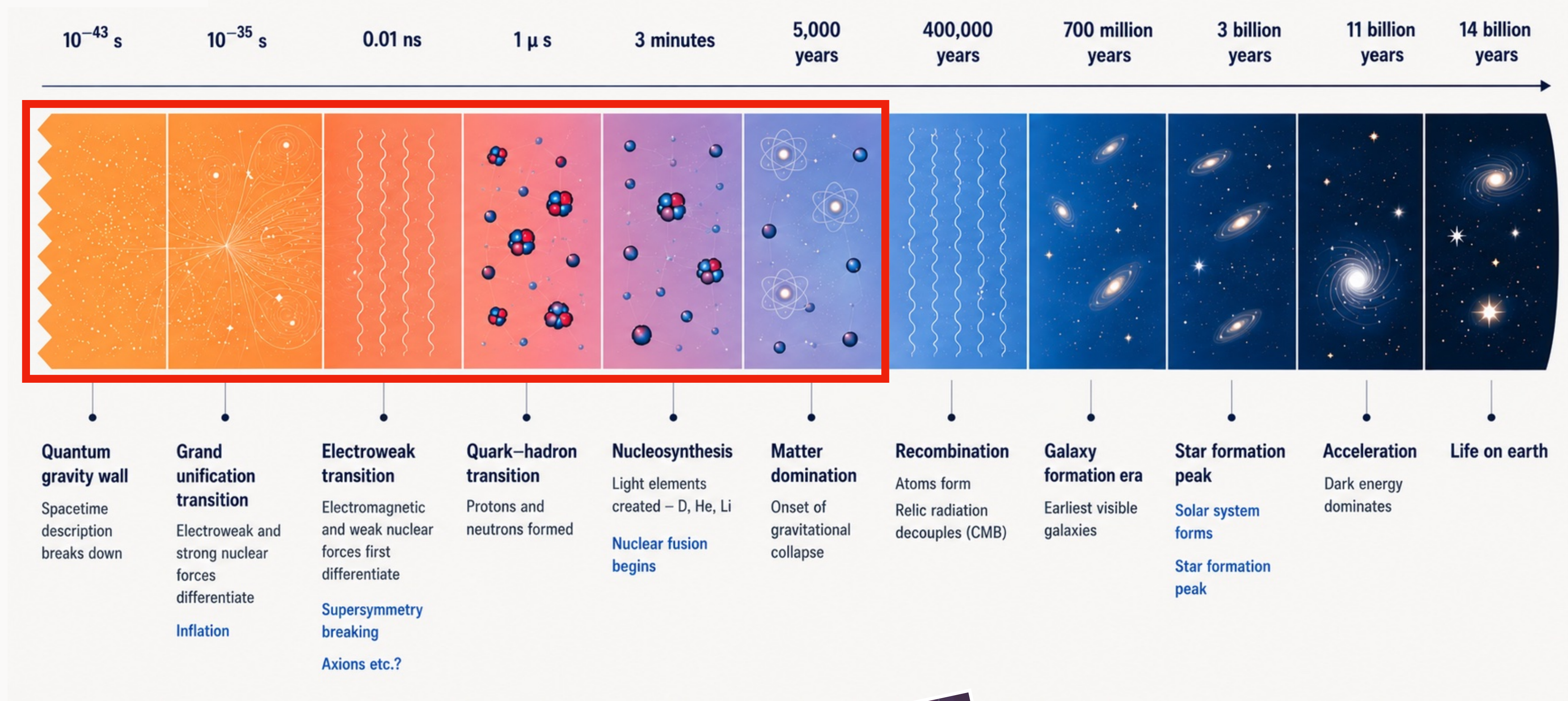


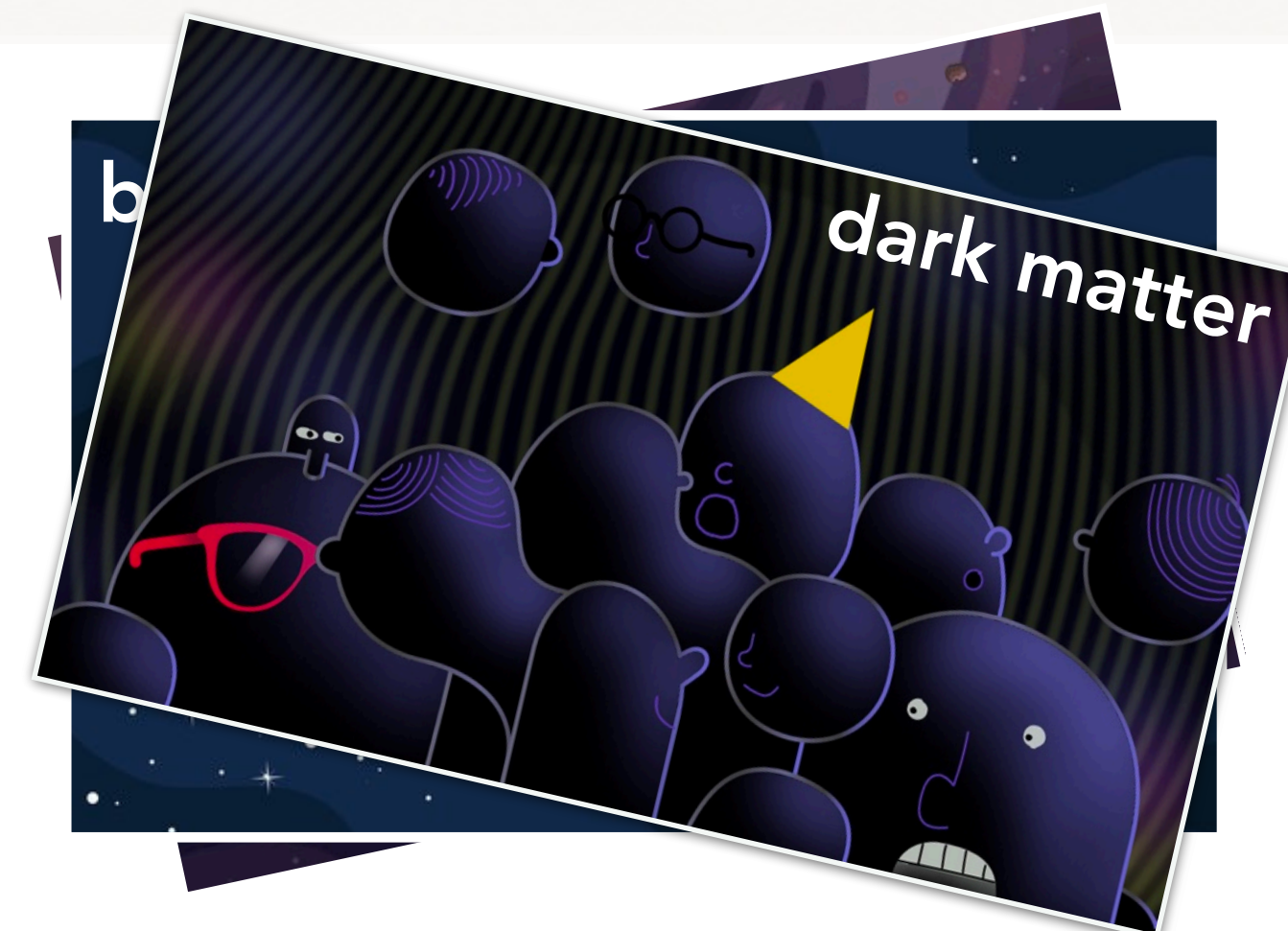
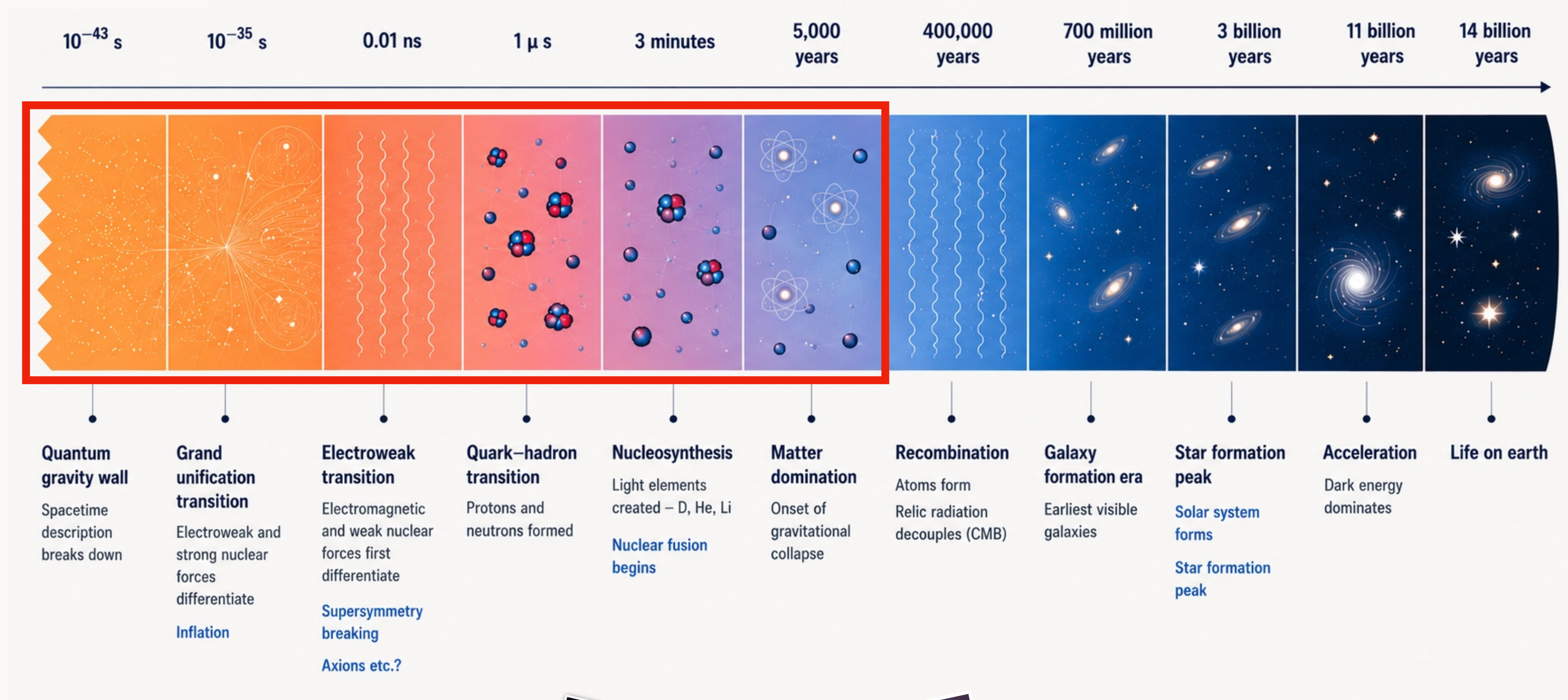
knock-knock-knocking on CMB doors





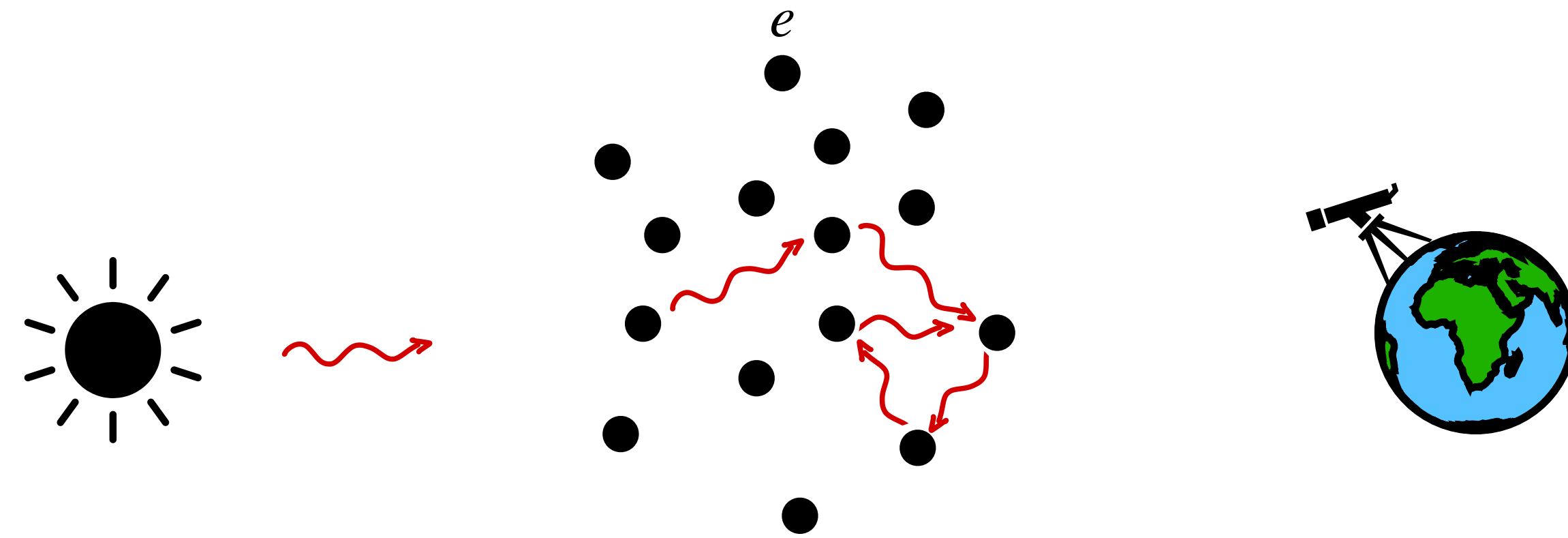






THE CMB WALL

photons are not the best messenger in environments with a high density of charged particles

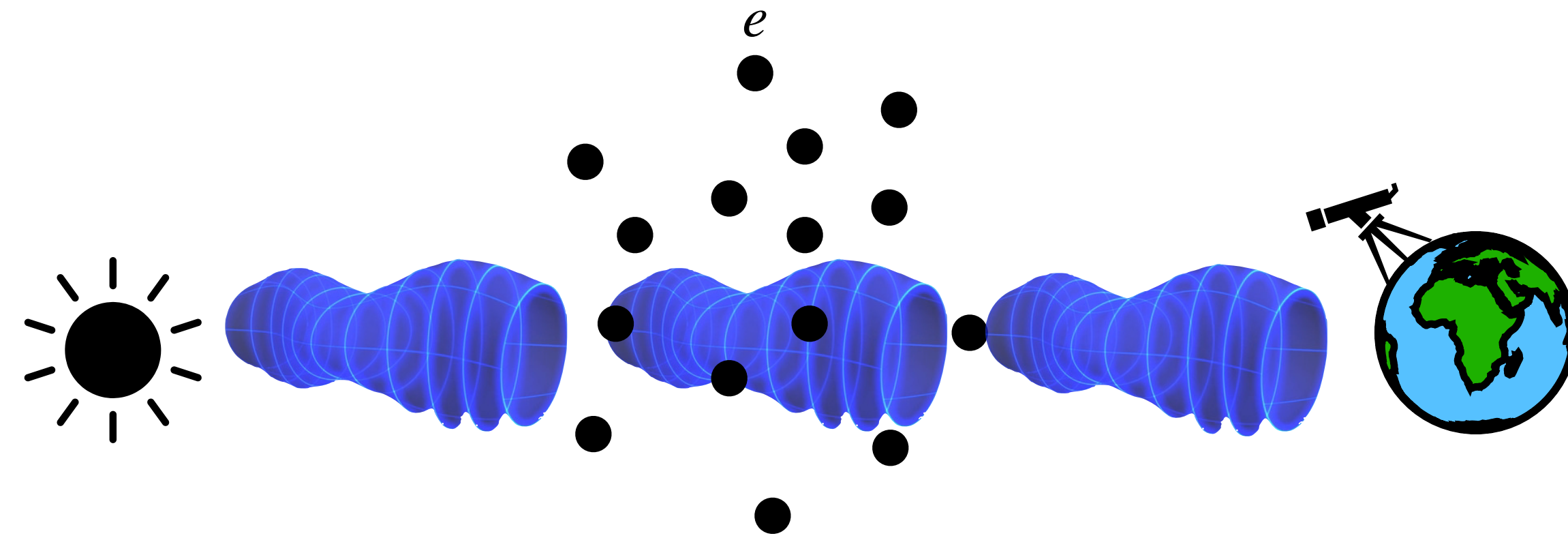


the mean free-path for photons in the primordial plasma is

$$\lambda_{\gamma} \sim \frac{1}{\sigma_T n_e} \sim \mu\text{m} \left(\frac{\text{MeV}}{T} \right)^3$$

any information encoded in EM radiation before recombination is lost in the electron bath

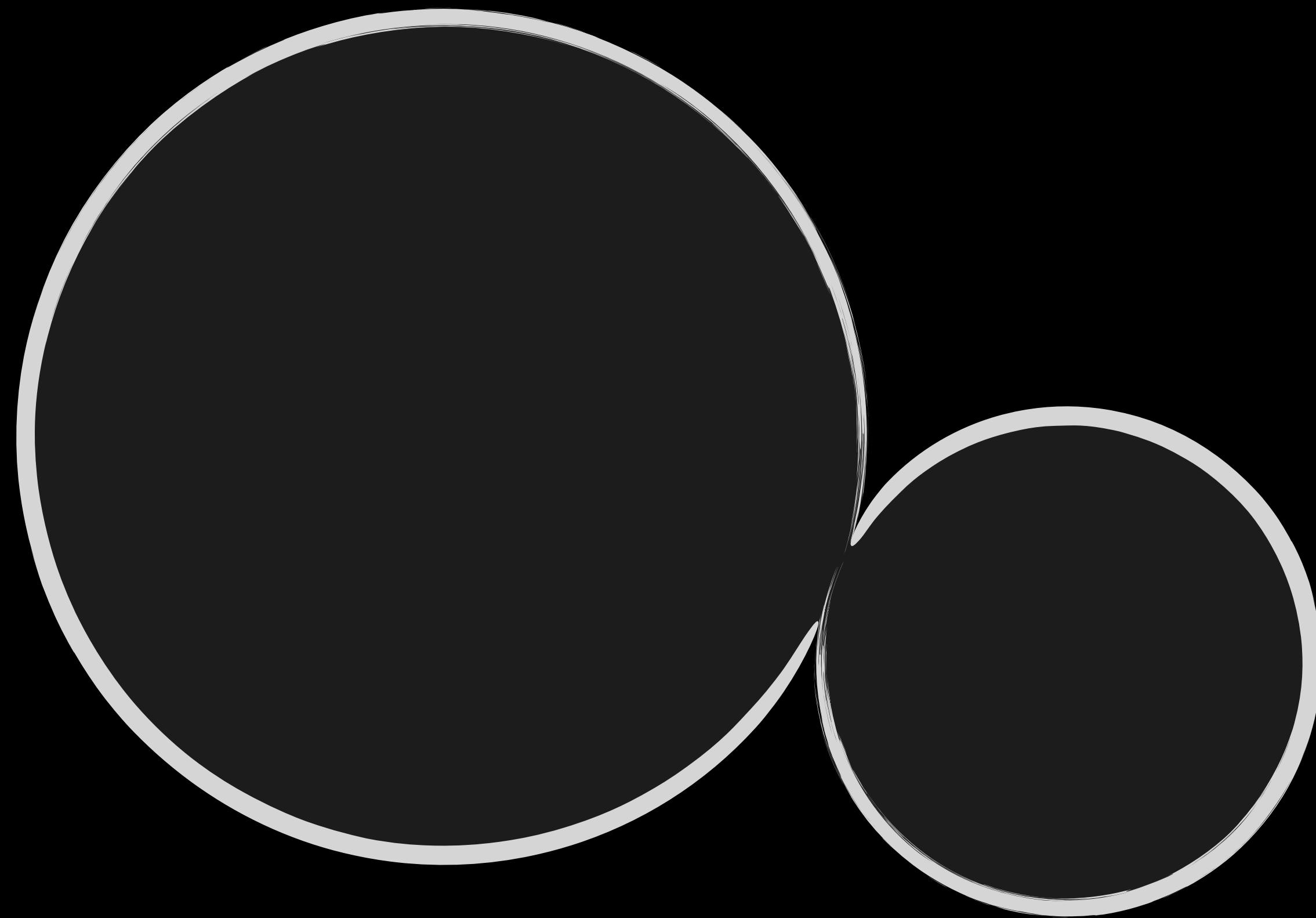
THE CMB WALL



the GW mean free-path in the primordial plasma is

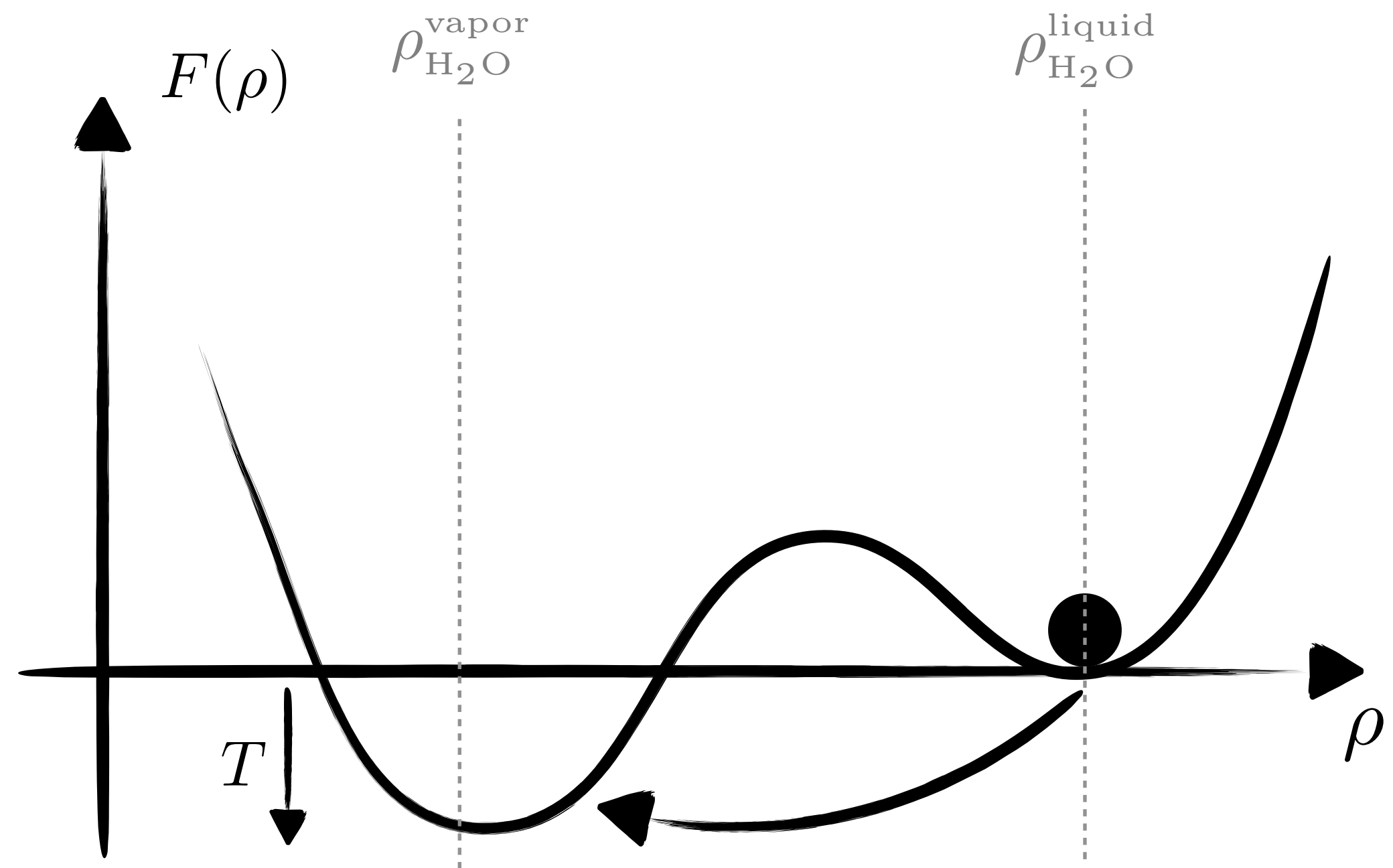
$$\lambda_{\text{GW}} \sim \left(\frac{Gm_e^2}{\alpha} \right)^{-2} \lambda_\gamma \sim \frac{1}{\sigma_T n_e} \sim 10^{79} \text{m} \left(\frac{\text{MeV}}{T} \right)^3$$

GW do not care about the bath and travel undisturbed through it



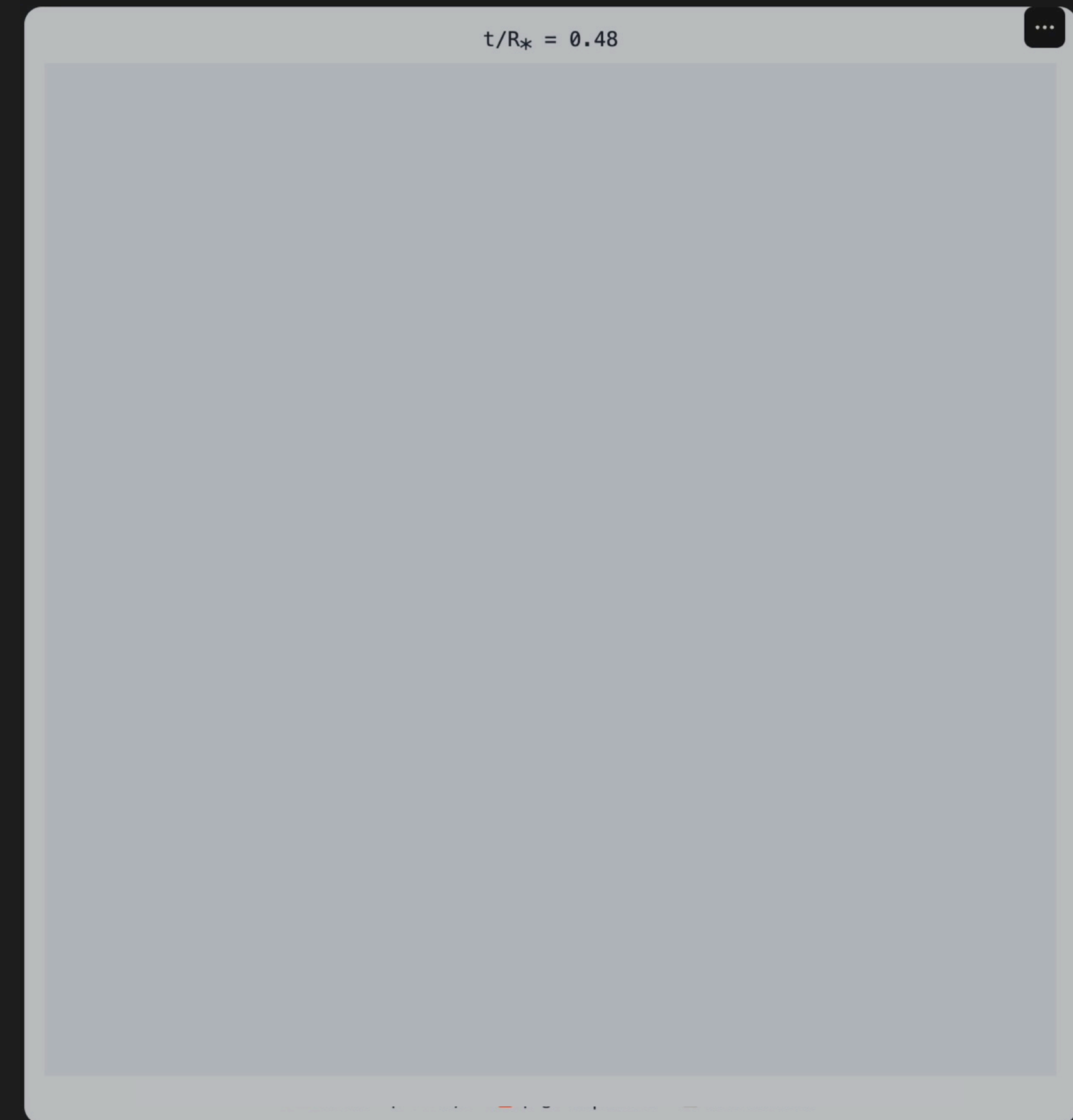
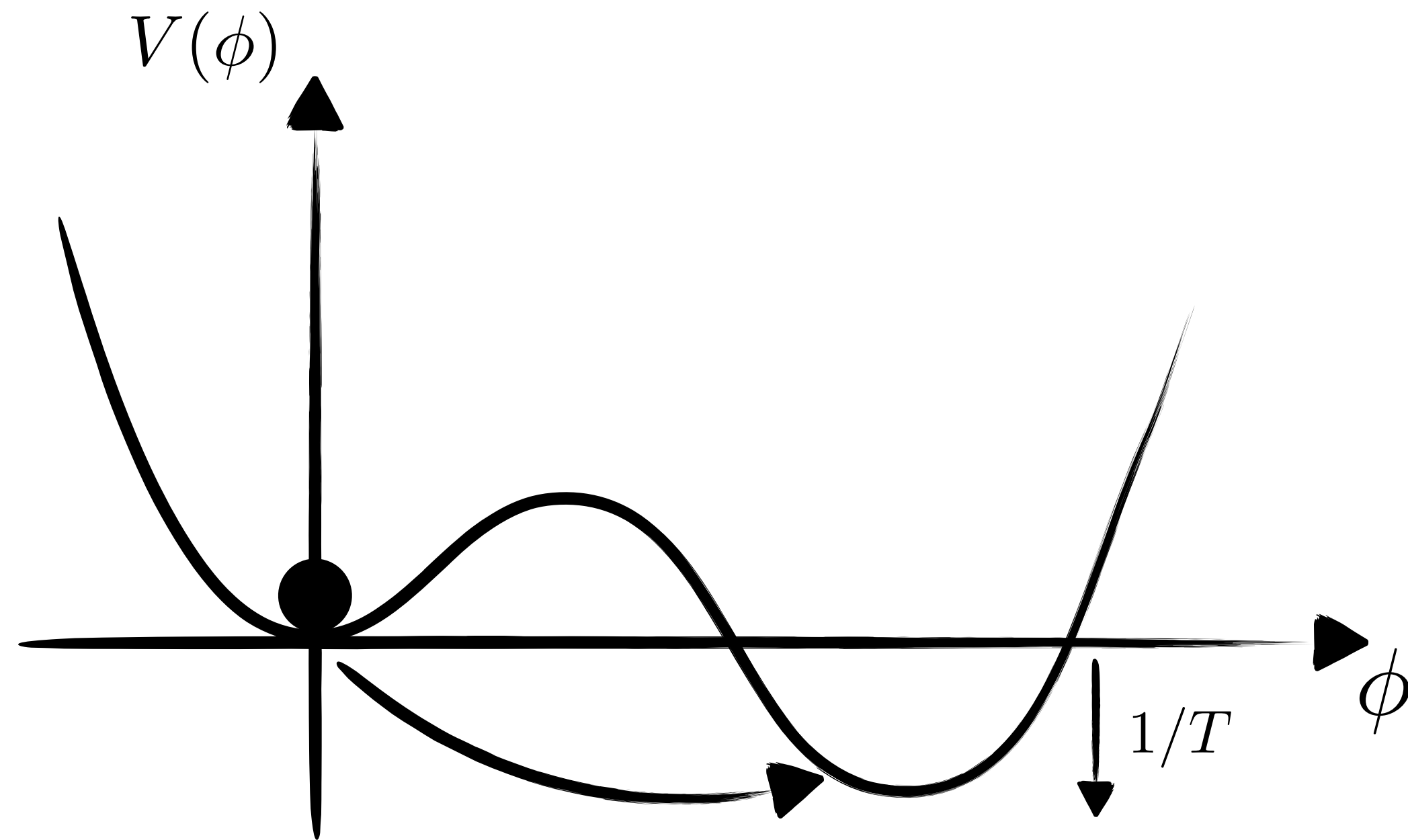
cosmological phase transitions

PHASE TRANSITIONS



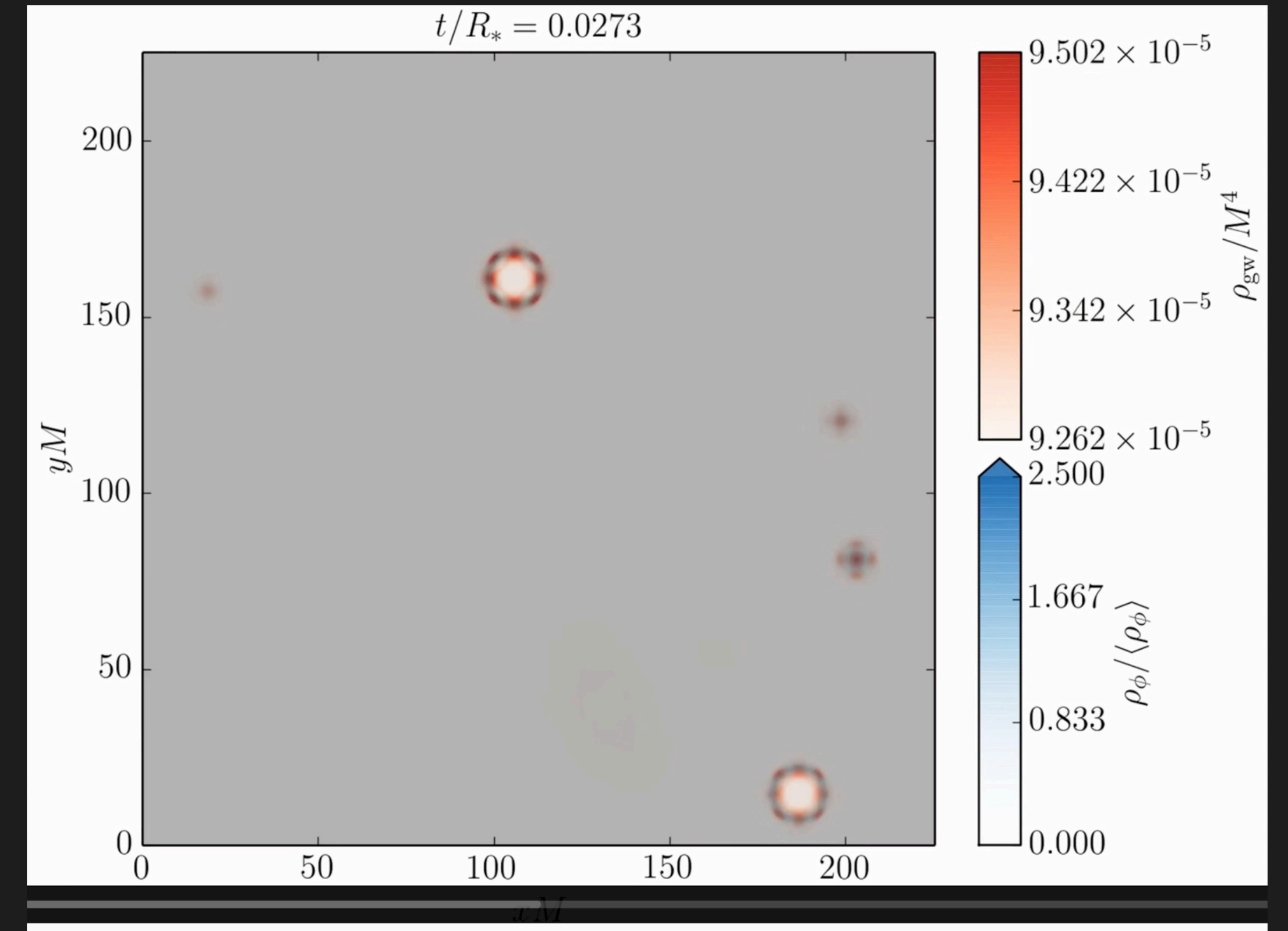
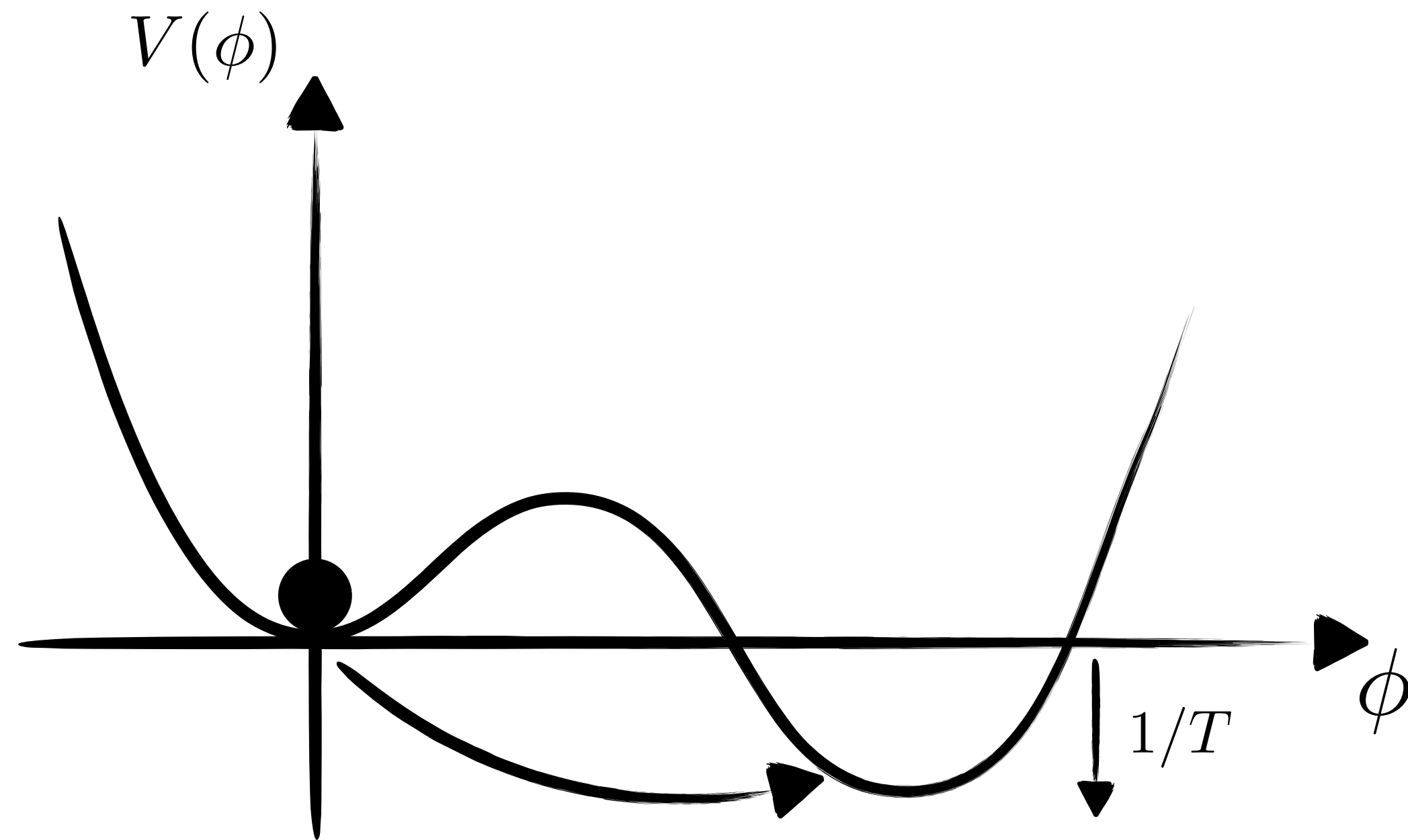
SLOW MOTION

PHASE TRANSITIONS



- bubble walls
- true vacuum
- GW production

PHASE TRANSITIONS



[arXiv: 1802.05712]

three contributions to GW production:

- bubble collisions
- sound waves collisions
- turbulence in the plasma

THE PTs SPECTRUM

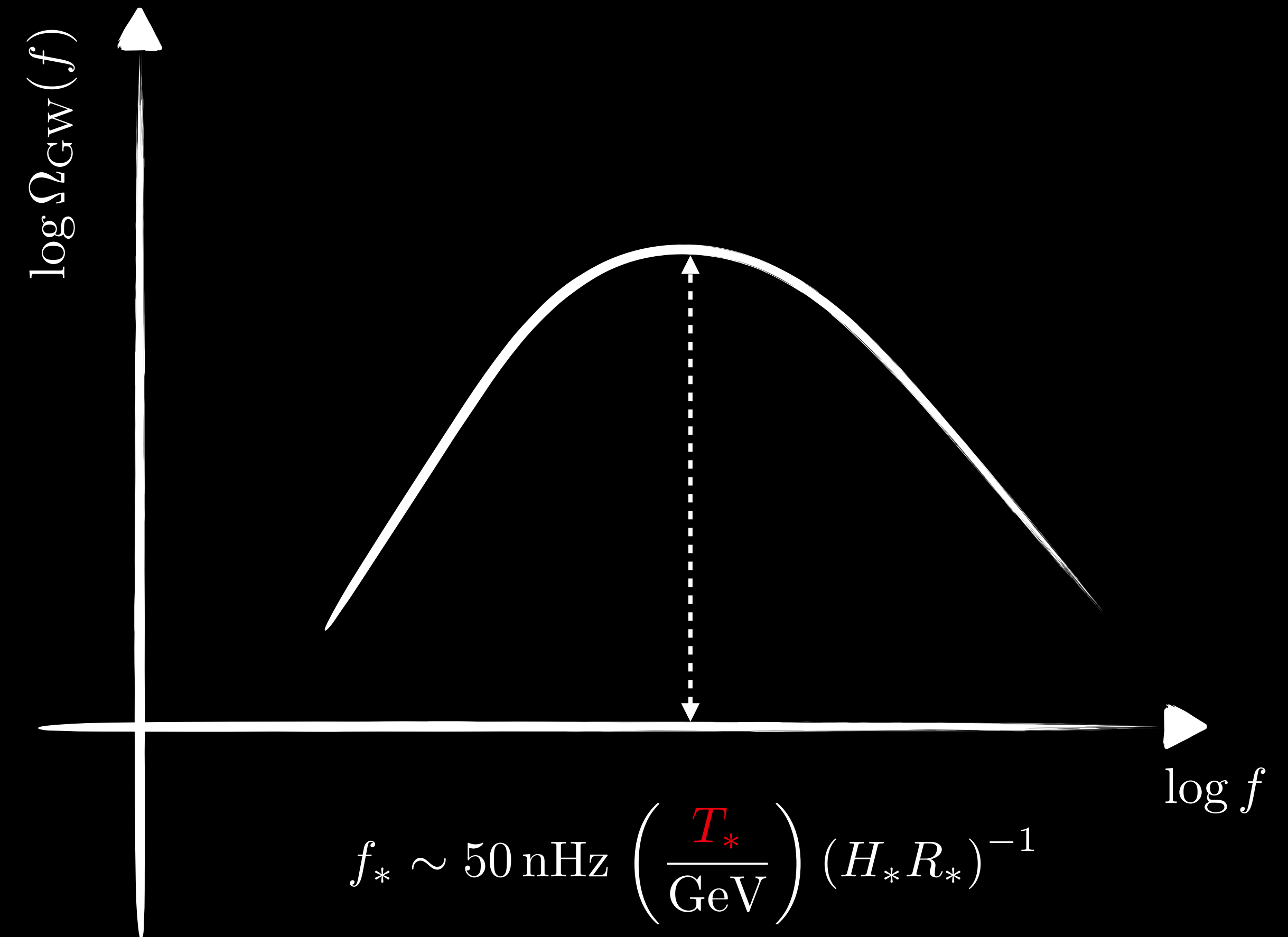
relevant parameters:

T_* : transition temperature

α_* : transition strength $\sim \Delta V / \rho_{\text{rad}}$

R_* : typical bubble size at nucleation

v_w : bubble wall speed



THE PTs SPECTRUM

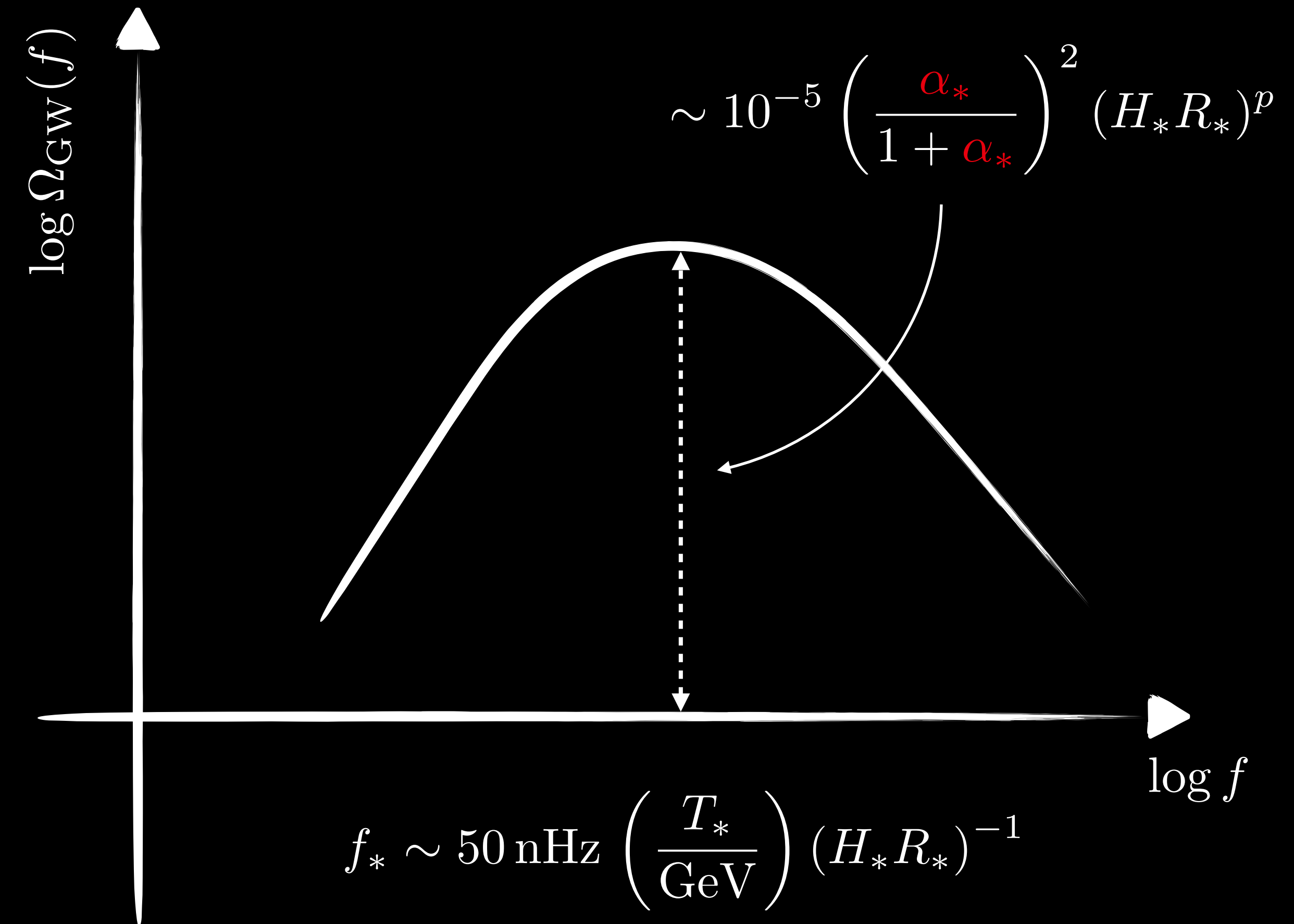
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THE PTs SPECTRUM

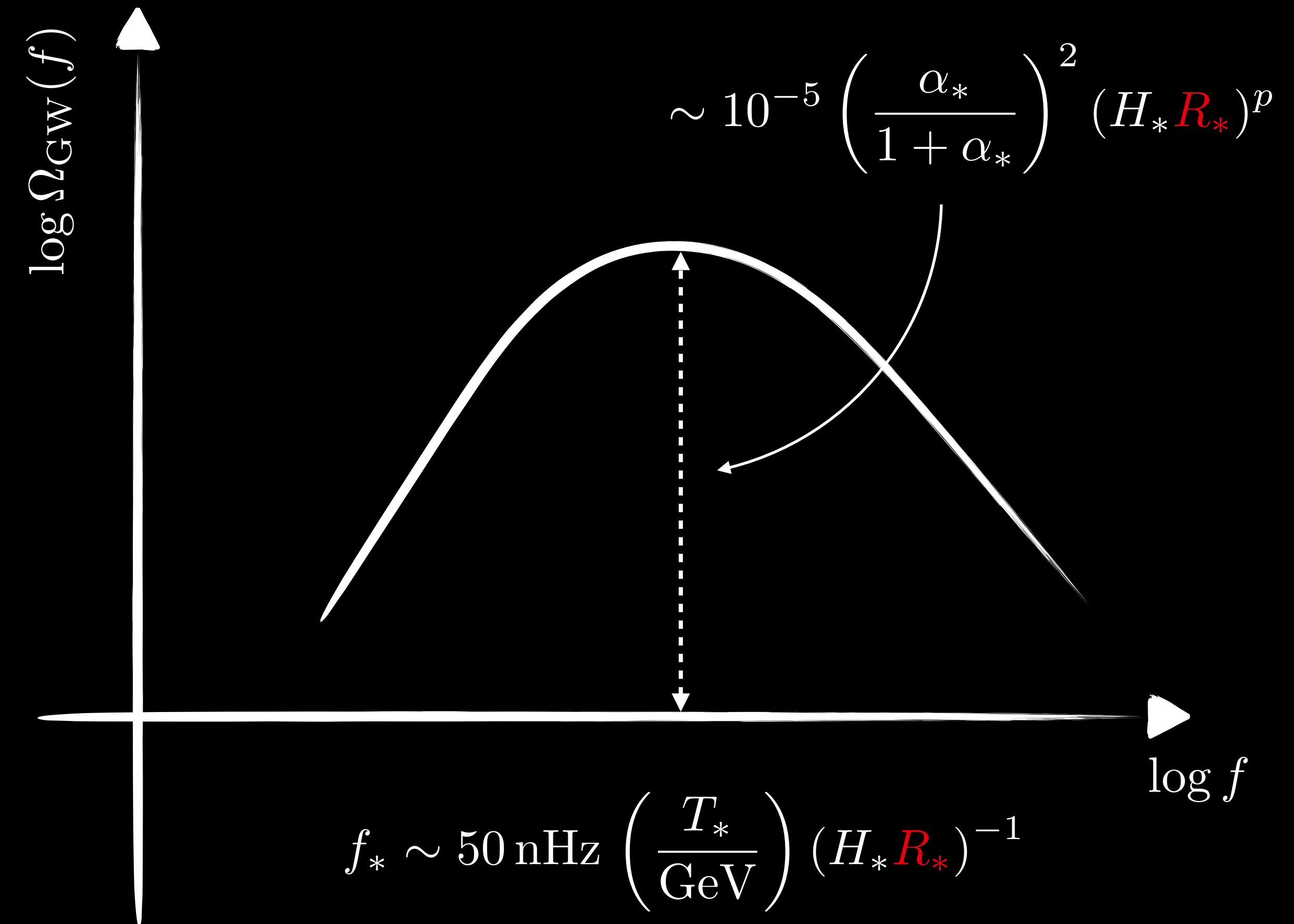
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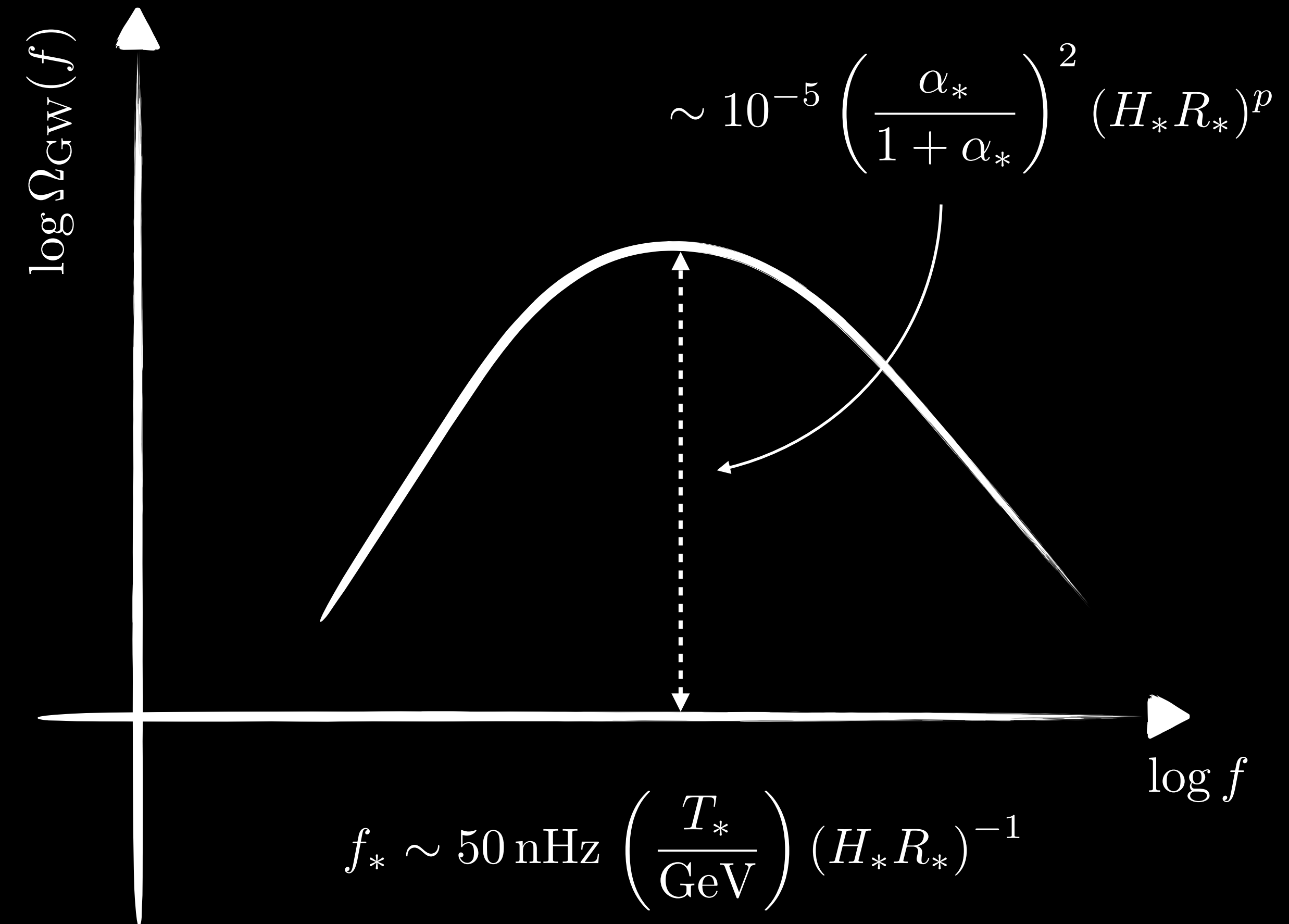
R_* : typical bubble size at nucleation

v_w : **bubble wall speed**

derived quantity of α_* and interactions between scalar field and plasma

we work in the simplifying (and optimistic) scenario where

$$v_w = 1$$



THE PTs SPECTRUM

relevant parameters:

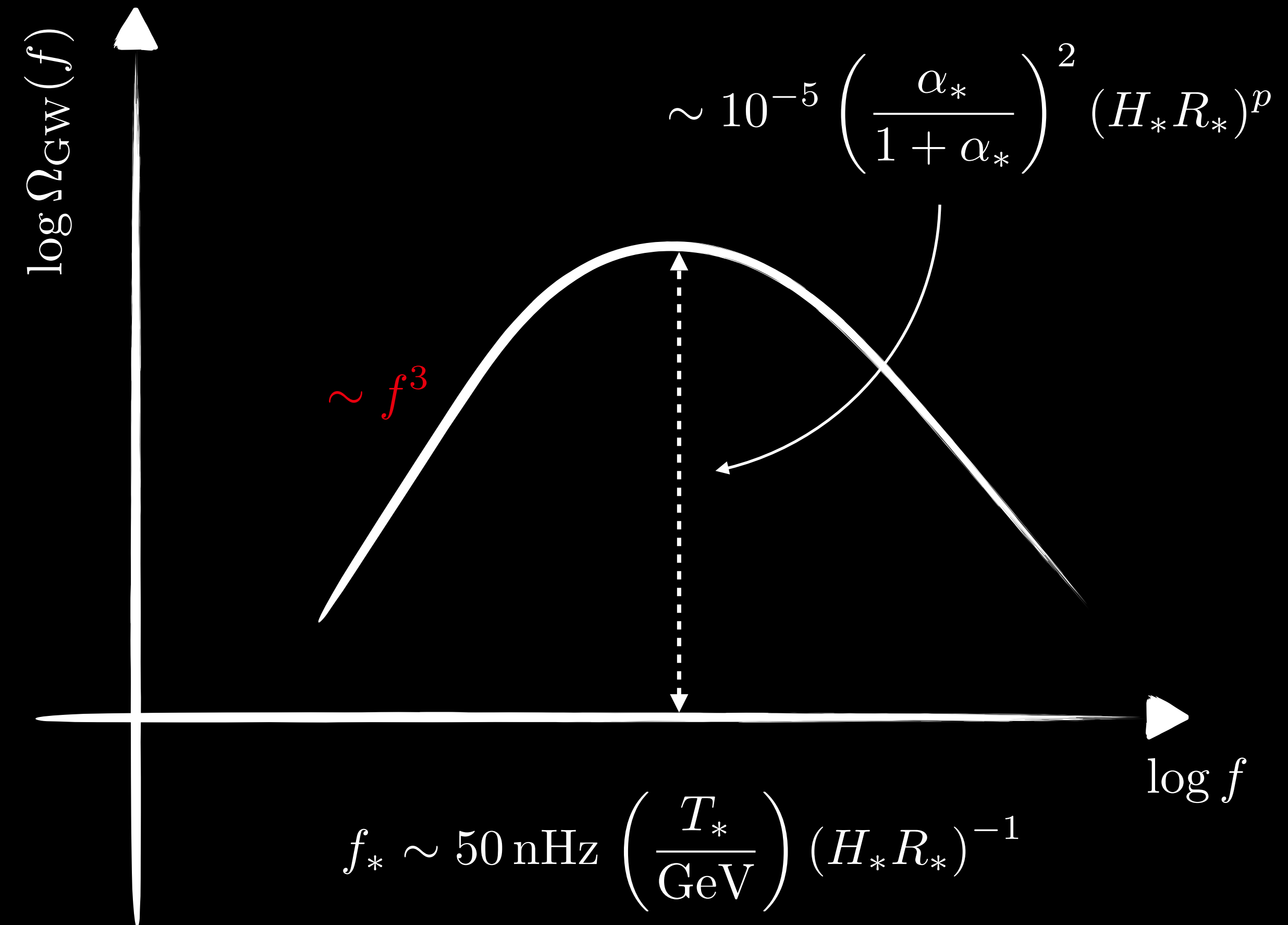
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- low-frequency slope is set by causality to f^3



THE PT_s SPECTRUM

relevant parameters:

T_* : transition temperature

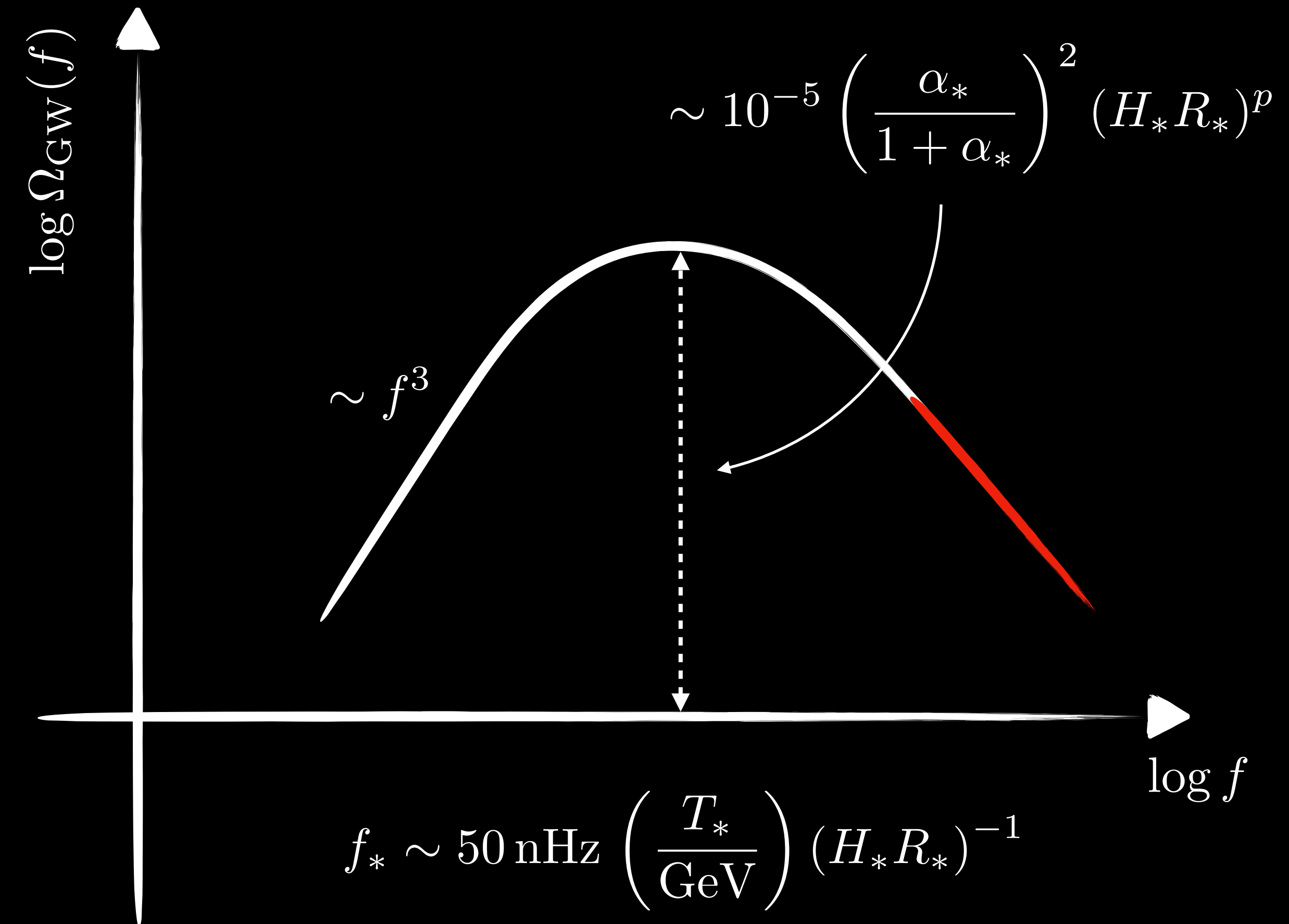
α_* : transition strength $\sim \Delta V / \rho_{\text{rad}}$

R_* : typical bubble size at nucleation

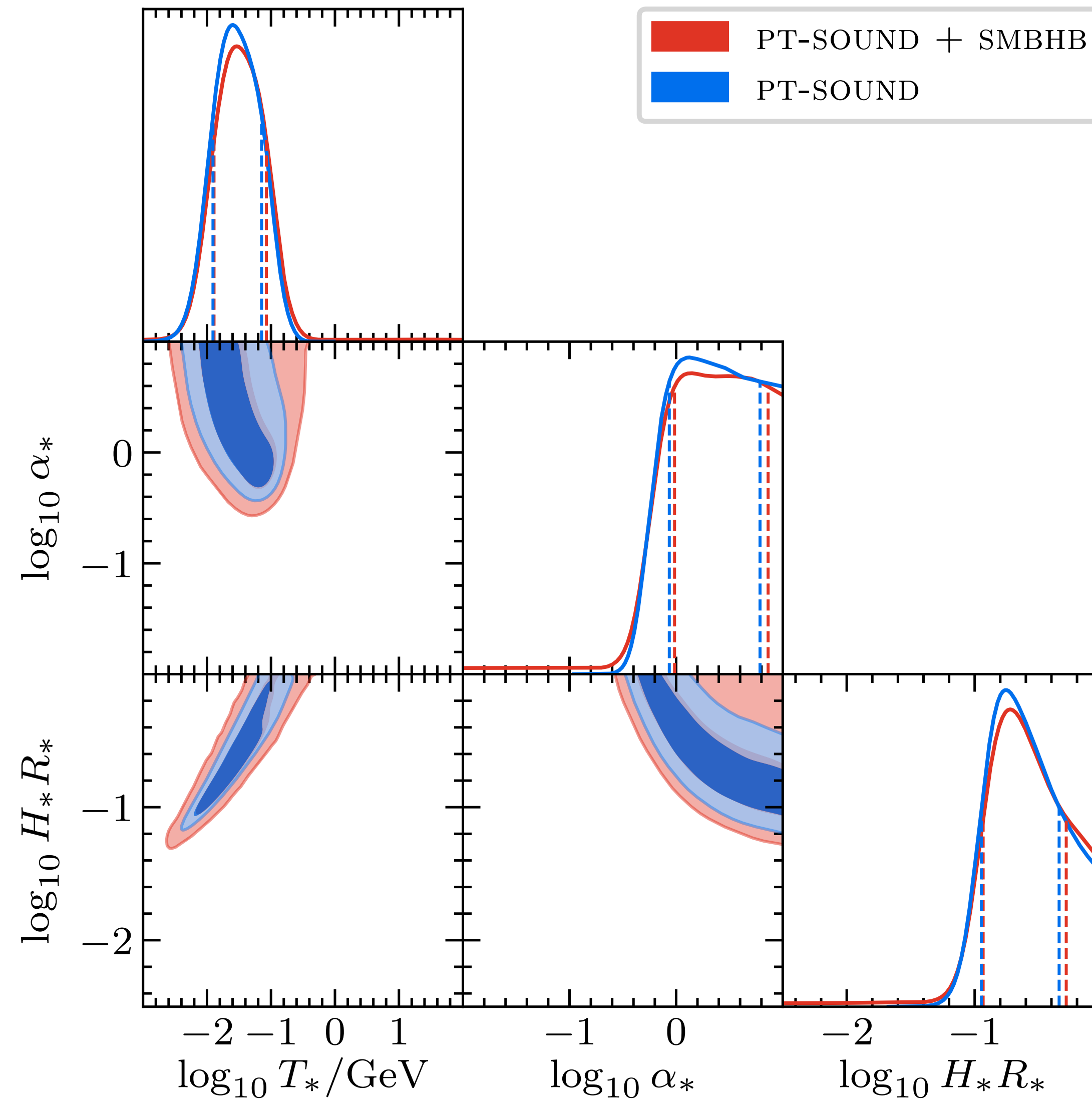
v_w : bubble wall speed

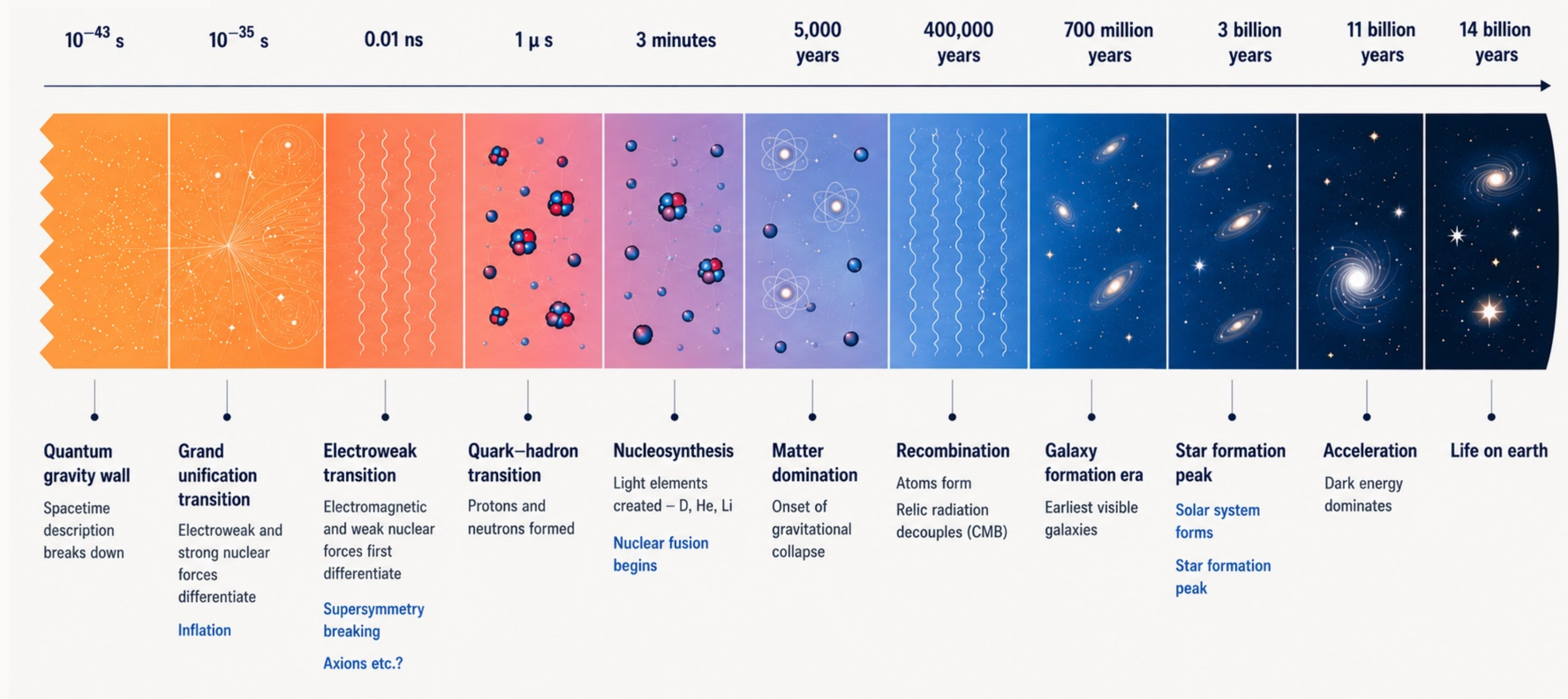
- low-frequency slope is set by causality to f^{-3}

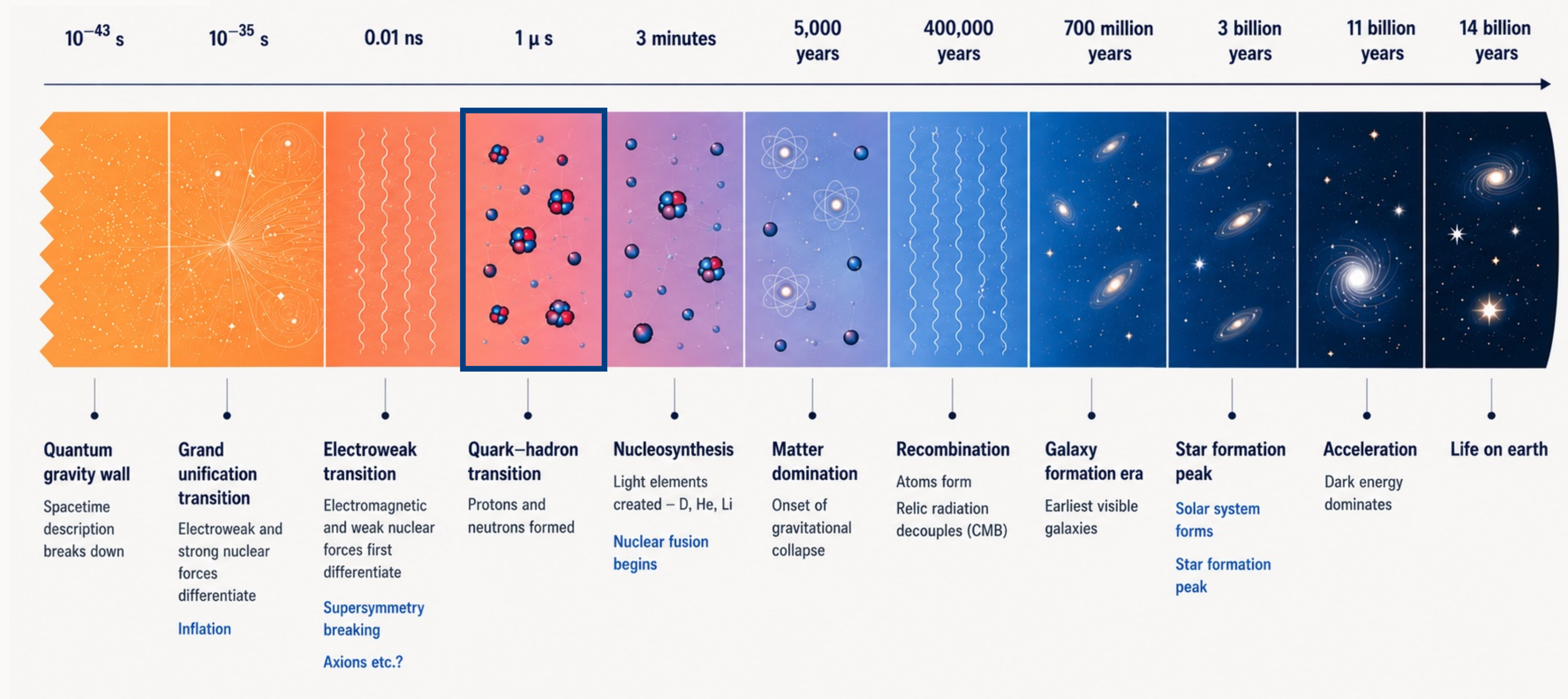
- **high-frequency slope is model-dependent**



NG15 POSTERIOR

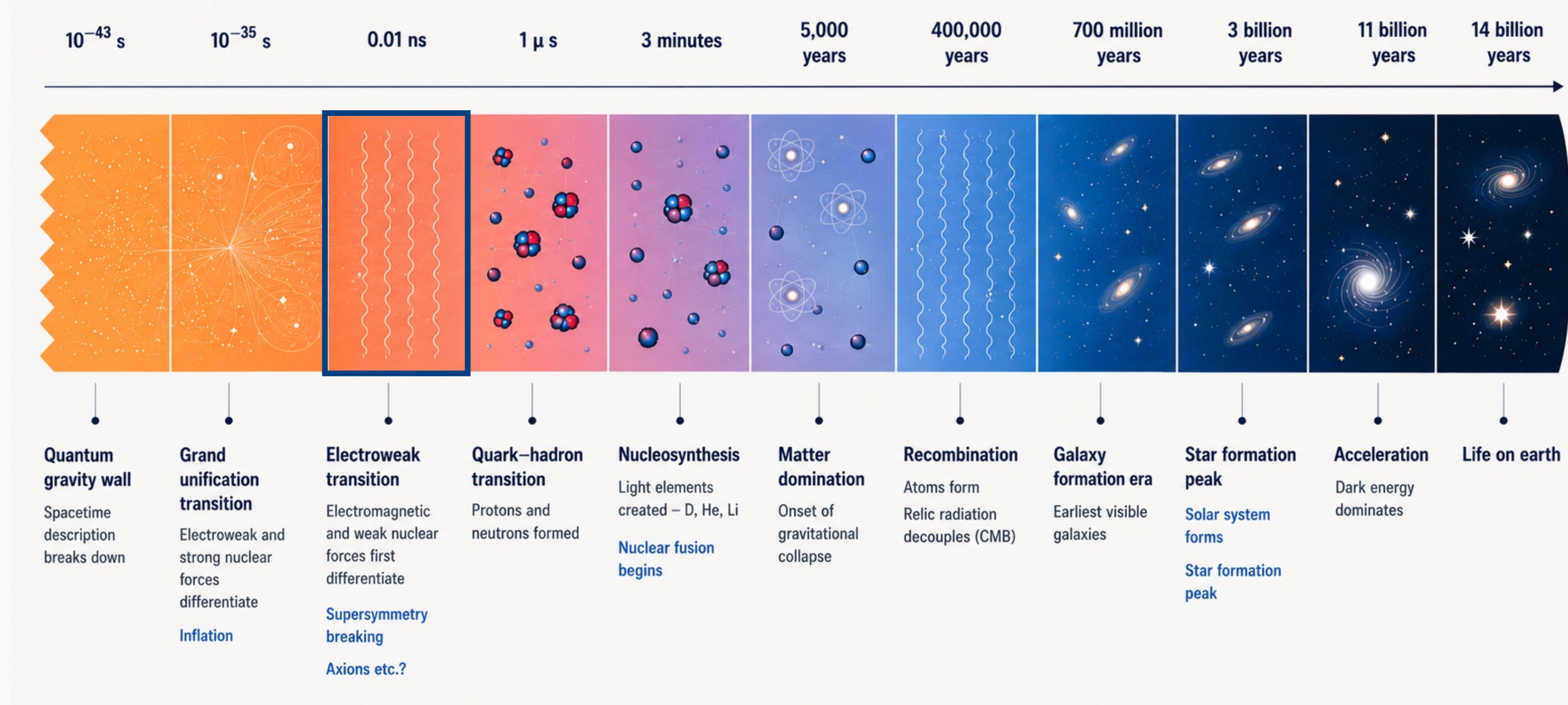






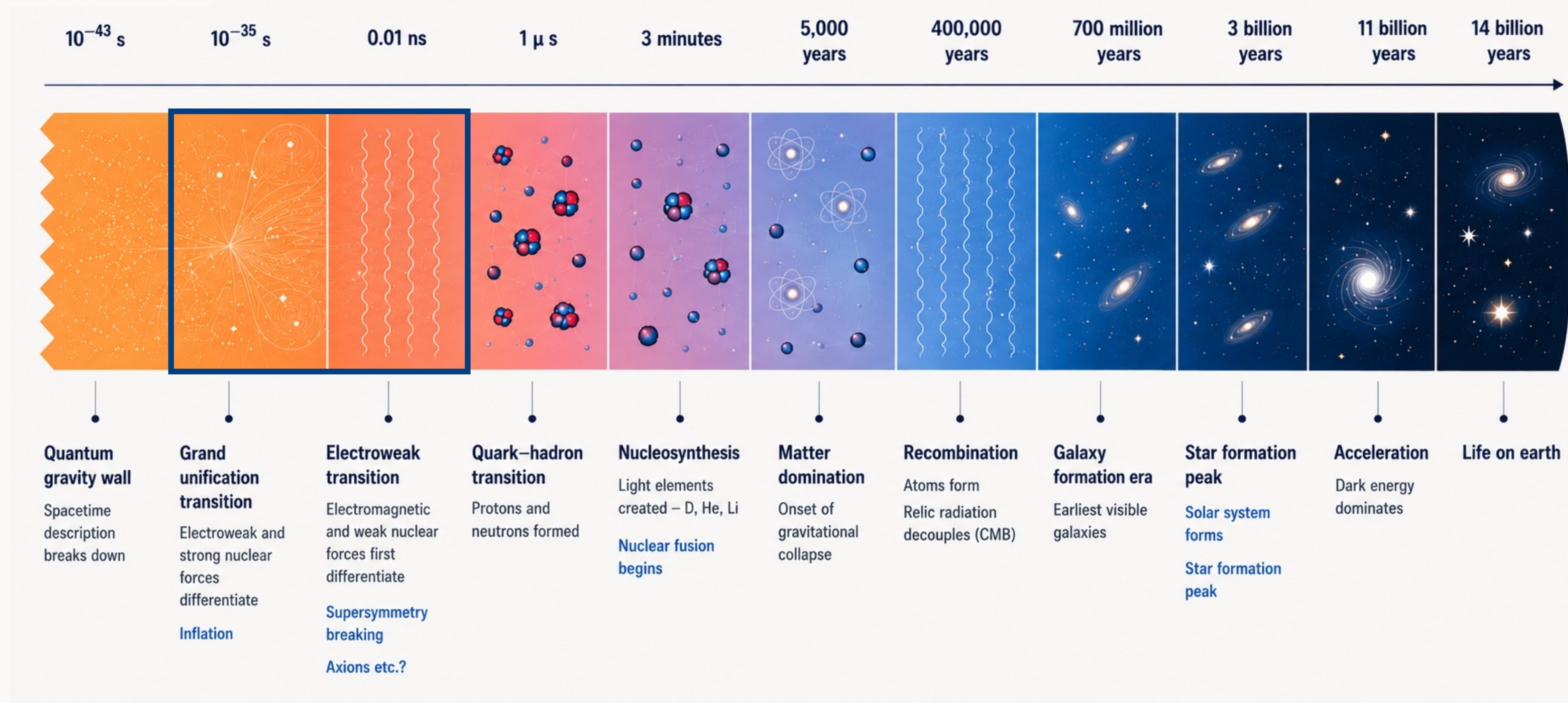
we know phase transitions have taken place in the early universe

- **QCD phase transition**: at $T \sim 150 \text{ MeV}$ quarks and gluons recombined into hadrons (proton, neutrons, ...)



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- **Electroweak phase transition**: at $T \sim 100 \text{ GeV}$ the Higgs field acquired a non-zero value giving particles a mass



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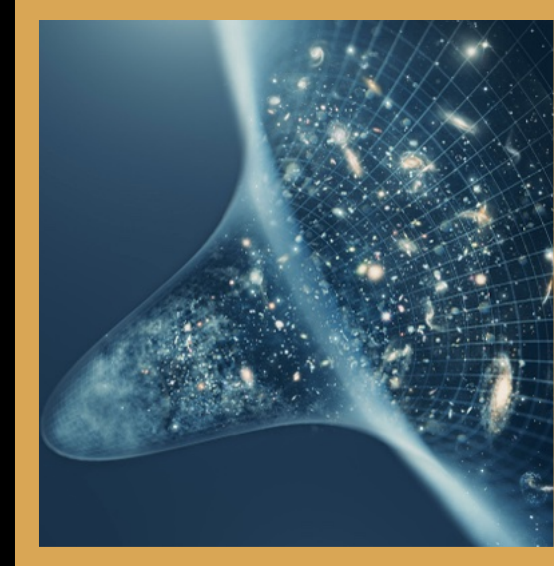
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many theories of Baryogenesis require a cosmological phase transition

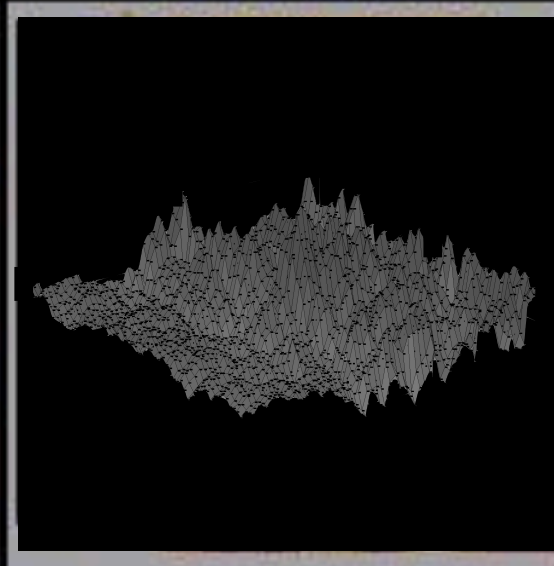
a quick survey of other primordial GWB sources

SELECT PLAYER

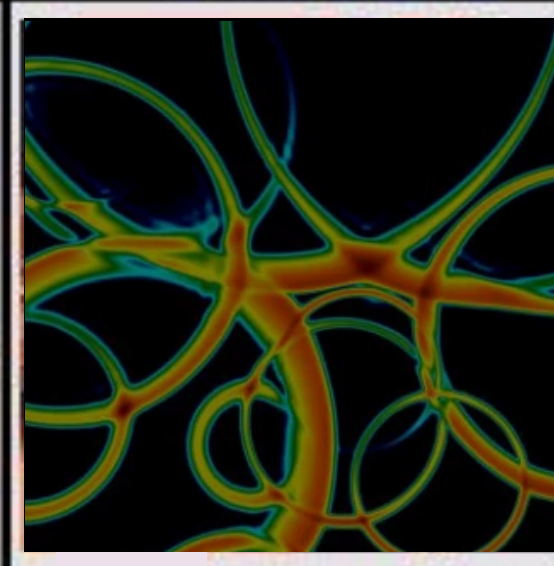
INFLATION



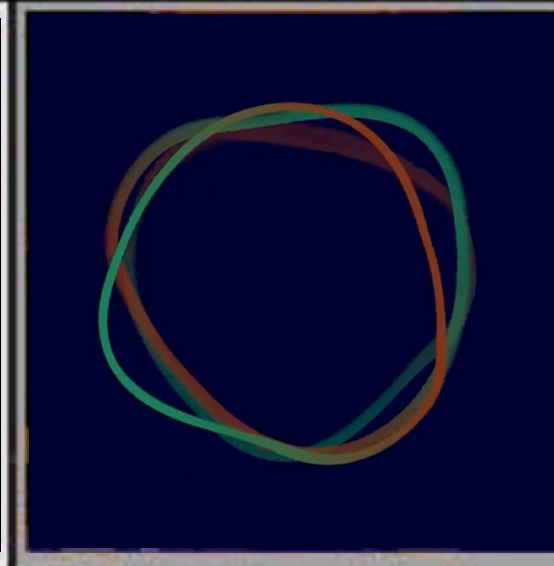
SIGW



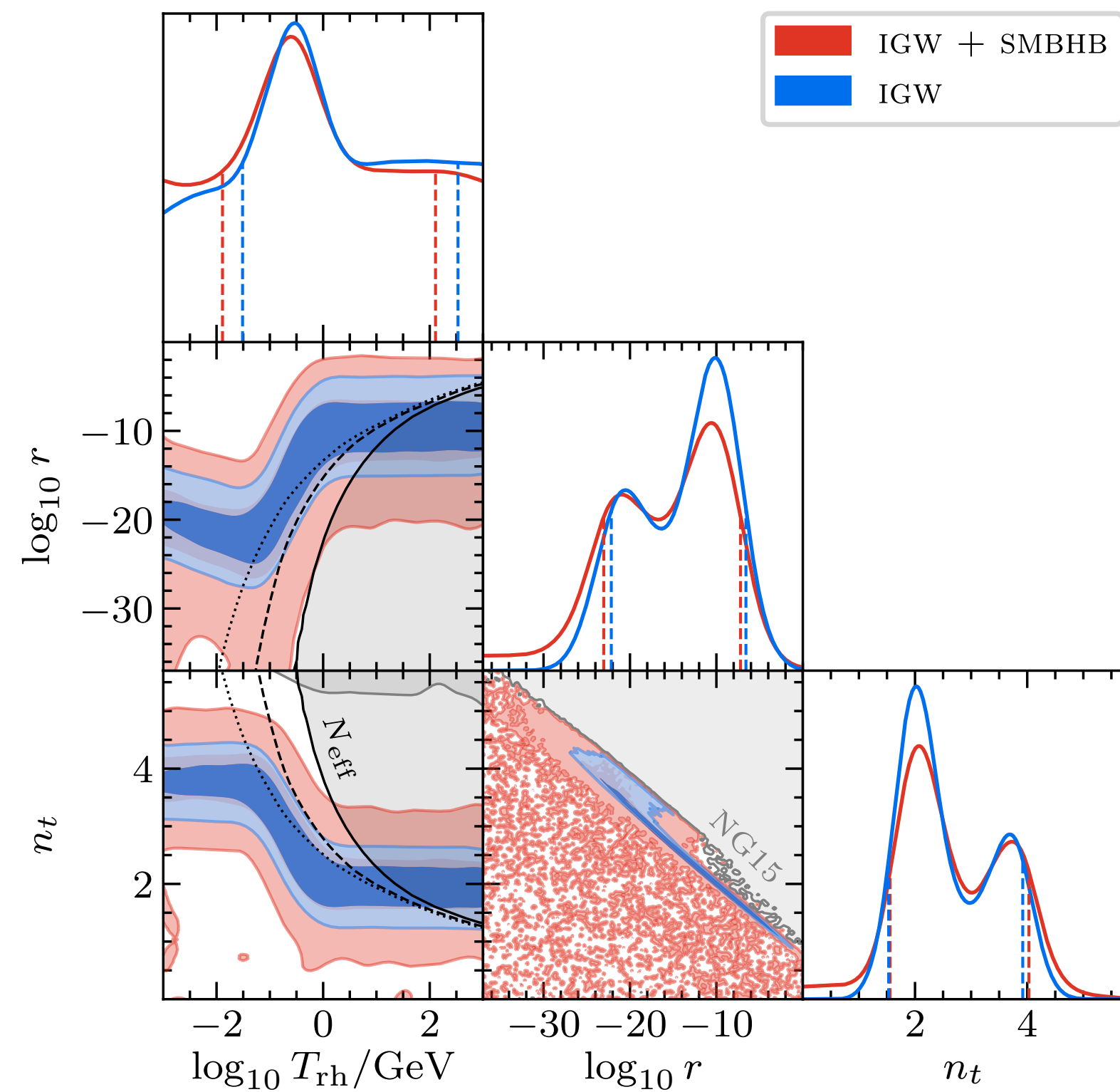
PT



STRINGS



Afzal et al. [2306.16219]



STRONG (BLUE) SPECTRAL TILT NEEDED

REHEATING TEMP. BELOW 100 GEV AVOIDS N_{eff} CONST.

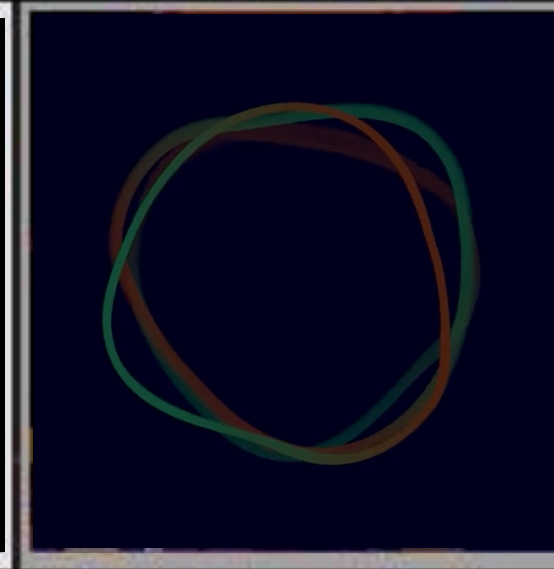
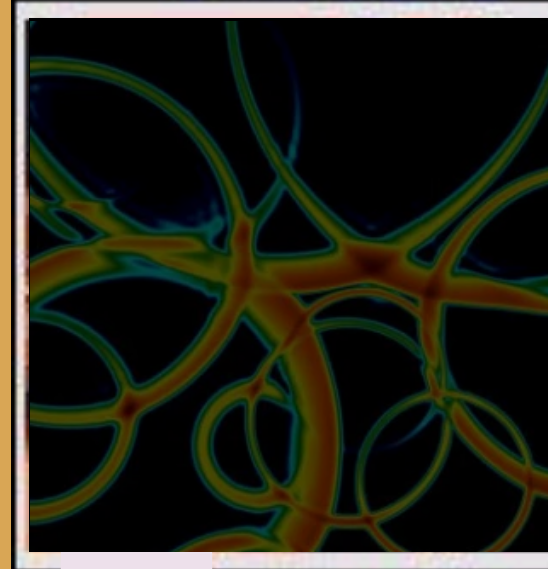
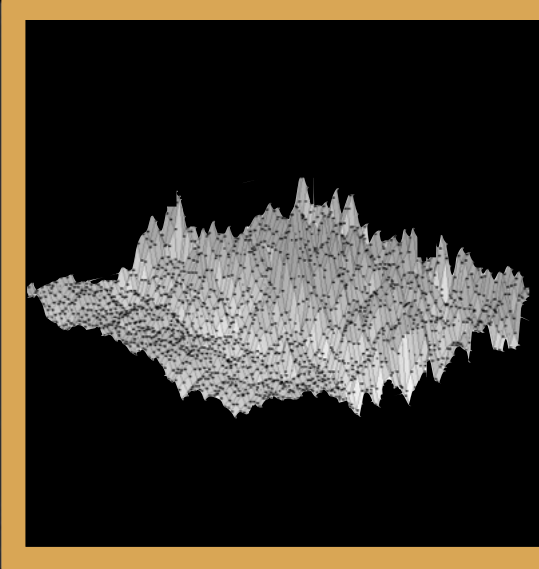
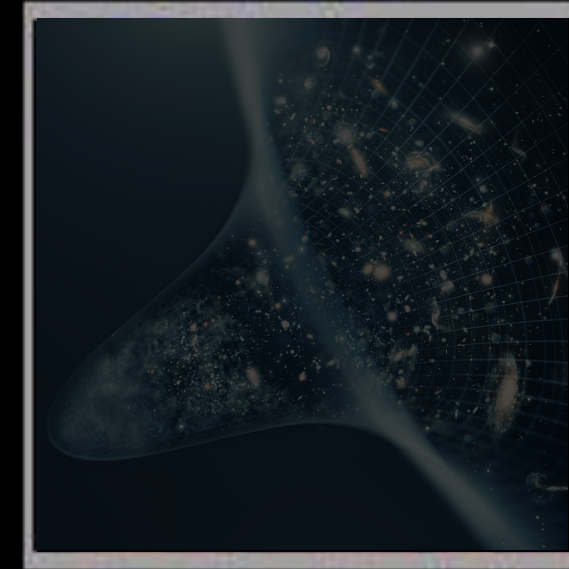
SELECT PLAYER

INFLATION

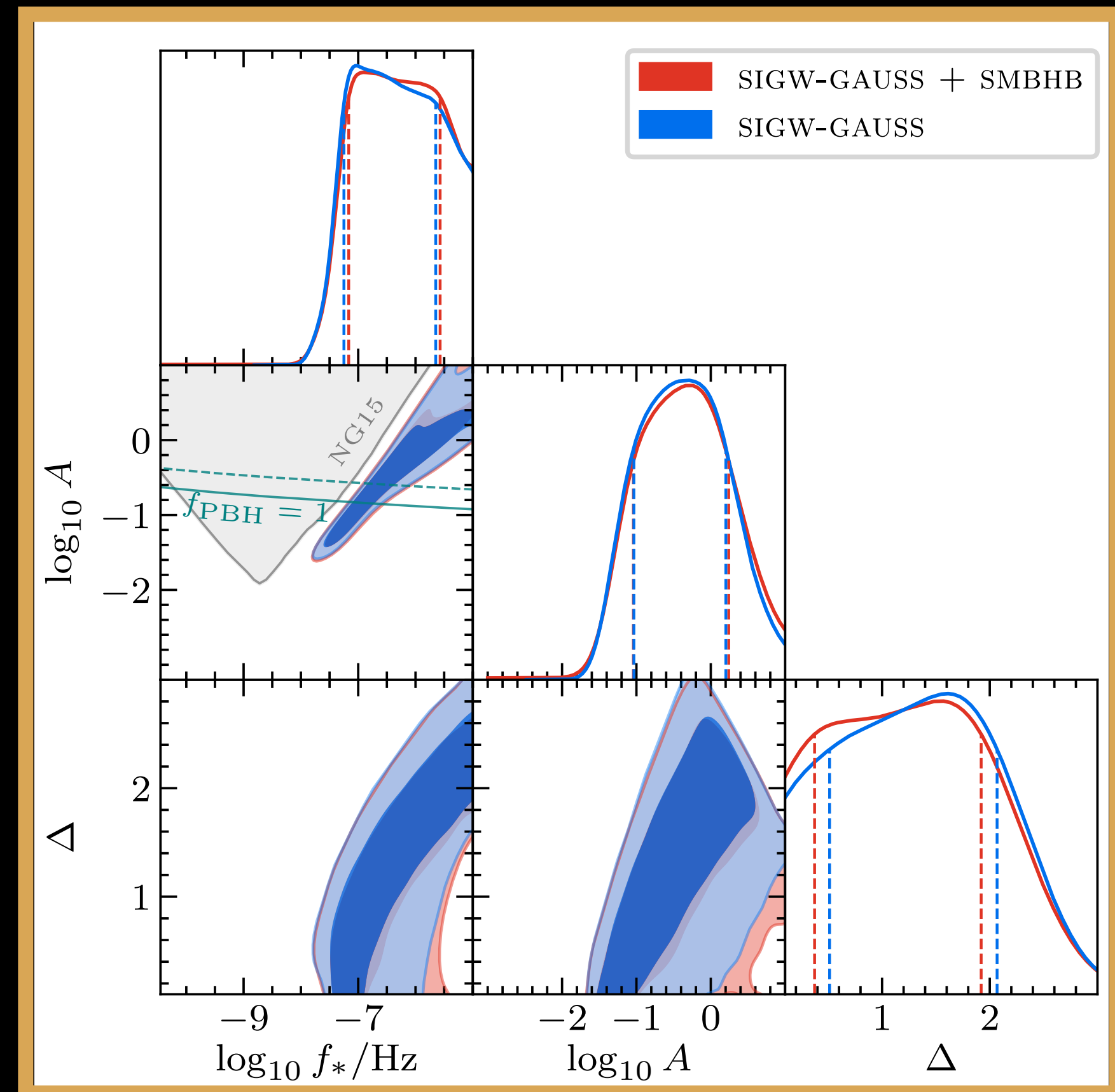
SIGW

PT

STRINGS



Afzal et al. [2306.16219]



LARGE FEATURE AT MPC SCALES NEEDED

RISK OF OVERPRODUCING PBH

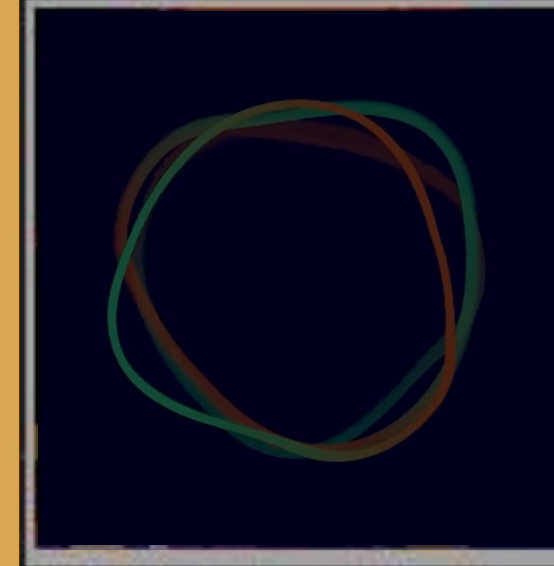
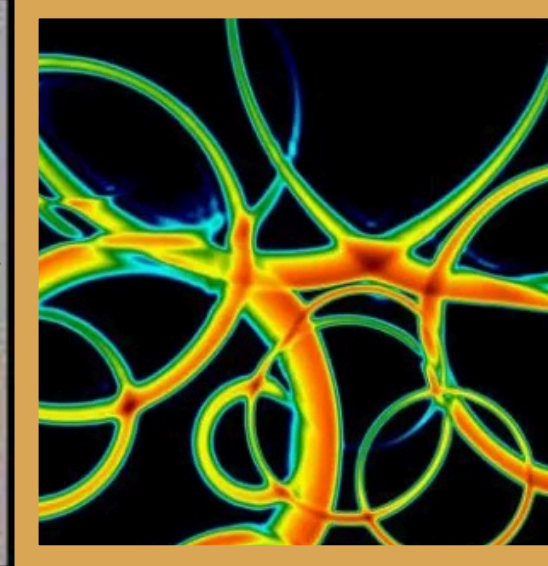
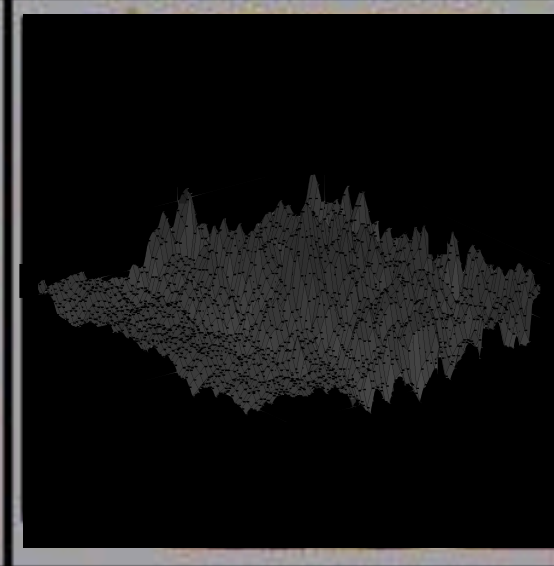
SELECT PLAYER

INFLATION

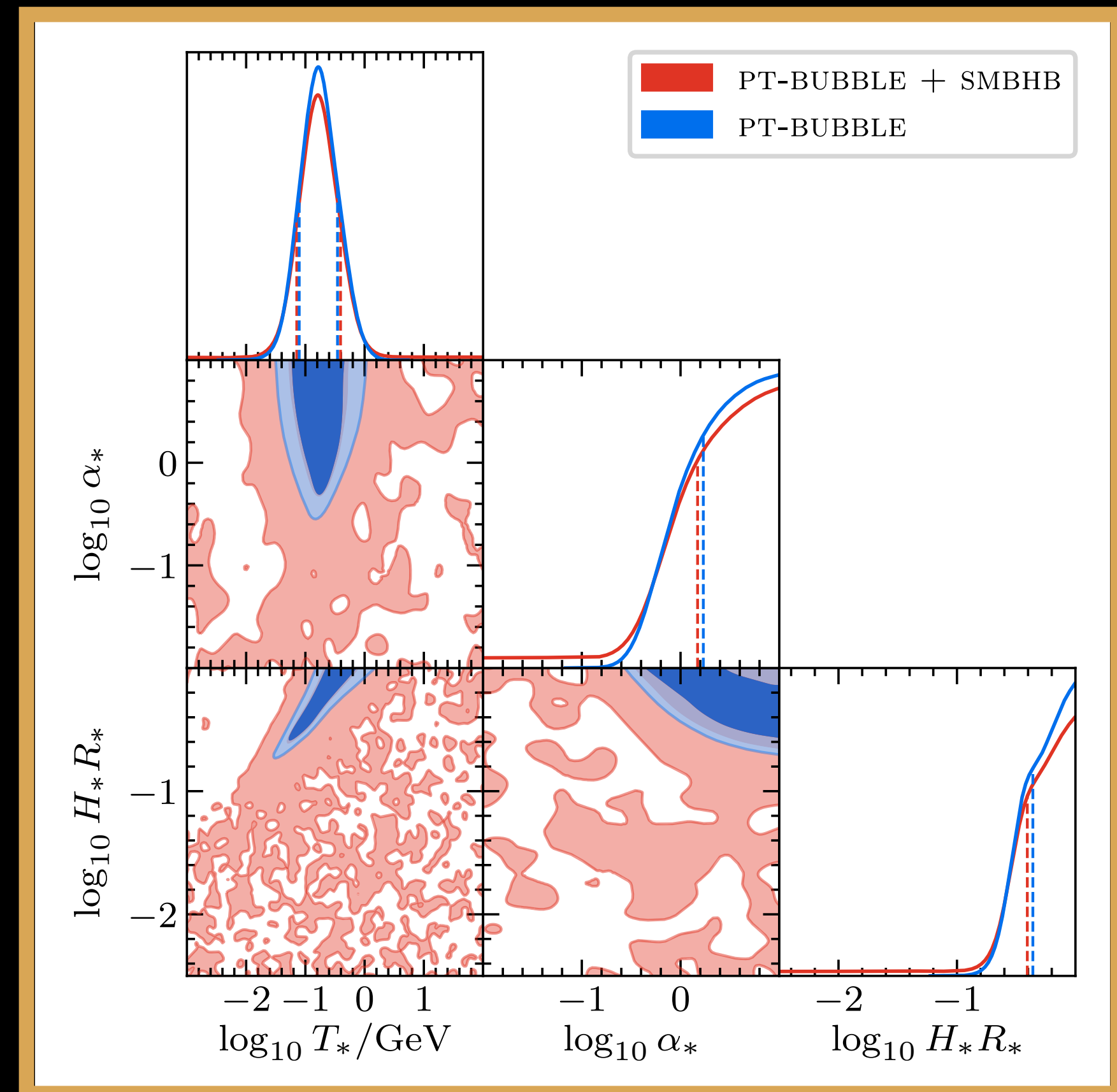
SIGW

PT

STRINGS



Afzal et al. [2306.16219]



10-100 MEV TRANSITION TEMP. NEEDED

SLOW TRANSITION NEEDED

EXTREMELY STRONG TRANSITION NEEDED

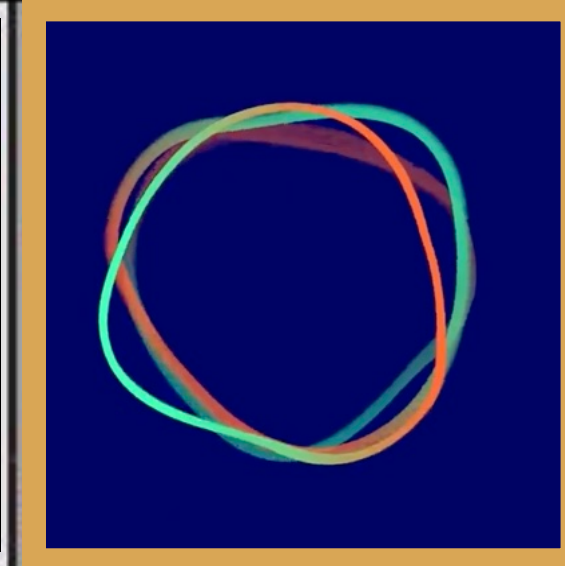
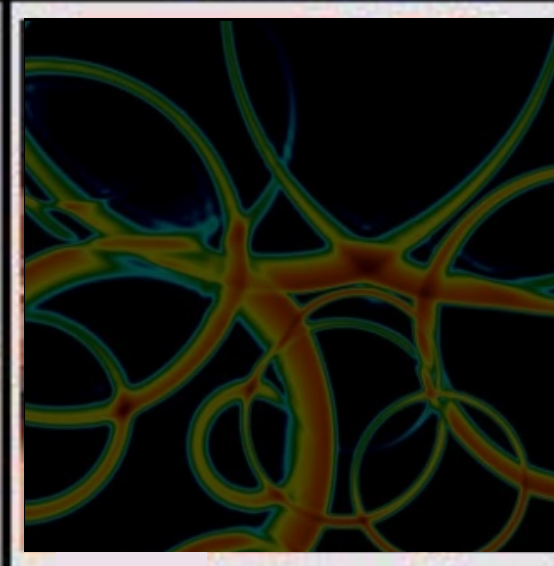
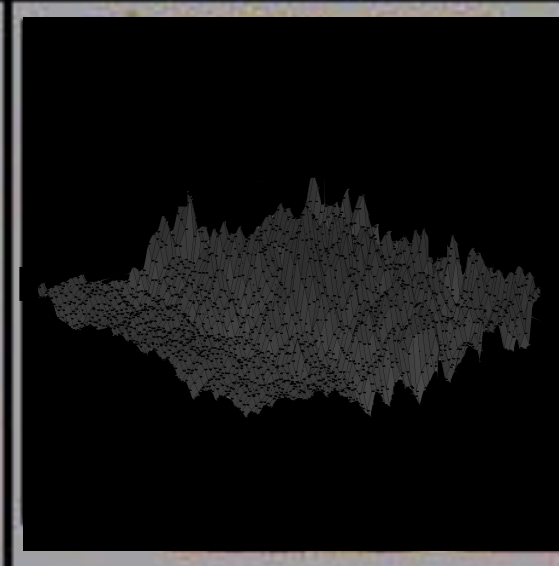
SELECT PLAYER

INFLATION

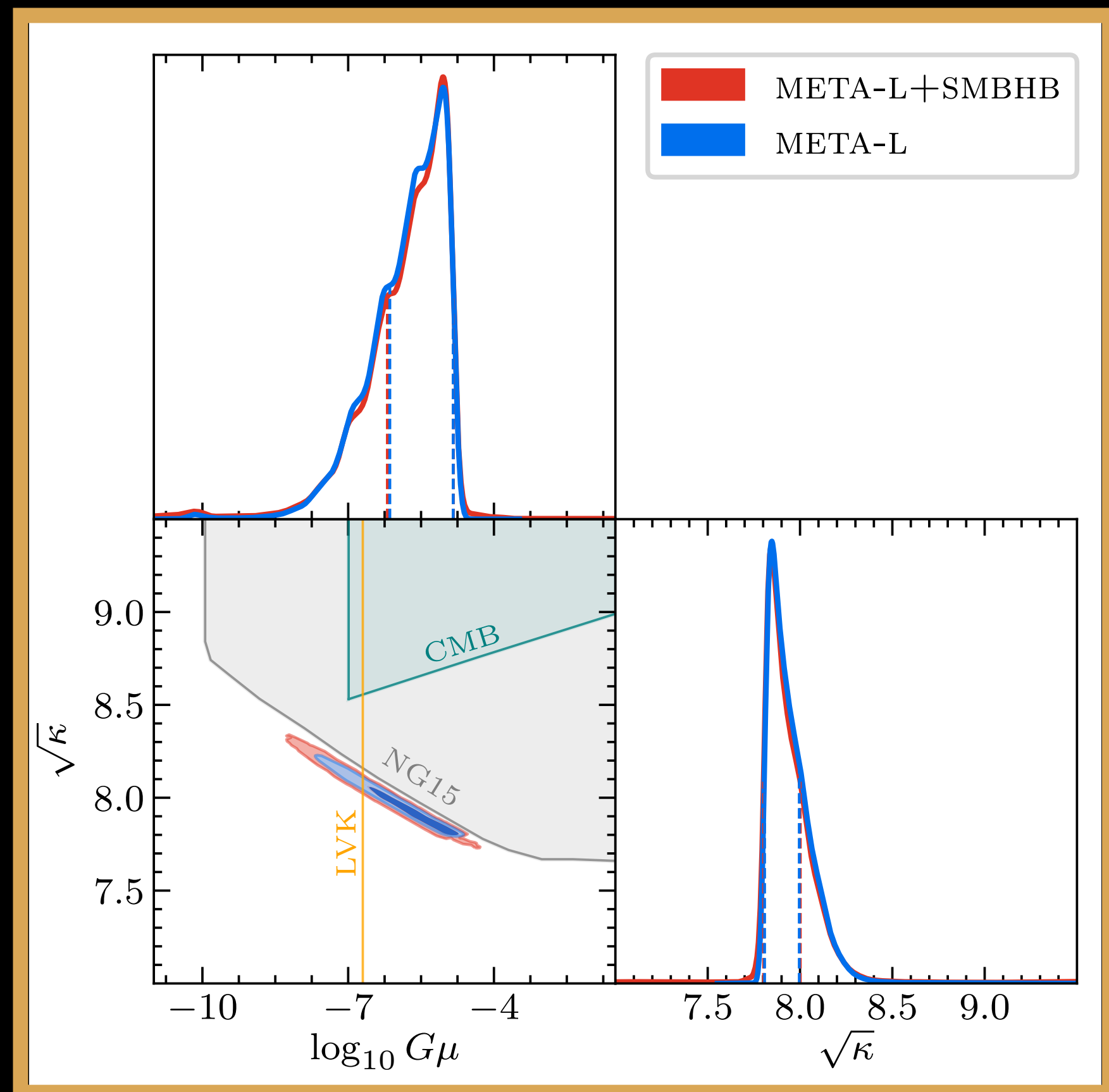
SIGW

PT

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STABLE STRING DO NOT WORK

METASTABLE STRING DO WORK

GUT SCALE STRING TENSION COULD WORK

questions?